

Project Data Page:





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Front end KOD (V6502 ABC)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Front end Flare KOD (V6502 ABC) 9%.pvdb

Date of Analysis : Sep 16,2026 3:25pm

PV Elite 28 U1, March 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 6500.000 mm. ID

Vessel Design Length, Tangent to Tangent 20500.00 mm.

Specified Datum Line Distance 0.00 mm.

Internal Design Temperature 415 °C

Internal Design Pressure 3.500 bars

External Design Temperature 415 °C

External Design Pressure 1.034 bars

Maximum Allowable Working Pressure 3.5 bars

External Max. Allowable Working Pressure 1.15 bars

Shop Test Pressure 9.545 bars

Field Test Pressure 6.648 bars

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -48.0 °C

Wind Design Code ASCE/SEI 7-16

Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
left Dish	3.544	1.03	4.790	1.6000	No	Yes
Shell-1	3.544	1.03	4.770	1.6000	N/A	Yes
Shell2	3.544	1.03	4.770	1.6000	N/A	Yes
Shell3	3.544	1.03	4.770	1.6000	N/A	Yes
Shell4	3.544	1.03	4.770	1.6000	N/A	Yes
Shell5	3.544	1.03	4.770	1.6000	N/A	Yes
Shell6	3.544	1.03	4.770	1.6000	N/A	Yes
shell7	3.544	1.03	4.770	1.6000	N/A	Yes
Right dish	3.544	1.03	4.790	1.6000	No	Yes

Liquid Level: 450.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.989**Stiffener Ring Specifications:**

Elevation mm. Selected Type Ring Description



3288.43	Bar 260.0 x 26.0	Ring:30 - 1
7168.50	Bar 260.0 x 26.0	Ring:40 - 2
11650.00	Bar 260.0 x 26.0	Ring:50 - 3
14928.82	Bar 260.0 x 26.0	Ring:60 - 4
18808.94	Bar 260.0 x 26.0	Ring:80 - 5

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	26.0	22.0	16.5	...	0.85	0.85
Cylinder	3050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	6050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	9050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	12050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	15050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	18050.0	3000.0	22.0	22.0	16.6	...	0.85	0.85
Cylinder	20450.0	2400.0	22.0	22.0	16.6	...	0.85	0.85
Ellipse	20500.0	50.0	26.0	22.0	16.5	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	22	20.5609	1.03421	1.2
20	30	22	19.9021	1.03421	1.35
30	Ring	22	19.9021	1.03421	1.35
Ring	40	22	20.0009	1.03421	1.33
40	Ring	22	20.0009	1.03421	1.33
Ring	50	22	21.1511	1.03421	1.15
50	Ring	22	21.1511	1.03421	1.15
Ring	60	22	18.7941	1.03421	1.59
60	Ring	22	18.7941	1.03421	1.59
Ring	70	22	20.001	1.03421	1.33
70	80	22	20.001	1.03421	1.33
80	Ring	22	20.001	1.03421	1.33
Ring	90	22	16.2249	1.03421	1.92
90	100	22	20.5609	1.03421	1.2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	3830.09	4939.52	No Calc	No Calc
30	Ring	3830.09	4939.52	No Calc	No Calc
Ring	40	3880.07	4939.69	9596.25	11119.4
40	Ring	3880.07	4939.69	No Calc	No Calc
Ring	50	4481.5	4940.2	10415.2	11119.4
50	Ring	4481.5	4940.2	No Calc	No Calc
Ring	60	3278.82	4940.33	9653.73	11119.4
60	Ring	3278.82	4940.33	No Calc	No Calc
Ring	70	3880.12	4939.69	8901.8	11119.4
70	80	3880.12	4939.69	No Calc	No Calc
80	Ring	3880.12	4939.69	No Calc	No Calc
Ring	90	2232.72	4939.88	7602.64	11119.4
90	100	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Analysis	Max Stress	High Stress	Pass
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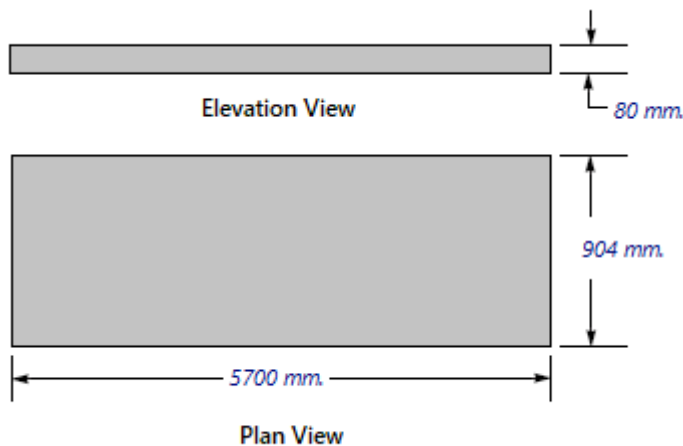
Description	Type	Ratio	Location	Fail
Trunnion	WRC-107/537	0.947	n/a	Passed
Trunnion	WRC-107/537	0.933	n/a	Passed
N1A-40"	WRC-297	1.213	Nozzle	Failed
N2-52"	WRC-297	1.096	Nozzle	Failed
N3-4"	WRC-297	0.474	Shell	Passed
N1B-40"	WRC-297	1.261	Nozzle	Failed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	350.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	3600.000 mm.
Wear Pad Width	800.000 mm.
Wear Pad Thickness	22.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	4100.000 mm.
Baseplate Length	5700.000 mm.
Baseplate Thickness	80.000 mm.
Baseplate Width	904.000 mm.
Number of Ribs (including outside ribs)	6
Rib Thickness	28.000 mm.
Web Thickness	28.000 mm.
Height of Center Web	226.000 mm.
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	110999.22 Kg
Maximum Transverse Saddle Shear Load	6061.83 Kg
Maximum Longitudinal Saddle Shear Load	12123.67 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	465407.25 Kg
Maximum Transverse Saddle Shear Load	1803.97 Kg
Maximum Longitudinal Saddle Shear Load	2183.97 Kg

Weights:

Fabricated - Bare w/o removable internals	130819.5 kg.
Shop Test - Fabricated + water (full)	882644.7 kg.
Shipping - Fab. + rem. intls.+ shipping app.	130819.5 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	171377.5 kg.
Empty - Fab. + intls. + details + wghts.	171377.5 kg.

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Vessel Design Summary:



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Operating - Empty + operating liquid (no ca)	192656.1 kg.
Field Test - Empty weight + test medium (full)	923202.6 kg.

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**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1-24"	3.5	OK	...	...		OK	Passed
K6-2"	3.5	OK	...	OK	6.40	OK	Passed
K6-2"	3.5	...	...	OK	6.40	OK	No Calc[*]
N1A-40"	3.5	OK	...	OK	9.93	OK	Passed
M2-24"	3.5	OK	...	...		OK	Passed
N2-52"	3.5	OK	...	OK	9.93	OK	Passed
N3-4"	3.5	OK	...	OK	6.86	OK	Passed
N1B-40"	3.5	OK	...	OK	9.93	OK	Passed
K1A	3.5	...	...	OK	6.40	OK	No Calc[*]
K2A	3.5	...	...	OK	6.40	OK	No Calc[*]
K3A	3.5	...	...	OK	6.40	OK	No Calc[*]
K4A	3.5	...	...	OK	6.40	OK	No Calc[*]
K5A	3.5	...	...	OK	6.40	OK	No Calc[*]
K1B	3.5	OK	...	OK	6.40	OK	Passed
K1B	3.5	...	...	OK	6.40	OK	No Calc[*]
K2B	3.5	OK	...	OK	6.40	OK	Passed
K2B	3.5	...	...	OK	6.40	OK	No Calc[*]
K3B	3.5	OK	...	OK	6.40	OK	Passed
K3B	3.5	...	...	OK	6.40	OK	No Calc[*]
K4B	3.5	OK	...	OK	6.40	OK	Passed
K4B	3.5	...	...	OK	6.40	OK	No Calc[*]
K5B	3.5	OK	...	OK	6.40	OK	Passed
K5B	3.5	...	...	OK	6.40	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3.5 Nozzle : K5B
 Minimum MAWP Shells/Flanges : 3.5 Element : shell17
 Minimum MAPnc Shells/Flanges : 7.91 Element : shell17

Computed Vessel M.A.W.P. : 3.5 bars

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	M1-24"	0.000	0.000	1185.600
10	K6-2"	0.000	180.000	125.200
20	N1A-40"	1550.000	0.000	1950.400
40	M2-24"	7850.000	90.000	1185.600
50	N2-52"	10549.999	0.000	2544.000
50	N3-4"	10549.999	180.000	196.062
70	N1B-40"	17000.000	0.000	1950.400
90	K1A	0.000	0.000	125.200
90	K2A	0.000	350.000	125.200
90	K3A	0.000	10.000	125.200
90	K4A	0.000	20.000	125.200
90	K5A	0.000	30.000	125.200
90	K1B	0.000	180.000	125.200
90	K2B	0.000	190.000	125.200
90	K3B	0.000	170.000	125.200
90	K4B	0.000	160.000	125.200
90	K5B	0.000	150.000	125.200

The nozzle spacing is computed by the following:



$= \sqrt{ll^2 + lc^2}$ where

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K6-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
K1A	2.000 in	Actual	LWN	3.323	16.800	...	...	240.45	300
K2A	2.000 in	Actual	LWN	3.323	16.800	...	...	240.45	300
K3A	2.000 in	Actual	LWN	3.323	16.800	...	...	240.45	300
K4A	2.000 in	Actual	LWN	3.323	16.800	...	...	240.45	300
K5A	2.000 in	Actual	LWN	3.323	16.800	...	...	240.45	300
K1B	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
K2B	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
K3B	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
K4B	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
K5B	2.000 in	Actual	LWN	3.323	16.800	...	...	250.76	300
N3-4"	4.000 in	120	WNF	4.500	11.125	314.30	22.00	428.50	300
M1-24"	24.000 in	Actual	WNF	24.000	10.000	909.60	26.00	279.92	150
M2-24"	24.000 in	Actual	WNF	24.000	10.000	1009.60	22.00	336.32	150
N1A-40"	40.000 in	Actual	WNF	40.000	22.000	1916.00	26.00	415.95	300
N1B-40"	40.000 in	Actual	WNF	40.000	22.000	1916.00	22.00	415.95	300
N2-52"	52.000 in	Actual	WNF	52.000	26.000	2320.80	28.00	495.80	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K6-2"	SA-105	22.000	10.000	...	...	...
K1A	SA-105	22.000	10.000	...	...	...
K2A	SA-105	22.000	10.000	...	...	...
K3A	SA-105	22.000	10.000	...	...	...
K4A	SA-105	22.000	10.000	...	...	...
K5A	SA-105	22.000	10.000	...	...	...
K1B	SA-105	22.000	10.000	...	...	...
K2B	SA-105	22.000	10.000	...	...	...
K3B	SA-105	22.000	10.000	...	...	...
K4B	SA-105	22.000	10.000	...	...	...
K5B	SA-105	22.000	10.000	...	...	...
N3-4"	SA-106 B	22.000	20.000	18.000	22.000	...
M1-24"	SA-516 70	10.000	10.000	14.000	26.000	...
M2-24"	SA-516 70	10.000	10.000	14.000	22.000	...
N1A-40"	SA-516 70	22.000	10.000	14.000	26.000	...
N1B-40"	SA-516 70	22.000	10.000	18.000	22.000	...
N2-52"	SA-516 70	22.000	20.000	18.000	28.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K6-2"	...	180.0	206.38	0.00	left Dish
K1A	...	0.0	206.38	0.00	Right dish

Front end KOD (V6502 ABC)

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K2A	...	350.0	206.38	0.00	Right dish
K3A	...	10.0	206.38	0.00	Right dish
K4A	...	20.0	206.38	0.00	Right dish
K5A	...	30.0	206.38	0.00	Right dish
K1B	...	180.0	206.38	0.00	Right dish
K2B	...	190.0	206.38	0.00	Right dish
K3B	...	170.0	206.38	0.00	Right dish
K4B	...	160.0	206.38	0.00	Right dish
K5B	...	150.0	206.38	0.00	Right dish
N3-4"	10549.999	180.0	400.00	0.00	Shell4
M1-24"	...	0.0	250.00	0.00	left Dish
M2-24"	7850.000	90.0	300.00	0.00	Shell3
N1A-40"	1550.000	0.0	350.00	0.00	Shell-1
N1B-40"	16999.998	0.0	350.00	0.00	Shell6
N2-52"	10549.999	0.0	400.00	0.00	Shell4

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**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 26.0mm. THK X 6500.0mm. ID X 50.0mm.	SA-516 70
2	INSULATION: 1675mm X 80mm THK	
1	PLATFORM: 1000mm X 6543mm WIDE	
1	LADDER: 7800.0mm	...
6	INSULATION: 3000mm X 80mm THK	
4	TRUNNIONS:	SA-516 70
4	TRUNNION PADS: 1500.0mm OD X 650.0mm ID X 22.0mm THK	
2	SADDLE: 350.0mm X 120 DEG	SA-516 70
5	BAR RING STIFFENER: 6544.0mm ID X 7064.0mm OD X 26.0mm	SA-516 70
1	INSULATION: 2400mm X 80mm THK	
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
13	CYLINDRICAL SEGMENT 22.0mm. THK X 3598.2mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm. THK X 2429.0mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 16821.8mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 10794.6mm. X 3000.0mm.	SA-516 70
15	CYLINDRICAL SEGMENT 22.0mm. THK X 3000.0mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 22.0mm. THK X 17420.1mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 22.0mm. THK X 11999.8mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 22.0mm. THK X 3598.2mm. X 2400.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm. THK X 16821.8mm. X 2400.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm. THK X 10794.6mm. X 2400.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
left Dish	[10]	B	-5	-48		0.404	22.000	0.85	No
left Dish	[7]	B	0	-104		0.339	26.000	0.85	No
Shell1	[8]	B	-5	-48		0.405	22.000	0.85	No
Shell2	[8]	B	-5	-48		0.405	22.000	0.85	No
Shell3	[8]	B	-5	-48		0.405	22.000	0.85	No
Shell4	[8]	B	-5	-48		0.405	22.000	0.85	No
Shell5	[8]	B	-5	-48		0.405	22.000	0.85	No
Shell6	[8]	B	-5	-48		0.405	22.000	0.85	No
shell7	[8]	B	-5	-48		0.405	22.000	0.85	No
Right dish	[10]	B	-5	-48		0.404	22.000	0.85	No
Right dish	[7]	B	0	-104		0.339	26.000	0.85	No
Ring:30 - 1	[23]	B	-5	-48		0.405	22.000	0.85	No
Ring:40 - 2	[23]	B	-5	-48		0.405	22.000	0.85	No
Ring:50 - 3	[23]	B	-5	-48		0.405	22.000	0.85	No
Ring:60 - 4	[23]	B	-5	-48		0.405	22.000	0.85	No
Ring:80 - 5	[23]	B	-5	-48		0.405	22.000	0.85	No
left saddle	[24]	B	-5	-48		0.405	22.000	0.85	No
Right saddle	[24]	B	-5	-48		0.405	22.000	0.85	No
M1-24"	[1]	B	-5	-48		0.364	22.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6-2"	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1A-40"	[1]	B	15	-104		0.063	22.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M2-24"	[1]	B	-5	-48		0.405	22.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2-52"	[1]	B	20	-104		0.069	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-4"	[1]	B	-8	-104		0.021	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1B-40"	[1]	B	15	-104		0.063	22.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3A	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4A	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5A	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3B	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4B	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5B	[1]	A	7	-104		0.005	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 20 -48

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.



- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (barsUniCode):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Front end KOD (V6502 ABC)

Design Internal Pressure (for Hydrotest) 3.5 bars
 Design Internal Temperature 415.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -12.0 °C
 Type of Construction Welded
 Special Service Air/Water/Steam
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(c)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13		IP+WE+EW	
Load Case 14		IP+WF+CW	
Load Case 15		IP+VO+OW	
Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	Km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm.
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm.
Distance Upwind of Crest	Lh	0	mm.
Distance from Crest to the Vessel	x	0	mm.
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm.
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.800	
Amplification Factor	Ap	1.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
M.A.W.P + Static Head to Nozzle		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	left Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	6500 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	26 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	415 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	415 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	90.702 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	178 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0009874 kg./cm3
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-1625 mm.
Height/Length of Insulation	1675 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	10
Detail Type	Nozzle
Detail ID	K6-2"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300



Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	10	
Detail Type	Weight	
Detail ID	Davit-1	
Dist. from "FROM" Node / Offset dist	-900	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	300	mm.

Element#	2/9	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6500	mm.
Element Thickness	22	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	415	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	415	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	0	cm2
Platform Weight	23106	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	6544	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	150	mm.
Platform Force Coefficient	0.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	7800	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	20500	mm.

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0009874	kg./cm3

Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	80	mm.



Density 0.000352 kg./cm3

Element From Node 20
 Detail Type Nozzle
 Detail ID N1A-40"
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 40 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70

Element From Node 20
 Detail Type Weight
 Detail ID N1-A insert pla
 Dist. from "FROM" Node / Offset dist 0 mm.
 Miscellaneous Weight 115 Kgf
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID N1-A elbow
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Miscellaneous Weight 800 Kgf
 Offset from Element Centerline 1000 mm.

Element# 3/9
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell2
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 6500 mm.
 Element Thickness 22 mm.
 Internal Corrosion Allowance 1.6 mm.
 Nominal Thickness 22 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 3.5 bars
 Design Temperature Internal Pressure 415 °C
 Design External Pressure 1.0342 bars
 Design Temperature External Pressure 415 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Weld is pre-Heated No

Element From Node 30
 Detail Type Saddle
 Detail ID left saddle
 Dist. from "FROM" Node / Offset dist 1050 mm.
 Width of Saddle 350 mm.
 Height of Saddle at Bottom 3600 mm.
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm.
 Width of Wear Plate 800 mm.
 Thickness of Wear Plate 22 mm.
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm.
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm.



Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0009874 kg./cm3

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	30
Detail Type	Ring
Detail ID	Ring:30 - 1
Dist. from "FROM" Node / Offset dist	238.43 mm.
Inside Diameter of Ring	6544 mm.
Thickness of Ring	26 mm.
Outside Diameter of Ring	7064 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Weight
Detail ID	Clip
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	110 Kg
Offset from Element Centerline	0 mm.

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell3
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	6500 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	415 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	415 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0009874 kg./cm3

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.



Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	40
Detail Type	Ring
Detail ID	Ring:40 - 2
Dist. from "FROM" Node / Offset dist	1118.5 mm.
Inside Diameter of Ring	6544 mm.
Thickness of Ring	26 mm.
Outside Diameter of Ring	7064 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	40
Detail Type	Nozzle
Detail ID	M2-24"
Dist. from "FROM" Node / Offset dist	1800 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	40
Detail Type	Weight
Detail ID	Manway davit
Dist. from "FROM" Node / Offset dist	1800 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	0 mm.

Element#	5/9
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell4
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	6500 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	415 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	415 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0009874 kg./cm3

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.



Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	50
Detail Type	Ring
Detail ID	Ring:50 - 3
Dist. from "FROM" Node / Offset dist	2600 mm.
Inside Diameter of Ring	6544 mm.
Thickness of Ring	26 mm.
Outside Diameter of Ring	7064 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	50
Detail Type	Nozzle
Detail ID	N2-52"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	52 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	50
Detail Type	Nozzle
Detail ID	N3-4"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	50
Detail Type	Weight
Detail ID	Clip wt
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	380 Kgf
Offset from Element Centerline	0 mm.

Element From Node	50
Detail Type	Weight
Detail ID	N2-Insert plate
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	270 Kgf
Offset from Element Centerline	0 mm.

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell15
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	6500 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	415 °C



Design External Pressure	1.0342	bars
Design Temperature External Pressure	415	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0009874	kg./cm3

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm3

Element From Node	60	
Detail Type	Ring	
Detail ID	Ring:60 - 4	
Dist. from "FROM" Node / Offset dist	2878.8	mm.
Inside Diameter of Ring	6544	mm.
Thickness of Ring	26	mm.
Outside Diameter of Ring	7064	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	

Element From Node	60	
Detail Type	Weight	
Detail ID	Impengment plat	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm.

Element#	7/9	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell6	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6500	mm.
Element Thickness	22	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	415	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	415	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	1350	mm.
Width of Saddle	350	mm.



Height of Saddle at Bottom	3600	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	800	mm.
Thickness of Wear Plate	22	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.40000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0009874	kg./cm3
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm3
Element From Node	70	
Detail Type	Nozzle	
Detail ID	N1B-40"	
Dist. from "FROM" Node / Offset dist	1950	mm.
Nozzle Diameter	40	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	70	
Detail Type	Weight	
Detail ID	N1B insertplate	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	115	Kgf
Offset from Element Centerline	0	mm.
Element From Node	70	
Detail Type	Weight	
Detail ID	N1B elbow	
Dist. from "FROM" Node / Offset dist	2000	mm.
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	1000	mm.

Element#	8/9
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	shell7
Distance "FROM" to "TO"	2400 mm.
Inside Diameter	6500 mm.
Element Thickness	22 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars



Design Temperature Internal Pressure	415	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	415	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid 80	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0009874	kg./cm3

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2400	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm3

Element From Node	80	
Detail Type	Ring	
Detail ID	Ring:80 - 5	
Dist. from "FROM" Node / Offset dist	758.94	mm.
Inside Diameter of Ring	6544	mm.
Thickness of Ring	26	mm.
Outside Diameter of Ring	7064	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	

Element#	9/9	
Element From Node	90	
Element To Node	100	
Element Type	Elliptical	
Description	Right dish	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	6500	mm.
Element Thickness	22	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	26	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	415	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	415	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid 90	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0009874	kg./cm3

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 90	



Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1675	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm ³

Element From Node	90	
Detail Type	Nozzle	
Detail ID	K1A	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	90	
Detail Type	Nozzle	
Detail ID	K2A	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	350.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	90	
Detail Type	Nozzle	
Detail ID	K3A	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	10.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	90	
Detail Type	Nozzle	
Detail ID	K4A	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	20.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	90	
Detail Type	Nozzle	
Detail ID	K5A	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	



Element From Node	90
Detail Type	Nozzle
Detail ID	K1B
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K2B
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	190.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K3B
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	170.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K4B
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	160.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	90
Detail Type	Nozzle
Detail ID	K5B
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	150.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
left Dish		50	...	50	...
Shell-1		3050	...	3000	...
Shell2		6050	...	3000	...
Shell3		9050	...	3000	...
Shell4		12050	...	3000	...
Shell5		15050	...	3000	...
Shell6		18050	...	3000	...
shell7		20450	...	2400	...
Right dish		20500	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
left Dish		3.5438	26	1.6	6500	77.097	1
Shell-1		3.5438	22	1.6	6500	77.097	2
Shell2		3.5438	22	1.6	6500	77.097	3
Shell3		3.5438	22	1.6	6500	77.097	4
Shell4		3.5438	22	1.6	6500	77.097	5
Shell5		3.5438	22	1.6	6500	77.097	6
Shell6		3.5438	22	1.6	6500	77.097	7
shell7		3.5438	22	1.6	6500	77.097	8
Right dish		3.5438	26	1.6	6500	77.097	9

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
left Dish		3.5	4.79	7.93	22	16.544	1
Shell-1		3.5	4.77	7.91	22	16.5883	2
Shell2		3.5	4.77	7.91	22	16.5883	3
Shell3		3.5	4.77	7.91	22	16.5883	4
Shell4		3.5	4.77	7.91	22	16.5883	5
Shell5		3.5	4.77	7.91	22	16.5883	6
Shell6		3.5	4.77	7.91	22	16.5883	7
shell7		3.5	4.77	7.91	22	16.5883	8
Right dish		3.5	4.79	7.93	22	16.544	9

Minimum

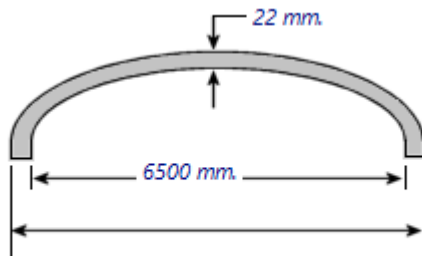
3.5

7.91

MAWP: 3.5 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: left Dish**

Elliptical Head Design Information:					
Design Pressure	P	3.544 bars	Int. Design Temperature		415 °C
Material		SA-516 70	Ext. Design Temperature		415 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	90.7 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6500.000 mm.	Actual Stress		66.5 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		26.000 mm.
Surface Area		475632.8 cm^2	SG of Contents		0.989
Element Volume		37614220.0 cm3	Weight of Contents		509.9 Kgf
Empty Weight		10478.2 Kgf	Operating Weight		10988.1 Kgf
			Comp. All. Ope. Stress		56.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 81.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 40.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.54 \cdot 6503 \cdot 1) / (2 \cdot 90.7 \cdot 0.85 - 0.2 \cdot 3.54)$$

$$= 14.9440 + 1.6000 = 16.5440 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 90.7 \cdot 0.85 \cdot 20.4) / (1 \cdot 6503 + 0.2 \cdot 20.4)$$

$$= 4.84 - 0.044 = 4.79 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 22) / (1 \cdot 6500 + 0.2 \cdot 22)$$

$$= 7.93 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.54 \cdot (1 \cdot 6503 + 0.2 \cdot 20.4)) / (2 \cdot 0.85 \cdot 20.4)$$

$$= 66.455 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54) + 1.6$$

$$= 16.588 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 24.4) / (3252 + 0.6 \cdot 24.4)$$

$$= 5.76 - 0.044 = 5.72 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (6503 / (2 \cdot 1627))^2) / 6$$

$$= 0.999344$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.744 %**MDMT Calculations in the Knuckle Portion:**

Govrn. thk, tg = 22, tr = 9.7, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.4$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-5 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 26, tr = 9.72, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.34$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

0 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.000 mm.	Basic MDMT	0 °C
Stress/Thk. Ratio	0.339	Temperature Difference	104 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 81.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 40.625 mm.

EXTERNAL PRESSURE RESULTS for: left Dish**Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 415 °C**

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
20.400	6544.00	320.78	0.0004330	34.57

$$MAEP = B / (K_0 \cdot Do / t) = 34.6 / (0.9 \cdot 321) = 1.2 \text{ bars}$$

Results for Required Thickness	Tca
--------------------------------	-----

Front end KOD (V6502 ABC).

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FileName : Front end Flare KOD (V6502 ABC) 9%

Element: left Dish

Step: 4

3:25pm

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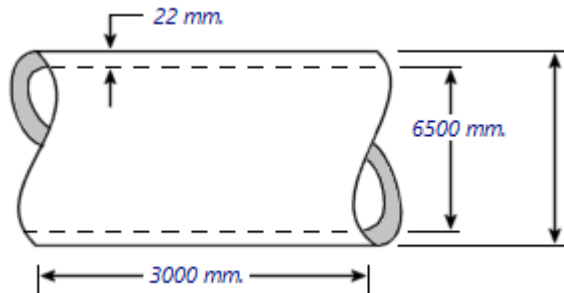


HEXAGON

Tca	Outer Dia	Do/t	Factor A	Factor B
18.961	6544.00	345.13	0.0004024	32.13

$$MAEP = B / (K0 * Do / t) = 32.1 / (0.9 * 345) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell-1**

Cylindrical Shell Design Information:			
Design Pressure P	3.544 bars	Int. Design Temperature	415 °C
Material	SA-516 70	Ext. Design Temperature	415 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	90.7 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	6500.000 mm.	Actual Stress	66.7 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm.	Spec. Nominal Thk.	22.000 mm.
Surface Area	616757.5 cm ²	SG of Contents	0.989
Element Volume	99567048.0 cm ³	Weight of Contents	2979.2 Kg
Empty Weight	10481.0 Kg	Operating Weight	13460.3 Kg
		Comp. All. Ope. Stress	56.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3830.09	320.78	0.5853	0.0004068	32.48

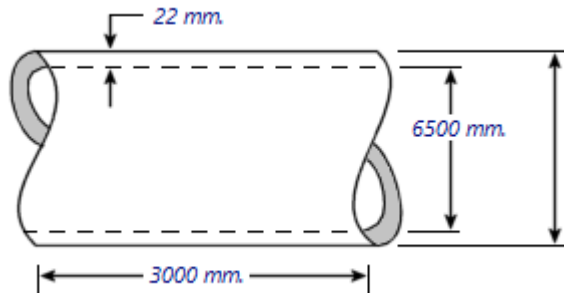
$$MAEP = (4*B)/(3*(Do/t)) = (4*32.5)/(3*321) = 1.35 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.302	6544.00	3830.09	357.55	0.5853	0.0003474	27.74

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.7)/(3*358) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.52	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell2**

Cylindrical Shell Design Information:			
Design Pressure P	3.544 bars	Int. Design Temperature	415 °C
Material	SA-516 70	Ext. Design Temperature	415 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	90.7 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	6500.000 mm.	Actual Stress	66.7 N./mm ²
Spec. Min. Thk t _e , or s _a	22.0000 mm.	Spec. Nominal Thk.	22.000 mm.
Surface Area	616757.5 cm ²	SG of Contents	0.989
Element Volume	99567048.0 cm ³	Weight of Contents	2979.2 Kg
Empty Weight	10481.0 Kg	Operating Weight	13460.3 Kg
		Comp. All. Ope. Stress	56.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 22$, $t_r = 9.72$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell2

Cylindrical Shell From 30 to Ring:30 - 1 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3830.09	320.78	0.5853	0.0004068	32.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*32.5)/(3*321) = 1.35 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.302	6544.00	3830.09	357.55	0.5853	0.0003474	27.74

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.7)/(3*358) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.52	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

Cylindrical Shell From Ring:30 - 1 to 40 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3880.07	320.78	0.5929	0.0004012	32.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*32)/(3*321) = 1.33 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.401	6544.00	3880.07	355.64	0.5929	0.0003455	27.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*356) = 1.03 \text{ bars}$$

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Element: Shell2

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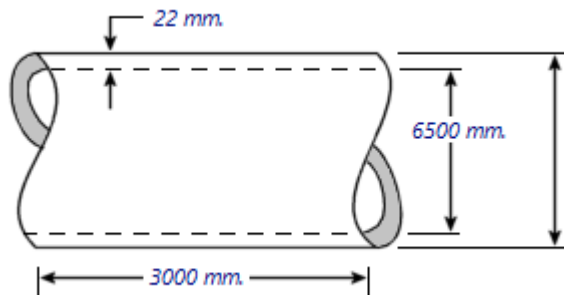


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Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.69	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell3**

Cylindrical Shell Design Information:					
Design Pressure	P	3.544 bars	Int. Design Temperature		415 °C
Material		SA-516 70	Ext. Design Temperature		415 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	90.7 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6500.000 mm.	Actual Stress		66.7 N./mm^2
Spec. Min. Thk t,e, or sa		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		616757.5 cm^2	SG of Contents		0.989
Element Volume		99567048.0 cm3	Weight of Contents		2979.2 Kg
Empty Weight		10481.0 Kg	Operating Weight		13460.3 Kg
			Comp. All. Ope. Stress		56.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 22$, $t_r = 9.72$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell3

Cylindrical Shell From 40 to Ring:40 - 2 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3880.07	320.78	0.5929	0.0004012	32.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*32)/(3*321) = 1.33 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.401	6544.00	3880.07	355.64	0.5929	0.0003455	27.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*356) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.69	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

Cylindrical Shell From Ring:40 - 2 to 50 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4481.50	320.78	0.6848	0.0003452	27.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*321) = 1.15 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.551	6544.00	4481.50	334.71	0.6848	0.0003252	25.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*26)/(3*335) = 1.03 \text{ bars}$$

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Element: Shell3

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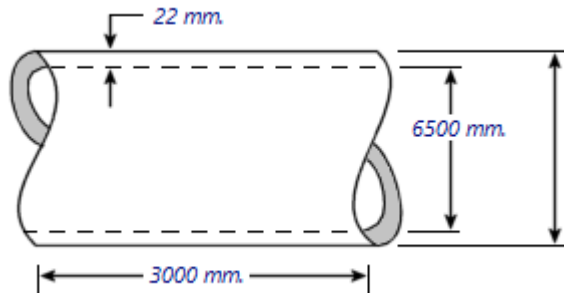


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Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4940.20	320.78	0.7549	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell4**

Cylindrical Shell Design Information:			
Design Pressure P	3.544 bars	Int. Design Temperature	415 °C
Material	SA-516 70	Ext. Design Temperature	415 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	90.7 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	6500.000 mm.	Actual Stress	66.7 N./mm ²
Spec. Min. Thk t,e, or sa	22.0000 mm.	Spec. Nominal Thk.	22.000 mm.
Surface Area	616757.5 cm ²	SG of Contents	0.989
Element Volume	99567048.0 cm ³	Weight of Contents	2979.2 Kg
Empty Weight	10481.0 Kg	Operating Weight	13460.3 Kg
		Comp. All. Ope. Stress	56.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54) \\
 &= 14.9883 + 1.6000 = 16.5883 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4) \\
 &= 4.82 - 0.044 = 4.77 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22) \\
 &= 7.91 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4) \\
 &= 66.707 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell4

Cylindrical Shell From 50 to Ring:50 - 3 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4481.50	320.78	0.6848	0.0003452	27.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*321) = 1.15 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.551	6544.00	4481.50	334.71	0.6848	0.0003252	25.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*26)/(3*335) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4940.20	320.78	0.7549	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

Cylindrical Shell From Ring:50 - 3 to 60 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3278.82	320.78	0.5010	0.0004796	38.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.3)/(3*321) = 1.59 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.194	6544.00	3278.82	380.60	0.5010	0.0003698	29.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*29.5)/(3*381) = 1.03 \text{ bars}$$

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Element: Shell4



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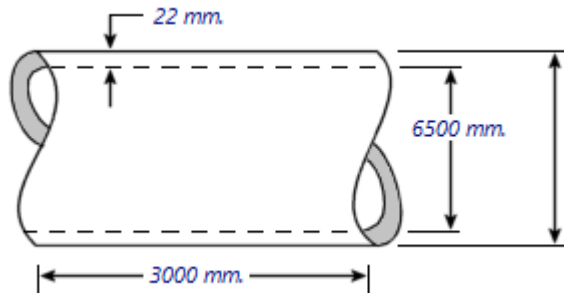


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Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4940.33	320.78	0.7549	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell5**

Cylindrical Shell Design Information:			
Design Pressure P	3.544 bars	Int. Design Temperature	415 °C
Material	SA-516 70	Ext. Design Temperature	415 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	90.7 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	6500.000 mm.	Actual Stress	66.7 N./mm ²
Spec. Min. Thk t, e, or sa	22.0000 mm.	Spec. Nominal Thk.	22.000 mm.
Surface Area	616757.5 cm ²	SG of Contents	0.989
Element Volume	99567048.0 cm ³	Weight of Contents	2979.2 Kg
Empty Weight	10481.0 Kg	Operating Weight	13460.3 Kg
		Comp. All. Ope. Stress	56.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, $t_g = 22$, $t_r = 9.72$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Shell5

Cylindrical Shell From 60 to Ring:60 - 4 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3278.82	320.78	0.5010	0.0004796	38.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.3)/(3*321) = 1.59 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.194	6544.00	3278.82	380.60	0.5010	0.0003698	29.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*29.5)/(3*381) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4940.33	320.78	0.7549	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

Cylindrical Shell From Ring:60 - 4 to 70 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3880.12	320.78	0.5929	0.0004012	32.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*32)/(3*321) = 1.33 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.401	6544.00	3880.12	355.63	0.5929	0.0003455	27.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*356) = 1.03 \text{ bars}$$

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Element: Shell5



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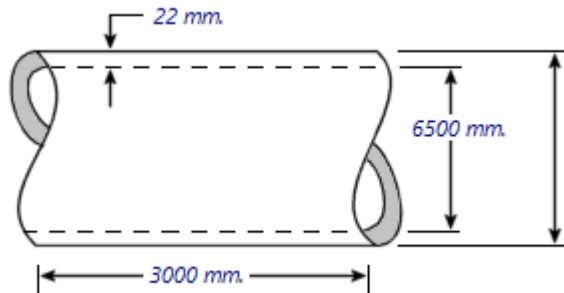


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Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.69	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Shell6**

Cylindrical Shell Design Information:					
Design Pressure	P	3.544 bars	Int. Design Temperature		415 °C
Material		SA-516 70	Ext. Design Temperature		415 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	90.7 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6500.000 mm.	Actual Stress		66.7 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		616757.5 cm^2	SG of Contents		0.989
Element Volume		99567048.0 cm3	Weight of Contents		2979.2 Kg
Empty Weight		10481.0 Kg	Operating Weight		13460.3 Kg
			Comp. All. Ope. Stress		56.4 N./mm^2
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: Shell6

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3880.12	320.78	0.5929	0.0004012	32.03

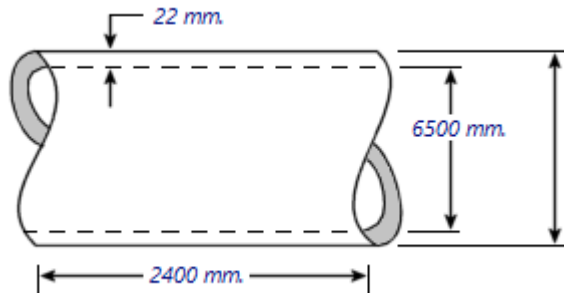
$$MAEP = (4*B)/(3*(Do/t)) = (4*32)/(3*321) = 1.33 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.401	6544.00	3880.12	355.63	0.5929	0.0003455	27.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*356) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.69	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: shell7**

Cylindrical Shell Design Information:					
Design Pressure	P	3.544 bars	Int. Design Temperature		415 °C
Material		SA-516 70	Ext. Design Temperature		415 °C
Length		2400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	90.7 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6500.000 mm.	Actual Stress		66.7 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		493406.0 cm^2	SG of Contents		0.989
Element Volume		79653648.0 cm3	Weight of Contents		2383.4 Kg
Empty Weight		8384.8 Kg	Operating Weight		10768.1 Kg
			Comp. All. Ope. Stress		56.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54)$$

$$= 14.9883 + 1.6000 = 16.5883 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 20.4) / (3252 + 0.6 \cdot 20.4)$$

$$= 4.82 - 0.044 = 4.77 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 22) / (3250 + 0.6 \cdot 22)$$

$$= 7.91 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.54 \cdot (3252 + 0.6 \cdot 20.4)) / (0.85 \cdot 20.4)$$

$$= 66.707 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.337 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.405	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell7

Cylindrical Shell From 80 to Ring:80 - 5 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	3880.12	320.78	0.5929	0.0004012	32.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*32)/(3*321) = 1.33 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.401	6544.00	3880.12	355.63	0.5929	0.0003455	27.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.6)/(3*356) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.69	320.78	0.7548	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

Cylindrical Shell From Ring:80 - 5 to 90 Ext. Chart: CS-2 at 415 °C

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	2232.72	320.78	0.3412	0.0007254	46.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*46.1)/(3*321) = 1.92 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.625	6544.00	2232.72	447.46	0.3412	0.0004348	34.71

$$MAEP = (4*B)/(3*(Do/t)) = (4*34.7)/(3*447) = 1.03 \text{ bars}$$

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Element: shell7

Step: 11

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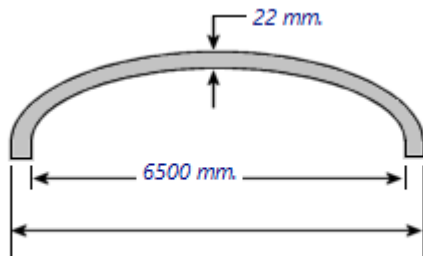
Results for Maximum Stiffened Length

Slen

Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.400	6544.00	4939.88	320.78	0.7549	0.0003119	24.90

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9)/(3*321) = 1.03 \text{ bars}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Right dish**

Elliptical Head Design Information:					
Design Pressure	P	3.544 bars	Int. Design Temperature		415 °C
Material		SA-516 70	Ext. Design Temperature		415 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	90.7 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6500.000 mm.	Actual Stress		66.5 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		26.000 mm.
Surface Area		475632.8 cm^2	SG of Contents		0.989
Element Volume		37614220.0 cm3	Weight of Contents		509.9 Kg
Empty Weight		10478.2 Kg	Operating Weight		10988.1 Kg
			Comp. All. Ope. Stress		56.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 81.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 40.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.54 \cdot 6503 \cdot 1) / (2 \cdot 90.7 \cdot 0.85 - 0.2 \cdot 3.54)$$

$$= 14.9440 + 1.6000 = 16.5440 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 90.7 \cdot 0.85 \cdot 20.4) / (1 \cdot 6503 + 0.2 \cdot 20.4)$$

$$= 4.84 - 0.044 = 4.79 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 22) / (1 \cdot 6500 + 0.2 \cdot 22)$$

$$= 7.93 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.54 \cdot (1 \cdot 6503 + 0.2 \cdot 20.4)) / (2 \cdot 0.85 \cdot 20.4)$$

$$= 66.455 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (3.54 \cdot 3252) / (90.7 \cdot 0.85 - 0.6 \cdot 3.54) + 1.6$$

$$= 16.588 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.044 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (90.7 \cdot 0.85 \cdot 24.4) / (3252 + 0.6 \cdot 24.4)$$

$$= 5.76 - 0.044 = 5.72 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (6503 / (2 \cdot 1627))^2) / 6$$

$$= 0.999344$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.744 %**MDMT Calculations in the Knuckle Portion:**

Govrn. thk, tg = 22, tr = 9.7, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.4$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-5 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 26, tr = 9.72, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.34$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

0 °C

Min Metal Temp. at Required thickness (UCS 66.1)

-104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.000 mm.	Basic MDMT	0 °C
Stress/Thk. Ratio	0.339	Temperature Difference	104 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 81.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 40.625 mm.

EXTERNAL PRESSURE RESULTS for: Right dish**Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 415 °C**

Elastic Modulus from Chart: CS-2 at 415 °C: 159665552 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
20.400	6544.00	320.78	0.0004330	34.57

$$MAEP = B / (K_0 \cdot Do / t) = 34.6 / (0.9 \cdot 321) = 1.2 \text{ bars}$$

Results for Required Thickness	Tca
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Front end KOD (V6502 ABC).

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FileName : Front end Flare KOD (V6502 ABC) 9%

Element: Right dish

Step: 12

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Tca	Outer Dia	Do/t	Factor A	Factor B
18.961	6544.00	345.13	0.0004024	32.13

$$MAEP = B/(K0*Do/t) = 32.1/(0.9 *345) = 1.03 \text{ bars}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.648 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.648 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9.545 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	5.625 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	6.392 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6.648 bars
App 27-4	= M.A.W.P.	3.500 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
left Dish	177.1	234.0	0.757	10.18
Shell-1	177.7	234.0	0.759	10.18
Shell2	177.7	234.0	0.759	10.18
Shell3	177.7	234.0	0.759	10.18
Shell4	177.7	234.0	0.759	10.18
Shell5	177.7	234.0	0.759	10.18
Shell6	177.7	234.0	0.759	10.18
shell7	177.7	234.0	0.759	10.18
Right dish	177.1	234.0	0.757	10.18

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	90.70	1.521
M1-24"	Pad	138.00	90.70	1.521
K6-2"	Nozzle	138.00	90.80	1.520
N1A-40"	Nozzle	138.00	90.70	1.521
N1A-40"	Pad	138.00	90.70	1.521
M2-24"	Nozzle	138.00	90.70	1.521
M2-24"	Pad	138.00	90.70	1.521
N2-52"	Nozzle	138.00	90.70	1.521
N2-52"	Pad	138.00	90.70	1.521
N3-4"	Nozzle	118.00	80.76	1.461
N3-4"	Pad	138.00	90.70	1.521
N1B-40"	Nozzle	138.00	90.70	1.521
N1B-40"	Pad	138.00	90.70	1.521
K1A	Nozzle	138.00	90.80	1.520
K2A	Nozzle	138.00	90.80	1.520
K3A	Nozzle	138.00	90.80	1.520
K4A	Nozzle	138.00	90.80	1.520
K5A	Nozzle	138.00	90.80	1.520
K1B	Nozzle	138.00	90.80	1.520
K2B	Nozzle	138.00	90.80	1.520
K3B	Nozzle	138.00	90.80	1.520
K4B	Nozzle	138.00	90.80	1.520
K5B	Nozzle	138.00	90.80	1.520

Minimum

1.461

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring:30 - 1	138.00	90.70	1.521
Ring:40 - 2	138.00	90.70	1.521
Ring:50 - 3	138.00	90.70	1.521



Ring:60 - 4	138.00	90.70	1.521
Ring:80 - 5	138.00	90.70	1.521

Minimum

1.521

Stress ratios for Pressurized Vessel Elements (N./mm2):

Description	Ambient	Operating	Ratio
left Dish	138.00	90.70	1.521
Shell-1	138.00	90.70	1.521
Shell2	138.00	90.70	1.521
Shell3	138.00	90.70	1.521
Shell4	138.00	90.70	1.521
Shell5	138.00	90.70	1.521
Shell6	138.00	90.70	1.521
shell7	138.00	90.70	1.521
Right dish	138.00	90.70	1.521

Minimum

1.521

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
M1-24"	30.63	234.00	0.131
K6-2"	2.15	225.00	0.010
N1A-40"	23.10	234.00	0.099
M2-24"	30.63	234.00	0.131
N2-52"	25.46	234.00	0.109
N3-4"	5.57	216.00	0.026
N1B-40"	23.10	234.00	0.099
K1A	2.15	225.00	0.010
K2A	2.15	225.00	0.010
K3A	2.15	225.00	0.010
K4A	2.15	225.00	0.010
K5A	2.15	225.00	0.010
K1B	2.15	225.00	0.010
K2B	2.15	225.00	0.010
K3B	2.15	225.00	0.010
K4B	2.15	225.00	0.010
K5B	2.15	225.00	0.010

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6.648 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6.648 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	9.545 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	5.625 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	6.392 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6.648 bars
App 27-4	= M.A.W.P.	3.500 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 3.5 • 1.46

= 6.648 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
left Dish	137.5	234.0	0.588	7.33
Shell-1	138.1	234.0	0.590	7.33
Shell2	138.1	234.0	0.590	7.33
Shell3	138.1	234.0	0.590	7.33
Shell4	138.1	234.0	0.590	7.33
Shell5	138.1	234.0	0.590	7.33
Shell6	138.1	234.0	0.590	7.33
shell7	138.1	234.0	0.590	7.33
Right dish	137.5	234.0	0.588	7.33

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-24"	Nozzle	138.00	90.70	1.521
M1-24"	Pad	138.00	90.70	1.521
K6-2"	Nozzle	138.00	90.80	1.520
N1A-40"	Nozzle	138.00	90.70	1.521
N1A-40"	Pad	138.00	90.70	1.521
M2-24"	Nozzle	138.00	90.70	1.521
M2-24"	Pad	138.00	90.70	1.521
N2-52"	Nozzle	138.00	90.70	1.521
N2-52"	Pad	138.00	90.70	1.521
N3-4"	Nozzle	118.00	80.76	1.461
N3-4"	Pad	138.00	90.70	1.521
N1B-40"	Nozzle	138.00	90.70	1.521
N1B-40"	Pad	138.00	90.70	1.521
K1A	Nozzle	138.00	90.80	1.520
K2A	Nozzle	138.00	90.80	1.520
K3A	Nozzle	138.00	90.80	1.520
K4A	Nozzle	138.00	90.80	1.520
K5A	Nozzle	138.00	90.80	1.520
K1B	Nozzle	138.00	90.80	1.520
K2B	Nozzle	138.00	90.80	1.520
K3B	Nozzle	138.00	90.80	1.520
K4B	Nozzle	138.00	90.80	1.520
K5B	Nozzle	138.00	90.80	1.520

Minimum

1.461

Stress ratios for Stiffening Ring Materials (N./mm²):



Description	Ambient	Operating	Ratio
Ring:30 - 1	138.00	90.70	1.521
Ring:40 - 2	138.00	90.70	1.521
Ring:50 - 3	138.00	90.70	1.521
Ring:60 - 4	138.00	90.70	1.521
Ring:80 - 5	138.00	90.70	1.521

Minimum

1.521

Stress ratios for Pressurized Vessel Elements (N./mm2):

Description	Ambient	Operating	Ratio
left Dish	138.00	90.70	1.521
Shell-1	138.00	90.70	1.521
Shell2	138.00	90.70	1.521
Shell3	138.00	90.70	1.521
Shell4	138.00	90.70	1.521
Shell5	138.00	90.70	1.521
Shell6	138.00	90.70	1.521
shell7	138.00	90.70	1.521
Right dish	138.00	90.70	1.521

Minimum

1.521

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
M1-24"	22.06	234.00	0.094
K6-2"	1.55	225.00	0.007
N1A-40"	16.64	234.00	0.071
M2-24"	22.06	234.00	0.094
N2-52"	18.34	234.00	0.078
N3-4"	4.01	216.00	0.019
N1B-40"	16.64	234.00	0.071
K1A	1.55	225.00	0.007
K2A	1.55	225.00	0.007
K3A	1.55	225.00	0.007
K4A	1.55	225.00	0.007
K5A	1.55	225.00	0.007
K1B	1.55	225.00	0.007
K2B	1.55	225.00	0.007
K3B	1.55	225.00	0.007
K4B	1.55	225.00	0.007
K5B	1.55	225.00	0.007

Elements Suitable for Test Pressure.

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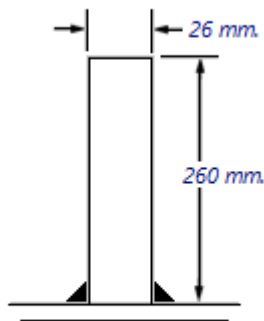
**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****MDMT Calculations for: Ring:30 - 1, Shell to Ring Junction, Curve: B**

Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring:30 - 1, SA-516 70 , Bar Ring: 260.0 x 26.0 mm.**Material treatment options:**

Effective Length of Shell: 402 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	81.990	10.2000	836.295
Ring :	67.600	150.4000	10167.041
Total:	149.590		11003.336

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	28.434	63.3568	329113.281
Ring :	3808.130	-76.8433	399170.312
Total:	3836.564		728283.625

Available Moment of Inertia, Ring + Shell: 0.111E+05 cm⁴Required Stress in Ring plus Shell Breq 22.91 N./mm²

Required Strain in Ring plus Shell Areq 0.0002860

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A)/10.9$$

$$= (6544^2 \cdot 3855(20.4 + 67.6/3855)0.000286)/10.9$$

$$= 9596 \text{ cm}^4$$

MDMT Calculations for: Ring:40 - 2, Shell to Ring Junction, Curve: B

Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85

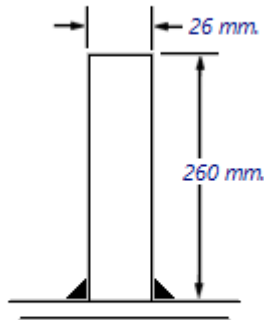
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.41$, Temp. Reduction = 49 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring:40 - 2, SA-516 70 , Bar Ring: 260.0 x 26.0 mm.

Material treatment options:



Effective Length of Shell: 402 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	81.990	10.2000	836.295
Ring :	67.600	150.4000	10167.041
Total:	149.590		11003.336

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	28.434	63.3568	329113.281
Ring :	3808.130	-76.8433	399170.312
Total:	3836.564		728283.625

Available Moment of Inertia, Ring + Shell: 0.111E+05 cm⁴

Required Stress in Ring plus Shell Breq 23.06 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002880

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6544^2 \cdot 4181(20.4 + 67.6/4181)0.000288) / 10.9$$

$$= 10415 \text{ cm}^4$$

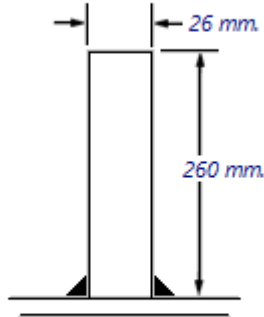
MDMT Calculations for: Ring:50 - 3, Shell to Ring Junction, Curve: B

Govern. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring:50 - 3, SA-516 70 , Bar Ring: 260.0 x 26.0 mm.

Material treatment options:



Effective Length of Shell: 402 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	81.990	10.2000	836.295
Ring :	67.600	150.4000	10167.041
Total:	149.590		11003.336

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	28.434	63.3568	329113.281
Ring :	3808.130	-76.8433	399170.312
Total:	3836.564		728283.625

Available Moment of Inertia, Ring + Shell: 0.111E+05 cm⁴

Required Stress in Ring plus Shell Breq 22.93 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002860

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6544^2 \cdot 3880(20.4 + 67.6/3880)0.000286) / 10.9$$

$$= 9654 \text{ cm}^4$$

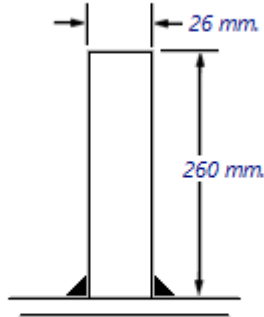
MDMT Calculations for: Ring:60 - 4, Shell to Ring Junction, Curve: B

Govern. thk, $t_g = 22$, $t_r = 9.72$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring:60 - 4, SA-516 70 , Bar Ring: 260.0 x 26.0 mm.

Material treatment options:



Effective Length of Shell: 402 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	81.990	10.2000	836.295
Ring :	67.600	150.4000	10167.041
Total:	149.590		11003.336

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	28.434	63.3568	329113.281
Ring :	3808.130	-76.8433	399170.312
Total:	3836.564		728283.625

Available Moment of Inertia, Ring + Shell: 0.111E+05 cm⁴

Required Stress in Ring plus Shell Breq 22.77 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002840

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6544^2 \cdot 3579(20.4 + 67.6/3579)0.000283999) / 10.9$$

$$= 8902 \text{ cm}^4$$

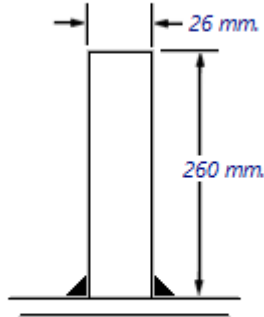
MDMT Calculations for: Ring:80 - 5, Shell to Ring Junction, Curve: B

Govern. thk, $t_g = 22$, $t_r = 9.72$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Stiffening Ring Calculations for: Ring:80 - 5, SA-516 70 , Bar Ring: 260.0 x 26.0 mm.

Material treatment options:



Effective Length of Shell: 402 mm.

	Area (cm2)	Distance (mm.)	Area*Dist
Shell:	81.990	10.2000	836.295
Ring :	67.600	150.4000	10167.041
Total:	149.590		11003.336

Centroid of Ring plus Shell: 74 mm.

	Inertia	Distance	A*Dist ²
Shell:	28.434	63.3568	329113.281
Ring :	3808.130	-76.8433	399170.312
Total:	3836.564		728283.625

Available Moment of Inertia, Ring + Shell: 0.111E+05 cm4

Required Stress in Ring plus Shell Breq 22.45 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002800

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6544^2 \cdot 3056(20.4 + 67.6/3056)0.000279999) / 10.9$$

$$= 7603 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	6544	20.4	0.00043297	34.5669
20	30	3830.09	6544	20.4	0.00040677	32.4755
30	Ring	3830.09	6544	20.4	0.00040677	32.4755
Ring	40	3880.07	6544	20.4	0.00040122	32.0322
40	Ring	3880.07	6544	20.4	0.00040122	32.0322
Ring	50	4481.5	6544	20.4	0.0003452	27.5595
50	Ring	4481.5	6544	20.4	0.0003452	27.5595
Ring	60	3278.82	6544	20.4	0.00047962	38.2914
60	Ring	3278.82	6544	20.4	0.00047962	38.2914
Ring	70	3880.12	6544	20.4	0.00040121	32.0318
70	80	3880.12	6544	20.4	0.00040121	32.0318
80	Ring	3880.12	6544	20.4	0.00040121	32.0318
Ring	90	2232.72	6544	20.4	0.00072541	46.1065
90	100	No Calc	6544	20.4	0.00043297	34.5669

**External Pressure Calculations:**

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	22	20.5609	1.03421	1.2
20	30	22	19.9021	1.03421	1.35
30	Ring	22	19.9021	1.03421	1.35
Ring	40	22	20.0009	1.03421	1.33
40	Ring	22	20.0009	1.03421	1.33
Ring	50	22	21.1511	1.03421	1.15
50	Ring	22	21.1511	1.03421	1.15
Ring	60	22	18.7941	1.03421	1.59
60	Ring	22	18.7941	1.03421	1.59
Ring	70	22	20.001	1.03421	1.33
70	80	22	20.001	1.03421	1.33
80	Ring	22	20.001	1.03421	1.33
Ring	90	22	16.2249	1.03421	1.92
90	100	22	20.5609	1.03421	1.2

Minimum

1.15

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	3830.09	4939.52	No Calc	No Calc
30	Ring	3830.09	4939.52	No Calc	No Calc
Ring	40	3880.07	4939.69	9596.25	11119.4
40	Ring	3880.07	4939.69	No Calc	No Calc
Ring	50	4481.5	4940.2	10415.2	11119.4
50	Ring	4481.5	4940.2	No Calc	No Calc
Ring	60	3278.82	4940.33	9653.73	11119.4
60	Ring	3278.82	4940.33	No Calc	No Calc
Ring	70	3880.12	4939.69	8901.8	11119.4
70	80	3880.12	4939.69	No Calc	No Calc
80	Ring	3880.12	4939.69	No Calc	No Calc
Ring	90	2232.72	4939.88	7602.64	11119.4
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	10478.2	37614220	9833.36	37668980	890.645
20	30	10481	99567048	9721.13	99665112	890.887
30	40	10481	99567048	9721.13	99665112	890.887
40	50	10481	99567048	9721.13	99665112	890.887
50	60	10481	99567048	9721.13	99665112	890.887
60	70	10481	99567048	9721.13	99665112	890.887
70	80	10481	99567048	9721.13	99665112	890.887
80	90	8384.75	79653648	7776.89	79732096	712.704
90	100	10478.2	37614220	9833.36	37668980	890.645
Total		92227	0.752E+09	85770	0.753E+09	7839

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	509.886	-541.667	3025	...	Liquid 10
10	Insl	1759.6	-787.5	...	...	Ins: 10
10	Nozl	455.997	-1625	...	...	M1-24"
10	Nozl	11.4939	-1038.33	-2500	...	K6-2"
10	Wght	50	-1038.33	...	300	Davit-1
20	Plat	25070.2	10250	3348.2	...	PLAT:[1 OF 1]
20	Liqd	2979.24	1500	3025	...	Liquid 20
20	Insl	1760.08	1500	...	...	Ins: 20
20	Nozl	1142.5	1500	3758	...	N1A-40"
20	Wght	115	...	...	...	N1-A insert pla
20	Wght	800	1500	1000	...	N1-A elbow
30	Sadl	7113.46	1050	3403	...	left saddle
30	Liqd	2979.24	1500	3025	...	Liquid 30
30	Insl	1760.08	1500	...	...	Ins: 30
30	Ring	1215.12	238.427	...	...	Ring:30 - 1
30	Wght	110	...	...	...	Clip
40	Liqd	2979.24	1500	3025	...	Liquid 40
40	Insl	1760.08	1500	...	...	Ins: 40
40	Ring	1215.12	1118.5	...	...	Ring:40 - 2
40	Nozl	479.76	1800	...	3554.8	M2-24"
40	Wght	50	1800	...	...	Manway davit
50	Liqd	2979.24	1500	3025	...	Liquid 50
50	Insl	1760.08	1500	...	...	Ins: 50
50	Ring	1215.12	2600	...	...	Ring:50 - 3
50	Nozl	1947.7	1500	3910.4	...	N2-52"
50	Nozl	37.6904	1500	-3307.15	...	N3-4"
50	Wght	380	...	...	...	Clip wt
50	Wght	270	...	...	...	N2-Insert plate
60	Liqd	2979.24	1500	3025	...	Liquid 60
60	Insl	1760.08	1500	...	...	Ins: 60
60	Ring	1215.12	2878.82	...	...	Ring:60 - 4
60	Wght	150	...	...	...	Impengment plat
70	Sadl	7113.46	1350	3403	...	Right saddle
70	Liqd	2979.24	1500	3025	...	Liquid 70
70	Insl	1760.08	1500	...	...	Ins: 70
70	Nozl	1072.79	1950	3758	...	N1B-40"
70	Wght	115	...	...	...	N1B insertplate
70	Wght	800	2000	1000	...	N1B elbow
80	Liqd	2383.39	1200	3025	...	Liquid 80
80	Insl	1408.07	1200	...	...	Ins: 80
80	Ring	1215.12	758.944	...	...	Ring:80 - 5
90	Liqd	509.886	591.667	3025	...	Liquid 90
90	Insl	1759.6	837.5	...	...	Ins: 90
90	Nozl	11.4939	1491.57	1500	...	K1A



90 Nozl	11.4939	1491.57	1477.21	-260.472	K2A
90 Nozl	11.4939	1491.57	1477.21	260.472	K3A
90 Nozl	11.4939	1491.57	1409.54	513.03	K4A
90 Nozl	11.4939	1491.57	1299.04	750	K5A
90 Nozl	11.4939	1088.33	-2500	...	K1B
90 Nozl	11.4939	1088.33	-2462.02	-434.12	K2B
90 Nozl	11.4939	1088.33	-2462.02	434.12	K3B
90 Nozl	11.4939	1088.33	-2349.23	855.05	K4B
90 Nozl	11.4939	1088.33	-2165.06	1250	K5B
30 Trun	1173.82	1050	...	...	Trunnions
70 Trun	1173.82	1350	...	...	Trunnions

Total Weight of Each Detail Type:

Saddles	14226.9
Platforms	25070.2
Liquid	21278.6
Insulation	15487.7
Stiffeners	6075.6
Nozzles	5262.9
Weights	2840.0
Trunnions	2347.6

Sum of the Detail Weights 92589.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	100066.5	100066.5	100066.5	100066.5	100066.5	100066.5
Saddles	14226.9	14226.9	14226.9	14226.9	14226.9	14226.9
Stif. Rings	6075.6	6075.6	6075.6	6075.6	6075.6	6075.6
Nozzles	5262.9	5262.9	5262.9	5262.9	5262.9	5262.9
Wld Weights	2840.0	2840.0	2840.0	2840.0	2840.0	2840.0
Trunnions	2347.6	2347.6	2347.6	2347.6	2347.6	2347.6
Platforms	...	...	...	25070.2	25070.2	25070.2
Insulation	...	...	...	15487.7	15487.7	15487.7
Ope. Liquid	...	...	...	...	...	21278.6
Test Liquid	...	751825.1	...	...	...	...
Totals	130819.5	882644.7	130819.5	171377.5	171377.5	192656.1

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 8.5 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	130819.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	882644.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	130819.5 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	171377.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	171377.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	192656.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	185650.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	923202.6 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
------	----	------------------------------	-----------------------------



10	20	475633	468194
20	30	616758	612611
30	40	616758	612611
40	50	616758	612611
50	60	616758	612611
60	70	616758	612611
70	80	616758	612611
80	90	493406	490089
90	100	475633	468194

Total 5145217.0 5102140.0 cm²

Total 5538.3 5491.9 ft²

**Nozzle Flange MAWP Results: (bars & °C)**

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1-24"	5.90	19.60	415	150	GR 1.1	...	...	...	...
K6-2"	31.06	51.00	415	300	GR 1.1	...	...	...	...
N1A-40"	31.06	51.00	415	300	GR 1.1	...	...	...	...
M2-24"	5.90	19.60	415	150	GR 1.1	...	...	...	...
N2-52"	31.06	51.00	415	300	GR 1.1	...	...	...	...
N3-4"	31.06	51.00	415	300	GR 1.1	38.208	8.386	31.062	8.386
N1B-40"	31.06	51.00	415	300	GR 1.1	...	...	...	...
K1A	31.06	51.00	415	300	GR 1.1	...	...	...	...
K2A	31.06	51.00	415	300	GR 1.1	...	...	...	...
K3A	31.06	51.00	415	300	GR 1.1	...	...	...	...
K4A	31.06	51.00	415	300	GR 1.1	...	...	...	...
K5A	31.06	51.00	415	300	GR 1.1	...	...	...	...
K1B	31.06	51.00	415	300	GR 1.1	...	...	...	...
K2B	31.06	51.00	415	300	GR 1.1	...	...	...	...
K3B	31.06	51.00	415	300	GR 1.1	...	...	...	...
K4B	31.06	51.00	415	300	GR 1.1	...	...	...	...
K5B	31.06	51.00	415	300	GR 1.1	...	...	...	...
Min Rating	5.900	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N3-4"	138.176	0.50	202	0	3.500	8.386	Pass

Evaluating Flange on Nozzle [N3-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(202) + 4 \cdot \text{abs}(0) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot [(31.062 - 3.500) + 500 \cdot 31.062]$$

$$= 0.32293E+04 \leq 0.36422E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(31.062, 0.500 \cdot 31.062 + 31.062 - 3229.3 / (\pi \cdot 138.176^3))$$

$$= \min(31.062, 8.386)$$

$$= 8.386 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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**Wind Load Calculation Results:****Input Values:**

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm.
Percent Wind for Hydrotect	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm.
Distance Upwind of Crest Lh	0	mm.
Distance from Crest to the Vessel x	0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.9)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.540
Structure Height to Diameter ratio	3.372

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 3600 + 300 = 3900 \text{ mm.}$$

$$= 12.8 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z $[Kz]$:

$$\text{Because } z (12.8 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Zg)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:

= 1, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.54 \cdot 102$$

$$= 750 \text{ lbs. } [340] \text{ Kgf}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
left Dish	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell1-1	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell12	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell13	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell14	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell15	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell16	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
shell17	3900.0	0.000	0.000	0.000	0.849	1.000	78.120
Right dish	3900.0	0.000	0.000	0.000	0.849	1.000	78.120

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT:[1 OF 1]	0.00	3900.00	78.12	0.00	0.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Davit-1	0.00	3900.00	78.12	0.00
N1-A insert p	0.00	3900.00	78.12	0.00
N1-A elbow	0.00	4900.00	78.12	0.00
Clip	0.00	3900.00	78.12	0.00
Manway davit	0.00	3900.00	78.12	0.00
Clip wt	0.00	3900.00	78.12	0.00
N2-Insert pla	0.00	3900.00	78.12	0.00
Impengment pl	0.00	3900.00	78.12	0.00
N1B insertpla	0.00	3900.00	78.12	0.00
N1B elbow	0.00	4900.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	3900	8044.8	94954.3	78.12	374.2
20	30	3900	8044.8	241344	78.12	951.099
30	40	3900	8044.8	241344	78.12	951.099
40	50	3900	8044.8	241344	78.12	951.099
50	60	3900	8044.8	241344	78.12	951.099
60	70	3900	8044.8	241344	78.12	951.099
70	80	3900	8044.8	241344	78.12	951.099
80	90	3900	8044.8	193075	78.12	760.879

Front end KOD (V6502 ABC)

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Wind Load Calculation:

Step: 18

3:25pm

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90	100	3900	8044.8	94954.3	78.12	374.2
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The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.800
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\
 SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(22.50.75)$
 $= 0.206$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.21, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1 \cdot 0.16 \cdot 178429 (1 + 2 \cdot 0.8) / (3/1.25) \\
 &= 12123.667 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.16 \cdot 1.25 \cdot 178429 = 55955 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 178429$
 $= 10491.637 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 12123.667 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 178429$
 $= 5595.5 \text{ Kgf}$

**Earthquake Load Calculation:**

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	3250	16220.8	1102.15
20	30	3250	16220.8	1102.15
30	40	3250	16220.8	1102.15
30	40	3250	16220.8	1102.15
40	50	3250	16220.8	1102.15
50	60	3250	16220.8	1102.15
60	70	3250	16220.8	1102.15
70	0	3250	16220.8	1102.15
70	80	3250	16220.8	1102.15
80	90	3250	16220.8	1102.15
90	100	3250	16220.8	1102.15

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.
Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	10250.000 mm.
Center of Gravity of the Platforms	10299.999 mm.
Center of Gravity of the Liquid	10250.001 mm.
Center of Gravity of the Insulation	10249.998 mm.
Center of Gravity of the Stiffening Rings	11168.938 mm.
Center of Gravity of the Nozzles	8829.046 mm.
Center of Gravity of the Added Weights (Operating)	8796.684 mm.
Center of Gravity of the Added Weights (Empty)	8796.684 mm.
Center of Gravity of the Trunnions	10249.998 mm.
Center of Gravity of Bare Shell New and Cold	10249.991 mm.
Center of Gravity of Bare Shell Corroded	10249.998 mm.
Vessel CG in the Operating Condition	10224.311 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	10203.956 mm.
Vessel CG in the Test Condition	10243.175 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	20500.00 mm.
Total vessel weight (No Liquid) Twt	171377.50 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	257066.25 Kgf
Distance to the Lug on the Left End	4100.00 mm.
Distance to the Lug on the Right End	16400.00 mm.
Design Reaction force at the Left End Trunnion	129495.44 Kgf
Design Reaction force at the Right End Trunnion	127570.81 Kgf
CG distance during field lifting operation	10203.96 mm.
CG distance from the left end lug/trunnion	6103.96 mm.
CG distance from the nearer lifting lug/trunnion	6103.96 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	2.69	4250.00	79.78 (UG-23)
Shear	1.66	3650.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	40507.4	0.9
1850.00	85371.0	1.1	57645.2	1.3
6650.00	91721.5	1.2	18588.1	0.4
11450.00	32410.8	0.4	18803.4	0.4
16250.00	158191.8	2.1	65542.7	1.4
20450.00	15416.5	0.2	3116.0	0.1

Unity Check (Actual Stress / Allowable Stress):

Front end KOD (V6502 ABC).

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FileName : Front end Flare KOD (V6502 ABC) 9%

Center of Gravity Calculation: Step: 20 3:25pm Sep 16,2026



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Maximum Unity Check is 0.034 at elevation 4250 mm. - Must be ≤ 1

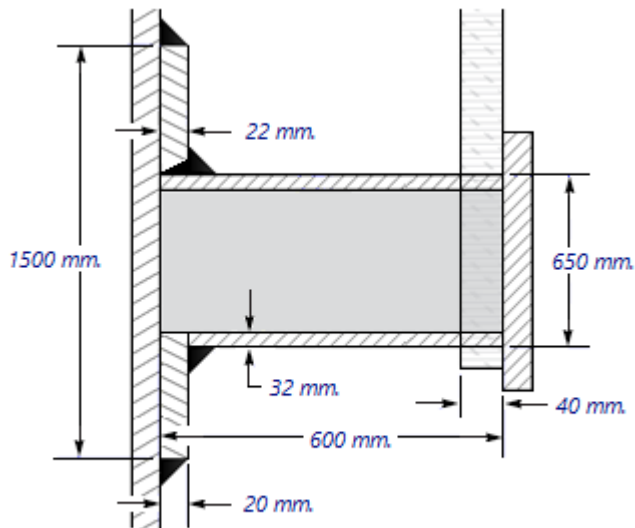
Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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**Trunnion Analysis: Left End of Vessel****Input Data:**

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 650.000 mm.
Trunnion Thickness	Tthk 32.0000 mm.
Trunnion Length	Tl 600.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1500.000 mm.
Pad Thickness	Thkp 22.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 20.0000 mm.
Bail/Sling Width	Tt 40.0000 mm.
Sling Angle from Horizontal	90.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 43165 • 1.5
 = 64747.7 Kgf

Trunnion Section Modulus [Z]:
 $= \pi (Tod^4 - Tid^4) / (32 \cdot Tod)$
 $= \pi (650^4 - 586^4) / (32 \cdot 650)$
 = 9150711.0 mm.³

Trunnion Bending Moment [Tmom]:
 $= (Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf$
 $= (600 - 0.5 \cdot 40) / 9.807 \cdot 64748$
 = 37554.4 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 $= Tmom \cdot 9.807 / Z$
 $= 37554 \cdot 9.807 / 9150711$
 = 40.25 N./mm [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (172.92)

Trunnion Cross Section Area [Txsa]:
 $= \pi / 4 (Tod^2 - Tid^2)$
 $= \pi / 4 (650^2 - 586^2)$



$$= 621.3 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= Tlf/Txsa$$

$$= 64748/621$$

$$= 10.22 \text{ N./mm [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= Sbc/Sba + Ssc/Ssa$$

$$= 40.2/173 + 10.2/105$$

$$= 0.330 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (T_{od}/2)^2$$

$$= \pi (650/2)^2$$

$$= 331830.5 \text{ mm.}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= T_{mom}/Sw$$

$$= 37554/331830$$

$$= 113.171 \text{ Kg/mm.}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, fs/(0.4 \cdot Sy))$$

$$= \max(10 \text{ mm}, 113/(0.4 \cdot 262))$$

$$= 10.590 \text{ mm.}$$

Trunnion Bearing Contact Area [Tbca]:

$$= T_{od} \cdot T_t$$

$$= 650 \cdot 40$$

$$= 260.0 \text{ cm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= Tlf/Tbca$$

$$= 64748/260$$

$$= 24.42 \text{ N./mm [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.3303 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	40.25	172.92	Pass
Shear Stress	10.22	104.80	Pass
Bearing Stress	24.42	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	6500.000 mm.
Vessel Thickness	Tv	22.000 mm.



Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	650.000	mm.
Nozzle Thickness	Tn	32.000	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	22.000	mm.
Diameter of Reinforcing Pad	Dpad	1540.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust		No	

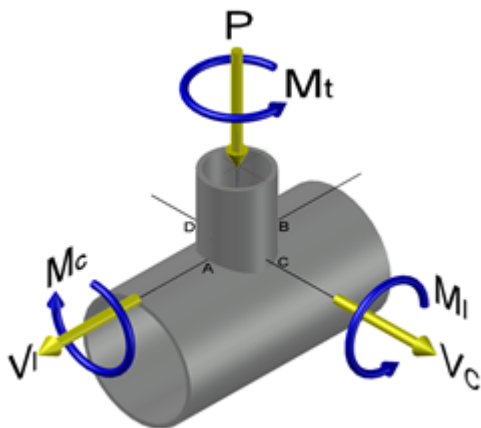
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	VL	0.0	Kgf
Circumferential Shear (SUS)	Vc	64747.7	Kgf
Circumferential Moment (SUS)	Mc	37554.4	Kg-m.
Longitudinal Moment (SUS)	ML	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 650.0 + 2 \cdot 1.65 \cdot \sqrt{3261.0 (22.0 - 0)}$$

$$= 1533.895 \text{ mm.}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	64747.7	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	37554.4	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.



Torsional Moment MT 0.0 Kg-m.

Dimensionless Parameters used : Gamma = 74.36

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.087	4C	12.493	(A,B)
N(PHI) / (P/Rm)	0.087	3C	10.470	(C,D)
M(PHI) / (P)	0.087	2C1	0.074	(A,B)
M(PHI) / (P)	0.087	1C	0.111	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.087	3A	2.393	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.087	1A	0.092	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.087	3B	8.494	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.087	1B	0.042	(A,B,C,D)
N(x) / (P/Rm)	0.087	3C	10.470	(A,B)
N(x) / (P/Rm)	0.087	4C	12.493	(C,D)
M(x) / (P)	0.087	1C1	0.114	(A,B)
M(x) / (P)	0.087	2C	0.075	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.087	4A	3.656	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.087	2A	0.050	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.087	4B	2.419	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.087	2B	0.061	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-21.5	-21.5	21.5	21.5
Circ. Bend. MC		0.0	0.0	0.0	0.0	-369.4	369.4	369.4	-369.4
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-390.9	347.8	390.9	-347.8
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-32.9	-32.9	32.9	32.9
Long. Bend. MC		0.0	0.0	0.0	0.0	-200.8	200.8	200.8	-200.8
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-233.7	167.9	233.7	-167.9
Shear VC		14.1	14.1	-14.1	-14.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		14.1	14.1	-14.1	-14.1	0.0	0.0	0.0	0.0
Str. Int.		28.3	28.3	28.3	28.3	390.9	347.8	390.9	347.8

Dimensionless Parameters used : Gamma = 148.23

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.207	4C	13.902	(A,B)
N(PHI) / (P/Rm)	0.207	3C	5.132	(C,D)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-21.5	-21.5	21.5	21.5
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-369.4	369.4	369.4	-369.4
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Long. P1 (SUS)	0.0	0.0	0.0	0.0	-32.9	-32.9	32.9	32.9
Long. Q (SUS)	0.0	0.0	0.0	0.0	-200.8	200.8	200.8	-200.8
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	14.1	14.1	-14.1	-14.1	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	28.3	28.3	28.3	28.3	32.9	32.9	32.9	32.9
Pm+P1+Q (Total)	28.3	28.3	28.3	28.3	390.9	347.8	390.9	347.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	32.89	207.00	Passed
Pm+P1+Q (TOTAL)	390.91	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. P1 (SUS)		0.0	0.0	0.0	0.0	-30.4	-30.4	30.4	30.4
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-361.5	361.5	361.5	-361.5
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)		0.0	0.0	0.0	0.0	-96.8	-96.8	96.8	96.8
Long. Q (SUS)		0.0	0.0	0.0	0.0	-143.3	143.3	143.3	-143.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)		11.9	11.9	-11.9	-11.9	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)		23.9	23.9	23.9	23.9	96.8	96.8	96.8	96.8
Pm+P1+Q (Total)		23.9	23.9	23.9	23.9	391.9	331.2	391.9	331.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	96.81	207.00	Passed
Pm+P1+Q (TOTAL)	391.92	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Front end KOD (V6502 ABC)

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FileName : Front end Flare KOD (V6502 ABC) 9%

Trunnion Calcs: Left End



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Step: 21 3:25pm Sep 16,2026



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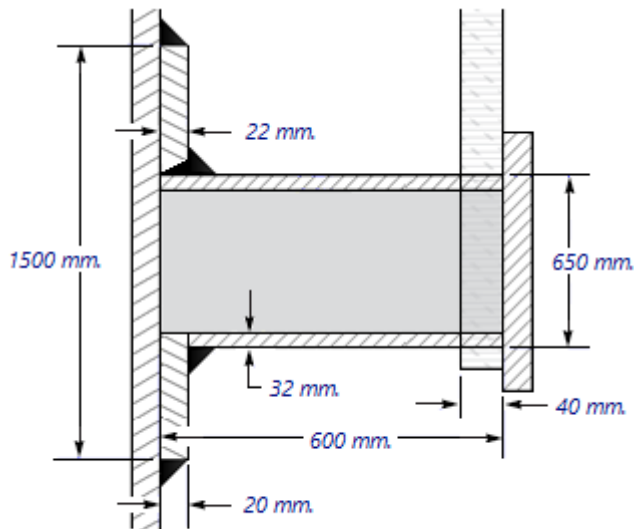
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Trunnion Analysis: Right End of Vessel

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 650.000 mm.
Trunnion Thickness	Tthk 32.0000 mm.
Trunnion Length	Tl 600.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1500.000 mm.
Pad Thickness	Thkp 22.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 20.0000 mm.
Bail/Sling Width	Tt 40.0000 mm.
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Sling Angle from Horizontal: 0.0 deg

Trunnion Lifting Load [Tlf]:

$$\begin{aligned}
 &= \text{Resultant Load} \cdot \text{Impact Factor} \\
 &= 42524 \cdot 1.5 \\
 &= 63785.4 \text{ Kg}
 \end{aligned}$$

Trunnion Section Modulus [Z]:

$$\begin{aligned}
 &= \pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od}) \\
 &= \pi (650^4 - 586^4) / (32 \cdot 650) \\
 &= 9150711.0 \text{ mm}^3
 \end{aligned}$$

Trunnion Bending Moment [Tmom]:

$$\begin{aligned}
 &= (Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf \\
 &= (600 - 0.5 \cdot 40) / 9.807 \cdot 63785 \\
 &= 36996.3 \text{ Kg-m}
 \end{aligned}$$

Trunnion Calculated Bending Stress [Sbc]:

$$\begin{aligned}
 &= Tmom \cdot 9.807 / Z \\
 &= 36996 \cdot 9.807 / 9150711 \\
 &= 39.65 \text{ N./mm [Passed]} \quad Sbc < \min(0.66 \cdot SY \cdot Occfac, 0.75 \cdot Sy) \quad (172.92)
 \end{aligned}$$

Trunnion Cross Section Area [Txa]:

$$= \pi/4 (T_{od}^2 - T_{id}^2)$$



$$= \pi/4 (650^2 - 586^2)$$

$$= 621.3 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= Tlf/Txsa$$

$$= 63785/621$$

$$= 10.07 \text{ N./mm [Passed] } Ssc < 0.4 \cdot Sy \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= Sbc/Sba + Ssc/Ssa$$

$$= 39.6/173 + 10.1/105$$

$$= 0.325 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (Tod/2)^2$$

$$= \pi (650/2)^2$$

$$= 331830.5 \text{ mm.}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= Tmom/Sw$$

$$= 36996/331830$$

$$= 111.489 \text{ Kg/mm.}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10\text{mm}, fs/(0.4 \cdot Sy))$$

$$= \max(10\text{mm}, 111/(0.4 \cdot 262))$$

$$= 10.433 \text{ mm.}$$

Trunnion Bearing Contact Area [Tbca]:

$$= Tod \cdot Tt$$

$$= 650 \cdot 40$$

$$= 260.0 \text{ cm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= Tlf/Tbca$$

$$= 63785/260$$

$$= 24.06 \text{ N./mm [Passed] } Sbec < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.3254 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	39.65	172.92	Pass
Shear Stress	10.07	104.80	Pass
Bearing Stress	24.06	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).**Input Echo, WRC107/537 Item 1, Description: Trunnion**

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	6500.000 mm.
Vessel Thickness	Tv	22.000 mm.



Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	650.000	mm.
Nozzle Thickness	Tn	32.000	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2
Thickness of Reinforcing Pad	Tpad	22.000	mm.
Diameter of Reinforcing Pad	Dpad	1540.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust	No		

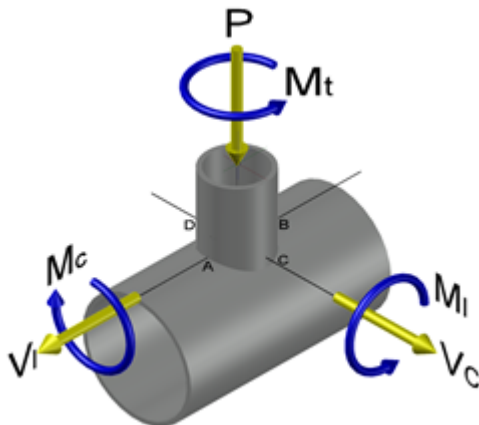
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	0.0	Kgf
Circumferential Shear (SUS)	Vc	63785.4	Kgf
Circumferential Moment (SUS)	Mc	36996.3	Kg-m.
Longitudinal Moment (SUS)	Ml	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 650.0 + 2 \cdot 1.65 \cdot \sqrt{(3261.0 (22.0 - 0))} \\
 &= 1533.895 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	63785.4	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	36996.3	Kg-m.



Longitudinal Moment ML 0.0 Kg-m.
Torsional Moment MT 0.0 Kg-m.

Dimensionless Parameters used : Gamma = 74.36

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.087	4C	12.493	(A,B)
N(PHI) / (P/Rm)	0.087	3C	10.470	(C,D)
M(PHI) / (P)	0.087	2C1	0.074	(A,B)
M(PHI) / (P)	0.087	1C	0.111	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.087	3A	2.393	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.087	1A	0.092	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.087	3B	8.494	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.087	1B	0.042	(A,B,C,D)
N(x) / (P/Rm)	0.087	3C	10.470	(A,B)
N(x) / (P/Rm)	0.087	4C	12.493	(C,D)
M(x) / (P)	0.087	1C1	0.114	(A,B)
M(x) / (P)	0.087	2C	0.075	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.087	4A	3.656	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.087	2A	0.050	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.087	4B	2.419	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.087	2B	0.061	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-21.2	-21.2	21.2	21.2
Circ. Bend. MC		0.0	0.0	0.0	0.0	-363.9	363.9	363.9	-363.9
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-385.1	342.7	385.1	-342.7
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-32.4	-32.4	32.4	32.4
Long. Bend. MC		0.0	0.0	0.0	0.0	-197.8	197.8	197.8	-197.8
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-230.2	165.4	230.2	-165.4
Shear VC		13.9	13.9	-13.9	-13.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		13.9	13.9	-13.9	-13.9	0.0	0.0	0.0	0.0
Str. Int.		27.8	27.8	27.8	27.8	385.1	342.7	385.1	342.7

Dimensionless Parameters used : Gamma = 148.23

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.207	4C	13.902	(A,B)



N(PHI) / (P/Rm)	0.207	3C	5.132	(C,D)
M(PHI) / (P)	0.207	2C1	0.012	(A,B)
M(PHI) / (P)	0.207	1C !	0.070	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.207	3A	3.987	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.207	1A	0.053	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.207	3B	8.737	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.207	1B	0.008	(A,B,C,D)
N(x) / (P/Rm)	0.207	3C	5.132	(A,B)
N(x) / (P/Rm)	0.207	4C	13.902	(C,D)
M(x) / (P)	0.207	1C1	0.031	(A,B)
M(x) / (P)	0.207	2C !	0.039	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.207	4A	12.706	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.207	2A	0.021	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.207	4B	4.181	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.207	2B	0.010	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-29.9	-29.9	29.9	29.9
Circ. Bend. MC		0.0	0.0	0.0	0.0	-356.2	356.2	356.2	-356.2
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-386.1	326.2	386.1	-326.2
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-95.4	-95.4	95.4	95.4
Long. Bend. MC		0.0	0.0	0.0	0.0	-141.2	141.2	141.2	-141.2
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-236.5	45.8	236.5	-45.8
Shear VC		11.8	11.8	-11.8	-11.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		11.8	11.8	-11.8	-11.8	0.0	0.0	0.0	0.0
Str. Int.		23.5	23.5	23.5	23.5	386.1	326.2	386.1	326.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-21.2	-21.2	21.2	21.2
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-363.9	363.9	363.9	-363.9



Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)	0.0	0.0	0.0	0.0	-32.4	-32.4	32.4	32.4
Long. Q (SUS)	0.0	0.0	0.0	0.0	-197.8	197.8	197.8	-197.8
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	13.9	13.9	-13.9	-13.9	0.0	0.0	0.0	0.0
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	27.8	27.8	27.8	27.8	32.4	32.4	32.4	32.4
Pm+P1+Q (Total)	27.8	27.8	27.8	27.8	385.1	342.7	385.1	342.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	32.40	207.00	Passed
Pm+P1+Q (TOTAL)	385.10	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. P1 (SUS)		0.0	0.0	0.0	0.0	-29.9	-29.9	29.9	29.9
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-356.2	356.2	356.2	-356.2
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. P1 (SUS)		0.0	0.0	0.0	0.0	-95.4	-95.4	95.4	95.4
Long. Q (SUS)		0.0	0.0	0.0	0.0	-141.2	141.2	141.2	-141.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)		11.8	11.8	-11.8	-11.8	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)		23.5	23.5	23.5	23.5	95.4	95.4	95.4	95.4
Pm+P1+Q (Total)		23.5	23.5	23.5	23.5	386.1	326.2	386.1	326.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+P1 (SUS)	95.37	207.00	Passed
Pm+P1+Q (TOTAL)	386.09	413.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Front end KOD (V6502 ABC)

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Trunnion Calcs: Right End



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Step: 22 3:25pm Sep 16,2026



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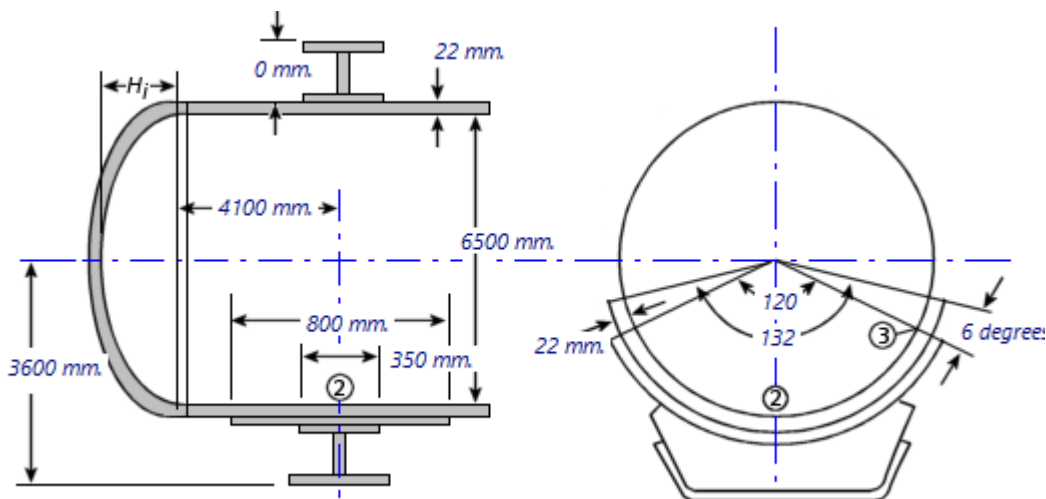


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.80	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1636.80	mm.
Shell Allowable Stress used in Calculation		90.70	N./mm ²
Head Allowable Stress used in Calculation		90.70	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	103885.77	Kgf
Pressure used in Analysis	P 3.500	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	26.93	90.70
Long. Stress at Bottom	of Midspan	29.04	90.70
Long. Stress at Top	of Saddles	41.44	90.70
Long. Stress at Bottom	of Saddles	20.52	90.70
Tangential Shear in Shell		9.72	72.56
Circ. Stress at Horn of Saddle		65.17	113.38
Circ. Compressive Stress in Shell		2.43	90.70

**Intermediate Results: Saddle Reaction Q due to Wind or Seismic:**

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (7216/2 + 0) \cdot 3600/5700$$

$$= 6836.1 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4368, 0) \cdot 3600/12300$$

$$= 1278.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(12124, 0) \cdot 3600/12300$$

$$= 3548.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (12124/2 + 0) \cdot 3600/5700$$

$$= 11485.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 92400 + \max(1278, 6836, 3548, 11486)$$

$$= 103885.8 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	110999.22	Kgf
Transverse Shear Load	6061.83	Kgf
Longitudinal Shear Load	12123.67	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 5700.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/(3L))]$$

$$= -103886 \cdot 4100 [1 - (1 - 4100/20500 + (3262^2 - 1637^2) / (2 \cdot 4100 \cdot 20500)) / (1 + (4 \cdot 1637^2)/(3 \cdot 20500))]$$

$$= -99744.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 103886 \cdot 20500 / 4 ((1 + 2(3262^2 - 1637^2)/(20500^2)) / (1 + (4 \cdot 1637^2)/(3 \cdot 20500)) - 4 \cdot 4100/20500)$$

$$= 73486.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) - 73487 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= 26.93 \text{ N./mm}^2$$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) + 73487 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= 29.04 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) - 99744 / (0.11 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 41.44 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) + 99744 / (0.19 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 20.52 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 103886 (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1637 / 3))$$

$$= 56334.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 56334 / (3262 \cdot 20.4)$$

$$= 9.72 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{3262 \cdot 20.4}$$

$$= 201.205 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{3262 \cdot 20.4}, 2 \cdot 4100)$$

$$= 752.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(90.7 / 90.7, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 103886 \cdot 0.1 / (752(20.4 + 1 \cdot 22))$$

$$= -2.43 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -103886 / (4(20.4 + 1 \cdot 22)752) -$$

$$12 \cdot 0.053 \cdot 103886 \cdot 3262 / (20500(20.4 + 1 \cdot 22)^2)$$

$$= -65.17 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 12300.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13963 \text{E-04} \cdot 12300 (415 - 21.1)$$

$$= 67.648 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C



Saddle Yield Stress	Sy	242.6	N./mm2
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3252 \cdot 20.4) + 800})^3) / 12$$

$$= 85.022 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 85 + 245 (199 - 10.2)^2$$

$$= 87915.656 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1201.8	20.4	10.2	245.2	250066.3	0.879E+05
Wearplate	800.0	22.0	31.4	176.0	552639.9	0.498E+05
Web	28.0	226.0	155.4	63.3	983373.4	0.392E+04
BasePlate	904.0	80.0	308.4	723.2	22303510.0	0.897E+05
Totals	...	...	...	1207.6	24089590.0	0.231E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24089590 / 1208$$

$$= 199.476 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 103886$$

$$= 21143.0059 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.1543 \text{ N./mm2}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm2}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3600 - 3252 \cdot \sin(120/2) / 1.05$$

$$= 910.948 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19260.5703 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.2901 \text{ N./mm2}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm2}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt[3]{(3 \cdot (103886 + 7113) \cdot 904 / (2 \cdot 5700 \cdot 162))}$$

$$= 40.014 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$



Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 110999 / (5700 \cdot 904)$$

$$= 2.154 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 5158 \cdot 2.15$$

$$= 11110.388 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.2 + 160$$

$$= 249.911 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 11110 / 250$$

$$= 4.360 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 28$$

$$= 322.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1927674 / 250$$

$$= 77.135 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.1, 350 - 77.1)$$

$$= 272.865 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(25545 / 250)}$$

$$= 101.101 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3600 - \cos(120/2) (3250 + 22 + 22) - 80$$

$$= 1873.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (3250 + 22 + 22)$$



= 5705.375 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 5700 - 50.8

= 5649.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (5705 - 5649)/2

= 28.087 mm.

Length of Outer Rib [L]:

= $\sqrt{L_v^2 + D_d^2}$

= $\sqrt{1873^2 + 28.1^2}$

= 1873.211 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 243)}$

= 127.551

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1873/101

= 18.528

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= $12124 / (2 \cdot 5700) \cdot 1141 \cdot 1873 / 2$

= 1136.602 Kg-m.

Compressive Allowable, KL/r < Cc (18.5 < 128) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$

= $(1 - (18.5)^2 / (2 \cdot 128^2)) 243 /$

$(5/3 + 3 \cdot (18.5) / (8 \cdot 128) - (18.5^3) / (8 \cdot 128^3))$

= 139.5 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= $4.36 / 139 + (1137 \cdot 273 / 255446192) / 162$

= 0.105

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 2151325/410

= 52.515 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(52.5, abs(350 - 52.5))

= 297.485 mm.

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$

= $3600 - \sqrt{(3294 + 22)^2 - (1141/2)^2} - 80$

= 275.786 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$



$$= \sqrt{29533/410}$$

$$= 84.907 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 276/84.9$$

$$= 3.248$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 12124 / (2 \cdot 5700)$$

$$= 1.063 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.06 \cdot 1141 \cdot 276$$

$$= 334.676 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 335/992772$$

$$= 3.306 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.25 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.25)^2 / (2 \cdot 128^2)) 243 /$$

$$(5/3 + 3 \cdot (3.25) / (8 \cdot 128) - (3.25^3) / (8 \cdot 128^3))$$

$$= 144.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 3.91/145 + (335 \cdot 297/29533)/162$$

$$= 0.047$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	570.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	22.1008 cm2
Saddle Load Q0 (Weight)	Q0	99513.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3548.4 Kg
Maximum Transverse Force	Ft	6061.8 Kg
Maximum Longitudinal Force	Fl	12123.7 Kg
Saddle Bolted to Steel Foundation	No	

Friction force due to thermal growth, [Ff]:

$$= 39495.2 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 39495 / (22.1 \cdot 4)$$

$$= 43.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6062 / (22.1 \cdot 4)$$

$$= 6.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarear]:



= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$

= $12124 / (82.8 \cdot 4)$

= 3.5898 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $3600 \cdot 6062 + 0$

= 21823.03 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $21823 / 99514$

= 219.29 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0, 3.59, 0]$

= 3.5898 cm²

MDMT Calculations for: left saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.41$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.80	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm2
Inside or Mean Depth of Head	Hi or hm	1636.80	mm.
Shell Allowable Stress used in Calculation		90.70	N./mm2
Head Allowable Stress used in Calculation		90.70	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		103110.16	Kgf
Pressure used in Analysis	P	3.500	bars

Horizontal Vessel Analysis Results:	Actual	Allowable
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	N./mm ²	N./mm ²
Long. Stress at Top of Midspan	26.93	90.70
Long. Stress at Bottom of Midspan	29.03	90.70
Long. Stress at Top of Saddles	41.34	90.70
Long. Stress at Bottom of Saddles	20.58	90.70
Tangential Shear in Shell	9.65	72.56
Circ. Stress at Horn of Saddle	64.68	113.38
Circ. Compressive Stress in Shell	2.41	90.70

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (7216/2 + 0) \cdot 3600/5700$$

$$= 6836.1 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4368, 0) \cdot 3600/12300$$

$$= 1278.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(12124, 0) \cdot 3600/12300$$

$$= 3548.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (12124/2 + 0) \cdot 3600/5700$$

$$= 11485.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 91625 + \max(1278, 6836, 3548, 11486)$$

$$= 103110.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 110223.62 Kgf

Transverse Shear Load 6061.83 Kgf

Longitudinal Shear Load 12123.67 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 5700.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -103110 \cdot 4100 [1 - (1 - 4100/20500 + (3262^2 \cdot 1637^2)/$$

$$(2 \cdot 4100 \cdot 20500))/(1 + (4 \cdot 1637)/(3 \cdot 20500))]$$



$$= -98999.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 103110 \cdot 20500 / 4 \cdot ((1 + 2(3262^2 - 1637^2) / (20500^2)) / (1 + (4 \cdot 1637) / (3 \cdot 20500)) - 4 \cdot 4100 / 20500)$$

$$= 72938.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) - 72938 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= 26.93 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) + 72938 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= 29.03 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) - 99000 / (0.11 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 41.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 3.5 \cdot 3262 / (2 \cdot 20.4) + 99000 / (0.19 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 20.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 103110 \cdot (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1637 / 3))$$

$$= 55913.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 55914 / (3262 \cdot 20.4)$$

$$= 9.65 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3262 \cdot 20.4)}$$

$$= 201.205 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(3262 \cdot 20.4)}, 2 \cdot 4100)$$

$$= 752.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr / S, 1)$$

$$= \min(90.7 / 90.7, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 103110 \cdot 0.1 / (752(20.4 + 1 \cdot 22))$$

$$= -2.41 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -103110 / (4(20.4 + 1 \cdot 22)752) - 12 \cdot 0.053 \cdot 103110 \cdot 3262 / (20500(20.4 + 1 \cdot 22)^2)$$

$$= -64.68 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.



Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	N./mm2
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3252 \cdot 20.4)} + 800) 20.4^3) / 12$$

$$= 85.022 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 85 + 245 (199 - 10.2)^2$$

$$= 87915.656 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	1201.8	20.4	10.2	245.2	250066.3	0.879E+05
Wearplate	800.0	22.0	31.4	176.0	552639.9	0.498E+05
Web	28.0	226.0	155.4	63.3	983373.4	0.392E+04
BasePlate	904.0	80.0	308.4	723.2	22303510.0	0.897E+05
Totals	...	...	...	1207.6	24089590.0	0.231E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24089590 / 1208$$

$$= 199.476 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 103110$$

$$= 20985.1543 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.1382 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3600 - 3252 \cdot \sin(120/2) / 1.05$$

$$= 910.948 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19116.7734 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.1685 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(103110 + 7113) 904 / (2 \cdot 5700 \cdot 162))}$$

$$= 39.874 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:



Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 110224 / (5700 \cdot 904)$$

$$= 2.139 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 5158 \cdot 2.14$$

$$= 11032.755 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.2 + 160$$

$$= 249.911 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 11033 / 250$$

$$= 4.329 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 28$$

$$= 322.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1927674 / 250$$

$$= 77.135 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.1, 350 - 77.1)$$

$$= 272.865 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(25545 / 250)}$$

$$= 101.101 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3600 - \cos(120/2) (3250 + 22 + 22) - 80$$



= 1873.000 mm.

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$

= $2 \cdot \sin(120/2) (3250 + 22 + 22)$

= 5705.375 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 5700 - 50.8

= 5649.200 mm.

Length of Outer Rib, step 4, [Dd]:

= $(Lw - Bd) / 2$

= $(5705 - 5649) / 2$

= 28.087 mm.

Length of Outer Rib [L]:

= $\sqrt{Lw^2 + Dd^2}$

= $\sqrt{1873^2 + 28.1^2}$

= 1873.211 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 243)}$

= 127.551

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 1873 / 101$

= 18.528

Bending Moment [Rm]:

= $Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$

= $12124 / (2 \cdot 5700) \cdot 1141 \cdot 1873 / 2$

= 1136.602 Kg-m.

Compressive Allowable, $KL/r < Cc$ ($18.5 < 128$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (18.5)^2 / (2 \cdot 128^2)) 243 /$

$(5/3 + 3 \cdot (18.5) / (8 \cdot 128) - (18.5^3) / (8 \cdot 128^3))$

= 139.5 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc / Sca + (Rm \cdot C1 / I) / Sba$

= $4.33 / 139 + (1137 \cdot 273 / 255446192) / 162$

= 0.105

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 2151325/410

= 52.515 mm.

Distance to Centroid [C1]:

= $\max(ytot, \text{abs}(\text{Saddle Width} - ytot))$

= $\max(52.5, \text{abs}(350 - 52.5))$

= 297.485 mm.

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$



$$= 3600 - \sqrt{(3294 + 22)^2 - (1141/2)^2} - 80$$

$$= 275.786 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{29533/410}$$

$$= 84.907 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 276/84.9$$

$$= 3.248$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 12124 / (2 \cdot 5700)$$

$$= 1.063 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.06 \cdot 1141 \cdot 276$$

$$= 334.676 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 335/992772$$

$$= 3.306 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.25 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.25)^2 / (2 \cdot 128^2)) 243 / (5/3 + 3 \cdot (3.25) / (8 \cdot 128) - (3.25)^3 / (8 \cdot 128^3))$$

$$= 144.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 3.88/145 + (335 \cdot 297/29533) / 162$$

$$= 0.047$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	570.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	22.1008 cm ²
Saddle Load Q0 (Weight)	Q0	98738.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	3548.4 Kg
Maximum Transverse Force	Ft	6061.8 Kg
Maximum Longitudinal Force	Fl	12123.7 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 12124 / (22.1 \cdot 4)$$

$$= 13.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6062 / (22.1 \cdot 4)$$

$$= 6.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 12124/(82.8 • 4)
= 3.5898 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 3600 • 6062 + 0
= 21823.03 Kg-m.

Eccentricity (e):
= Rmom / QO
= 21823/98738
= 221.02 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 3.59, 0]
= 3.5898 cm²

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 22, tr = 9.72, c = 1.6 mm., E* = 0.85
Thickness Ratio = tr * E*/(tg - c) = 0.41, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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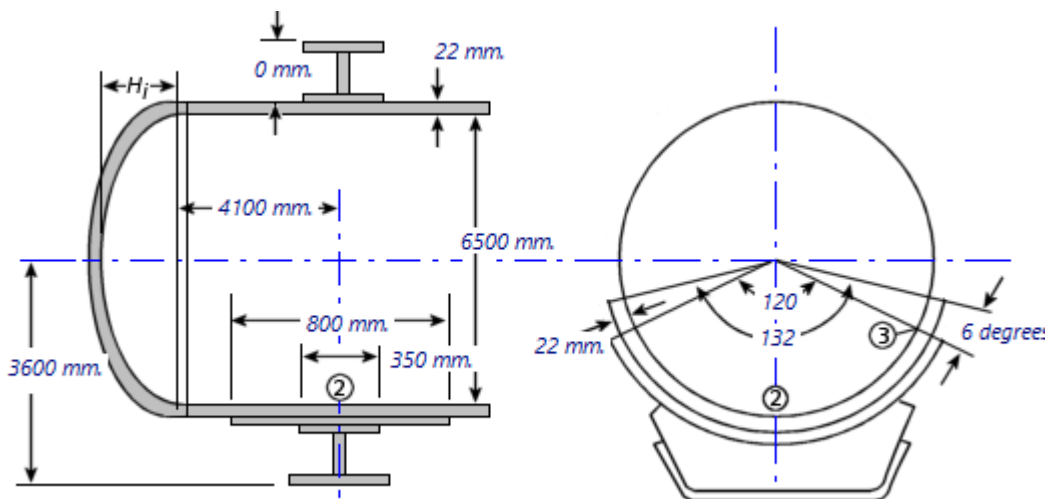


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.00	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1636.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	458293.78	Kgf
Pressure used in Analysis	P 6.968	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	47.32	234.00
Long. Stress at Bottom	of Midspan	55.97	234.00
Long. Stress at Top	of Saddles	106.71	234.00
Long. Stress at Bottom	of Saddles	21.12	234.00
Tangential Shear in Shell		39.77	138.58
Circ. Stress at Horn of Saddle		267.47	350.99
Circ. Compressive Stress in Shell		10.11	234.00

**Intermediate Results: Saddle Reaction Q due to Wind or Seismic:**

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (3608/2 + 0) \cdot 3600/5700$$

$$= 3418.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind; Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(2184, 0) \cdot 3600/12300$$

$$= 639.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 454876 + \max(639, 3418, 0, 0)$$

$$= 458293.8 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	465407.25	Kgf
Transverse Shear Load	1803.97	Kgf
Longitudinal Shear Load	2183.97	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 5700.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -458294 \cdot 4100 [1 - (1 - 4100/20500 + (3261^2 - 1636^2)/(2 \cdot 4100 \cdot 20500))/(1 + (4 \cdot 1636)/(3 \cdot 20500))]$$

$$= -439982.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 458294 \cdot 20500/4((1 + 2(3261^2 - 1636^2)/(20500^2))/(1 + (4 \cdot 1636)/(3 \cdot 20500)) - 4 \cdot 4100/20500)$$

$$= 324265.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 6.97 \cdot 3261/(2 \cdot 22) - 324265/(\pi \cdot 3261^2 \cdot 22)$$

$$= 47.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 6.97 \cdot 3261/(2 \cdot 22) + 324265/(\pi \cdot 3261^2 \cdot 22)$$

$$= 55.97 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 6.97 \cdot 3261/(2 \cdot 22) - 439982/(0.11 \cdot \pi \cdot 3261^2 \cdot 22)$$

$$= 106.71 \text{ N./mm}^2$$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 6.97 \cdot 3261 / (2 \cdot 22) + 439982 / (0.19 \cdot \pi \cdot 3261^2 \cdot 22)$$

$$= 21.12 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 458294 (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1636 / 3))$$

$$= 248531.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 248531 / (3261 \cdot 22)$$

$$= 39.77 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3261 \cdot 22)}$$

$$= 208.921 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(3261 \cdot 22)}, 2 \cdot 4100)$$

$$= 767.84 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 458294 \cdot 0.1 / (768 (22 + 1 \cdot 22))$$

$$= -10.11 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -458294 / (4(22 + 1 \cdot 22)768) - 12 \cdot 0.053 \cdot 458294 \cdot 3261 / (20500(22 + 1 \cdot 22)^2)$$

$$= -267.47 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3250 \cdot 22)} + 800)^3) / 12$$

$$= 108.000 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 108 + 268 (197 - 11)^2$$

$$= 93168.273 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	1217.1	22.0	11.0	267.8	294546.9	0.932E+05
Wearplate	800.0	22.0	33.0	176.0	580799.9	0.477E+05



Web	28.0	226.0	157.0	63.3	993498.3	0.373E+04
BasePlate	904.0	80.0	310.0	723.2	22419222.0	0.955E+05
Totals	...	...	...	1230.3	24288068.0	0.240E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24288068/1230$$

$$= 197.424 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 458294$$

$$= 93272.7266 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 9.5037 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3600 - 3250 \cdot \sin(120/2) / 1.05$$

$$= 912.271 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 85091.7266 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 68.6257 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(458294 + 7113) \cdot 904 / (2 \cdot 5700 \cdot 173)}$$

$$= 79.147 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 465407 / (5700 \cdot 904)$$

$$= 9.032 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 5158 \cdot 9.03$$

$$= 46584.609 \text{ Kgf}$$



Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 90.2 + 160

= 249.911 cm²

Compressive Stress [Sc]:

= P/Ar

= 46585/250

= 18.280 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 350 - 28

= 322.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A

= 1927674/250

= 77.135 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)

= max(77.1, 350 - 77.1)

= 272.865 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{25545/250}$

= 101.101 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 3600 - cos(120/2)(3250 + 22 + 22) - 80

= 1873.000 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(3250 + 22 + 22)

= 5705.375 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 5700 - 50.8

= 5649.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (5705 - 5649)/2

= 28.087 mm.

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1873^2 + 28.1^2}$

= 1873.211 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1873/101$$

$$= 18.528$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2184 / (2 \cdot 5700) \cdot 1141 \cdot 1873/2$$

$$= 204.749 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.5 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.5)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (18.5) / (8 \cdot 123) - (18.5)^3 / (8 \cdot 123^3))$$

$$= 149.2 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 18.3/149 + (205 \cdot 273/255446192) / 173$$

$$= 0.135$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2151325/410$$

$$= 52.515 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.5, \text{abs}(350 - 52.5))$$

$$= 297.485 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 3600 - \sqrt{(3294 + 22)^2 - (1141/2)^2} - 80$$

$$= 275.786 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(29533/410)}$$

$$= 84.907 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 276/84.9$$

$$= 3.248$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2184 / (2 \cdot 5700)$$

$$= 0.192 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.19 \cdot 1141 \cdot 276$$

$$= 60.289 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 60.3/992772$$



$$= 0.596 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.25 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.25)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (3.25) / (8 \cdot 123) - (3.25)^3 / (8 \cdot 123^3))$$

$$= 155 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 16.4 / 155 + (60.3 \cdot 297 / 29533) / 173$$

$$= 0.109$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	570.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	22.1008 cm2
Saddle Load Q0 (Weight)	Q0	461989.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	639.2 Kg
Maximum Transverse Force	Ft	1804.0 Kg
Maximum Longitudinal Force	Fl	2184.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2184 / (22.1 \cdot 4)$$

$$= 2.4 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1804 / (22.1 \cdot 4)$$

$$= 2.0 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2184 / (68.4 \cdot 4)$$

$$= 0.7828 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 3600 \cdot 1804 + 0$$

$$= 6494.41 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 6494 / 461989$$

$$= 14.06 \text{ mm.} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0, 0.78, 0]$$



= 0.7828 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.00	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1636.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		457518.25	Kgf
Pressure used in Analysis	P	6.968	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	47.32	234.00
Long. Stress at Bottom of Midspan	55.96	234.00
Long. Stress at Top of Saddles	106.62	234.00
Long. Stress at Bottom of Saddles	21.18	234.00
Tangential Shear in Shell	39.70	138.58
Circ. Stress at Horn of Saddle	267.01	350.99
Circ. Compressive Stress in Shell	10.10	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (3608/2 + 0) \cdot 3600/5700$$

$$= 3418.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(2184, 0) \cdot 3600/12300$$

$$= 639.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 454100 + \max(639, 3418, 0, 0)$$

$$= 457518.2 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

Vertical Load (including saddle weight)	464631.72	Kgf
Transverse Shear Load	1803.97	Kgf
Longitudinal Shear Load	2183.97	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 5700.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h_m)}{3L} \right) \right]$$

$$= -457518 \cdot 4100 \left[1 - \left(1 - \frac{4100}{20500} + \frac{(3261^2 - 1636^2)}{(2 \cdot 4100 \cdot 20500)} \right) / \left(1 + \frac{(4 \cdot 1636)}{(3 \cdot 20500)} \right) \right]$$

$$= -439237.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{1 + 2(R_m^2 - h_m^2)/(L^2)}{(1 + (4h_m^2)/(3L)) - 4a/L} \right)$$

$$= 457518 \cdot 20500 / 4 \left(\frac{1 + 2(3261^2 - 1636^2)/(20500^2)}{(1 + (4 \cdot 1636)/(3 \cdot 20500)) - 4 \cdot 4100/20500} \right)$$

$$= 323716.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 6.97 \cdot 3261 / (2 \cdot 22) - 323717 / (\pi \cdot 3261^2 \cdot 22)$$

$$= 47.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 6.97 \cdot 3261 / (2 \cdot 22) + 323717 / (\pi \cdot 3261^2 \cdot 22)$$

$$= 55.96 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 6.97 \cdot 3261 / (2 \cdot 22) - 439238 / (0.11 \cdot \pi \cdot 3261^2 \cdot 22)$$

$$= 106.62 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 6.97 \cdot 3261 / (2 \cdot 22) + 439238 / (0.19 \cdot \pi \cdot 3261^2 \cdot 22)$$

$$= 21.18 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 457518 (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1636 / 3))$$

$$= 248110.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 248110 / (3261 \cdot 22)$$

$$= 39.70 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3261 \cdot 22)}$$

$$= 208.921 \text{ mm.}$$



Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{ 3261 \cdot 22 }, 2 \cdot 4100)$$

$$= 767.84 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

= 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 457518 \cdot 0.1 / (768 (22 + 1 \cdot 22))$$

$$= -10.10 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -457518 / (4(22 + 1 \cdot 22)768) -$$

$$12 \cdot 0.053 \cdot 457518 \cdot 3261 / (20500(22 + 1 \cdot 22)^2)$$

$$= -267.01 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 3250 \cdot 22 } + 800) 22^3) / 12$$

$$= 108.000 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 108 + 268 (197 - 11)^2$$

$$= 93168.273 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	1217.1	22.0	11.0	267.8	294546.9	0.932E+05
Wearplate	800.0	22.0	33.0	176.0	580799.9	0.477E+05
Web	28.0	226.0	157.0	63.3	993498.3	0.373E+04
BasePlate	904.0	80.0	310.0	723.2	22419222.0	0.955E+05
Totals	...	...	...	1230.3	24288068.0	0.240E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24288068 / 1230$$

$$= 197.424 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:



$$= K1 \cdot Q$$

$$= 0.2 \cdot 457518$$

$$= 93114.8828 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 9.4876 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

$$\text{Saddle Splitting Dimension [d]:}$$

$$= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$$

$$= 3600 - 3250 \cdot \sin(120/2)/1.05$$

$$= 912.271 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 84947.7266 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 68.5096 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(457518 + 7113) \cdot 904) / (2 \cdot 5700 \cdot 173)}$$

$$= 79.081 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\text{Web Length Dimension [Web Length]:}$$

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

$$\text{Distance between Ribs [e]:}$$

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$

$$\text{Baseplate Pressure Area [Ap]:}$$

$$= e \cdot \text{Bpwid} / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

$$\text{Bearing Pressure [Bp]:}$$

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 464632 / (5700 \cdot 904)$$

$$= 9.017 \text{ Kgf/cm}^2$$

$$\text{Axial Load [P]:}$$

$$= Ap \cdot Bp$$

$$= 5158 \cdot 9.02$$

$$= 46506.984 \text{ Kgf}$$

$$\text{Area of the Rib and Web [Ar]:}$$

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.2 + 160$$

$$= 249.911 \text{ cm}^2$$

$$\text{Compressive Stress [Sc]:}$$

$$= P/Ar$$

$$= 46507/250$$

$$= 18.250 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05



Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 350 - 28 \\
 &= 322.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 1927674/250 \\
 &= 77.135 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(y_{tot}, \text{Saddle Width} - y_{tot}) \\
 &= \max(77.1, 350 - 77.1) \\
 &= 272.865 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 25545/250 } \\
 &= 101.101 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 3600 - \cos(120/2) (3250 + 22 + 22) - 80 \\
 &= 1873.000 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120/2) (3250 + 22 + 22) \\
 &= 5705.375 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 5700 - 50.8 \\
 &= 5649.200 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (5705 - 5649) / 2 \\
 &= 28.087 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{ L_v^2 + D_d^2 } \\
 &= \sqrt{ 1873^2 + 28.1^2 } \\
 &= 1873.211 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199943008/260 } \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1873/101 \\
 &= 18.528
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\
 &= 2184 / (2 \cdot 5700) \cdot 1141 \cdot 1873/2 \\
 &= 204.749 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($18.5 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (18.5)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (18.5) / (8 \cdot 123) - (18.5)^3 / (8 \cdot 123^3)) \\
 &= 149.2 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 18.2/149 + (205 \cdot 273/255446192) / 173
 \end{aligned}$$



= 0.135

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 2151325/410

= 52.515 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(52.5, abs(350 - 52.5))

= 297.485 mm.

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 3600 - $\sqrt{(3294 + 22)^2 - (1141/2)^2}$ - 80

= 275.786 mm.

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(29533/410)}$

= 84.907 mm.

Slenderness ratio [KL/r]:

= KL/r

= 1 • 276/84.9

= 3.248

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 2184/(2 • 5700)

= 0.192 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.19 • 1141 • 276

= 60.289 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 60.3/992772

= 0.596 N./mm²

Compressive Allowable, KL/r < Cc (3.25 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (3.25)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (3.25) / (8 \cdot 123) - (3.25^3) / (8 \cdot 123^3))$

= 155 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 16.4/155 + (60.3 • 297/29533)/173

= 0.109

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.



Distance from Bolts to Edge	Edgedis	570.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	22.1008 cm2
Saddle Load Q0 (Weight)	Q0	461213.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	639.2 Kgf
Maximum Transverse Force	Ft	1804.0 Kgf
Maximum Longitudinal Force	Fl	2184.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2184 / (22.1 \cdot 4)$$

$$= 2.4 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1804 / (22.1 \cdot 4)$$

$$= 2.0 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 2184 / (68.4 \cdot 4)$$

$$= 0.7828 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 3600 \cdot 1804 + 0$$

$$= 6494.41 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 6494 / 461214$$

$$= 14.08 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.78, 0]$$

$$= 0.7828 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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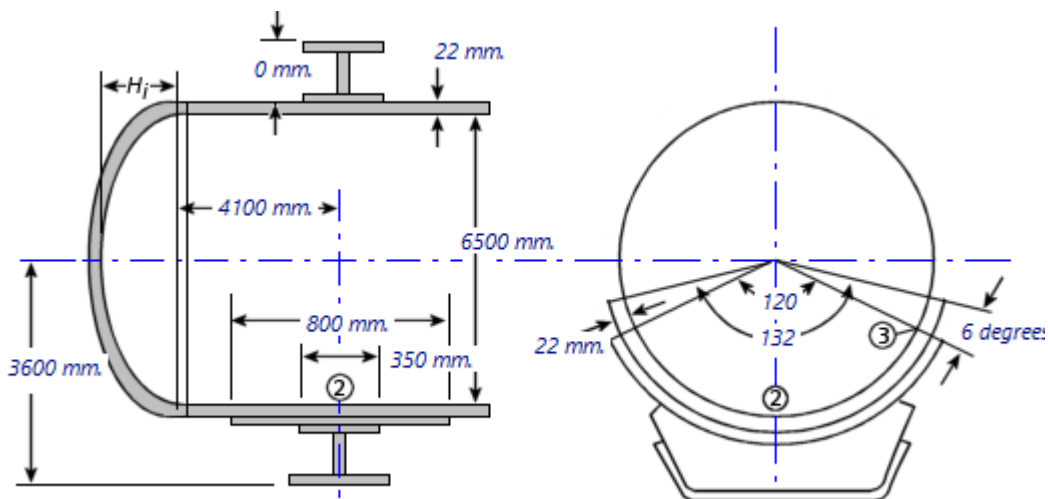


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.80	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1636.80	mm.
Shell Allowable Stress used in Calculation		90.70	N./mm ²
Head Allowable Stress used in Calculation		90.70	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	103885.77	Kgf
Pressure used in Analysis	P -1.034	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-9.33	-79.15
Long. Stress at Top of Midspan	9.33	90.70
Long. Stress at Bottom of Midspan	-7.21	-79.15
Long. Stress at Bottom of Midspan	7.21	90.70
Long. Stress at Top of Saddles	5.19	90.70
Long. Stress at Bottom of Saddles	-15.73	-79.15
Long. Stress at Bottom of Saddles	15.73	90.70
Tangential Shear in Shell	9.72	72.56
Circ. Stress at Horn of Saddle	65.17	113.38



Circ. Compressive Stress in Shell

2.43

90.70

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (7216/2 + 0) \cdot 3600/5700$$

$$= 6836.1 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(4368, 0) \cdot 3600/12300$$

$$= 1278.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(12124, 0) \cdot 3600/12300$$

$$= 3548.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (12124/2 + 0) \cdot 3600/5700$$

$$= 11485.6 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 92400 + \max(1278, 6836, 3548, 11486)$$

$$= 103885.8 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	110999.22	Kgf
Transverse Shear Load	6061.83	Kgf
Longitudinal Shear Load	12123.67	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 5700.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -103886 \cdot 4100 [1 - (1 - 4100/20500 + (3262^2 - 1637^2)/(2 \cdot 4100 \cdot 20500))/(1 + (4 \cdot 1637)/(3 \cdot 20500))]$$

$$= -99744.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 103886 \cdot 20500/4((1 + 2(3262^2 - 1637^2)/(20500^2))/(1 + (4 \cdot 1637)/(3 \cdot 20500)) - 4 \cdot 4100/20500)$$

$$= 73486.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:



$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) - 73487 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= -9.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) + 73487 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= -7.21 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) - 99744 / (0.11 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 5.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) + 99744 / (0.19 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= -15.73 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 103886 (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1637/3))$$

$$= 56334.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 56334 / (3262 \cdot 20.4)$$

$$= 9.72 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3262 \cdot 20.4)}$$

$$= 201.205 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(3262 \cdot 20.4)}, 2 \cdot 4100)$$

$$= 752.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(90.7/90.7, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 103886 \cdot 0.1 / (752(20.4 + 1 \cdot 22))$$

$$= -2.43 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -103886 / (4(20.4 + 1 \cdot 22)752) -$$

$$12 \cdot 0.053 \cdot 103886 \cdot 3262 / (20500(20.4 + 1 \cdot 22)^2)$$

$$= -65.17 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	N./mm2
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3252 \cdot 20.4)} + 800) 20.4^3) / 12$$

$$= 85.022 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 85 + 245 (199 - 10.2)^2$$

$$= 87915.656 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1201.8	20.4	10.2	245.2	250066.3	0.879E+05
Wearplate	800.0	22.0	31.4	176.0	552639.9	0.498E+05
Web	28.0	226.0	155.4	63.3	983373.4	0.392E+04
BasePlate	904.0	80.0	308.4	723.2	22303510.0	0.897E+05
Totals	...	...	...	1207.6	24089590.0	0.231E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24089590 / 1208$$

$$= 199.476 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 103886$$

$$= 21143.0059 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.1543 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.5600 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3600 - 3252 \cdot \sin(120/2) / 1.05$$

$$= 910.948 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19260.5703 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 16.2901 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 161.7334 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt[3]{(3(103886 + 7113) 904 / (2 \cdot 5700 \cdot 162))}$$

$$= 40.014 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 110999 / (5700 \cdot 904)$$

$$= 2.154 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 5158 \cdot 2.15$$

$$= 11110.388 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 90.2 + 160$$

$$= 249.911 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 11110 / 250$$

$$= 4.360 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 28$$

$$= 322.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1927674 / 250$$

$$= 77.135 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(77.1, 350 - 77.1)$$

$$= 272.865 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 25545 / 250 }$$

$$= 101.101 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3600 - \cos(120 / 2) (3250 + 22 + 22) - 80$$

$$= 1873.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120 / 2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 5700 - 50.8$$



$$= 5649.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (L_w - B_d) / 2 \\ &= (5705 - 5649) / 2 \\ &= 28.087 \text{ mm.} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{(L_v^2 + D_d^2)} \\ &= \sqrt{(1873^2 + 28.1^2)} \\ &= 1873.211 \text{ mm.} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 243)} \\ &= 127.551 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 1873 / 101 \\ &= 18.528 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\ &= 12124 / (2 \cdot 5700) \cdot 1141 \cdot 1873 / 2 \\ &= 1136.602 \text{ Kg-m.} \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($18.5 < 128$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (18.5)^2 / (2 \cdot 128^2)) 243 / \\ &\quad (5/3 + 3 \cdot (18.5) / (8 \cdot 128) - (18.5)^3 / (8 \cdot 128^3)) \\ &= 139.5 \text{ N/mm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\ &= 4.36 / 139 + (1137 \cdot 273 / 255446192) / 162 \\ &= 0.105 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 2151325 / 410 \\ &= 52.515 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot})) \\ &= \max(52.5, \text{abs}(350 - 52.5)) \\ &= 297.485 \text{ mm.} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk} \\ &= 3600 - \sqrt{(3294 + 22)^2 - (1141/2)^2} - 80 \\ &= 275.786 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(29533 / 410)} \\ &= 84.907 \text{ mm.} \end{aligned}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



$$= 1 \cdot 276/84.9$$

$$= 3.248$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 12124 / (2 \cdot 5700)$$

$$= 1.063 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.06 \cdot 1141 \cdot 276$$

$$= 334.676 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 335/992772$$

$$= 3.306 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($3.25 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (3.25)^2 / (2 \cdot 128^2)) 243 / (5/3 + 3 \cdot (3.25) / (8 \cdot 128) - (3.25^3) / (8 \cdot 128^3))$$

$$= 144.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 3.91/145 + (335 \cdot 297/29533) / 162$$

$$= 0.047$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	3261.80	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	20500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	4100.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	800.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	22.0	mm.
Wear Plate Allowable Stress	fw or Sr	90.70	N./mm2
Inside or Mean Depth of Head	Hi or hm	1636.80	mm.
Shell Allowable Stress used in Calculation		90.70	N./mm2
Head Allowable Stress used in Calculation		90.70	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3600.00	mm.
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		103110.16	Kgf
Pressure used in Analysis	P	-1.034	bars

Horizontal Vessel Analysis Results:		Actual N./mm^2	Allowable N./mm^2
Long. Stress at Top	of Midspan	-9.32	-79.15
Long. Stress at Top	of Midspan	9.32	90.70
Long. Stress at Bottom	of Midspan	-7.22	-79.15
Long. Stress at Bottom	of Midspan	7.22	90.70
Long. Stress at Top	of Saddles	5.09	90.70



Long. Stress at Bottom of Saddles	-15.67	-79.15
Long. Stress at Bottom of Saddles	15.67	90.70
Tangential Shear in Shell	9.65	72.56
Circ. Stress at Horn of Saddle	64.68	113.38
Circ. Compressive Stress in Shell	2.41	90.70

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (7216/2 + 0) \cdot 3600/5700$$

$$= 6836.1 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(4368, 0) \cdot 3600/12300$$

$$= 1278.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(12124, 0) \cdot 3600/12300$$

$$= 3548.4 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (12124/2 + 0) \cdot 3600/5700$$

$$= 11485.6 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 91625 + \max(1278, 6836, 3548, 11486)$$

$$= 103110.2 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(6.7 \cdot 1.2)^2)$$

$$= 42834.184 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 110223.62 Kg

Transverse Shear Load 6061.83 Kg

Longitudinal Shear Load 12123.67 Kg

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 5700.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/(3L))]$$

$$= -103110 \cdot 4100 [1 - (1 - 4100/20500 + (3262^2 - 1637^2) /$$

$$(2 \cdot 4100 \cdot 20500)) / (1 + (4 \cdot 1637) / (3 \cdot 20500))]$$

$$= -98999.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$



$$= 103110 \cdot 20500 / 4 \left((1 + 2(3262^2 - 1637^2) / (20500^2)) / (1 + (4 \cdot 1637 / (3 \cdot 20500)) - 4 \cdot 4100 / 20500) \right)$$

$$= 72938.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) - 72938 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= -9.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) + 72938 / (\pi \cdot 3262^2 \cdot 20.4)$$

$$= -7.22 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) - 99000 / (0.11 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= 5.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.03 \cdot 3262 / (2 \cdot 20.4) + 99000 / (0.19 \cdot \pi \cdot 3262^2 \cdot 20.4)$$

$$= -15.67 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 103110 (20500 - 2 \cdot 4100) / (20500 + (4 \cdot 1637 / 3))$$

$$= 55913.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 55914 / (3262 \cdot 20.4)$$

$$= 9.65 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3262 \cdot 20.4)}$$

$$= 201.205 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(3262 \cdot 20.4)}, 2 \cdot 4100)$$

$$= 752.41 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(90.7 / 90.7, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 103110 \cdot 0.1 / (752(20.4 + 1 \cdot 22))$$

$$= -2.41 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -103110 / (4(20.4 + 1 \cdot 22) 752) -$$

$$12 \cdot 0.053 \cdot 103110 \cdot 3262 / (20500(20.4 + 1 \cdot 22)^2)$$

$$= -64.68 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5700.0000	mm.
Baseplate Thickness	Bpthk	80.0000	mm.
Baseplate Width	Bpwid	904.0001	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	



Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	242.6	N./mm ²
Height of Web at Center	Hw,c	226.0	mm.
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3252 \cdot 20.4)} + 800)^3) / 12$$

$$= 85.022 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 85 + 245 (199 - 10.2)^2$$

$$= 87915.656 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1201.8	20.4	10.2	245.2	250066.3	0.879E+05
Wearplate	800.0	22.0	31.4	176.0	552639.9	0.498E+05
Web	28.0	226.0	155.4	63.3	983373.4	0.392E+04
BasePlate	904.0	80.0	308.4	723.2	22303510.0	0.897E+05
Totals	...	...	...	1207.6	24089590.0	0.231E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 24089590 / 1208$$

$$= 199.476 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 103110$$

$$= 20985.1543 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	2.1382	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	145.5600	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 3600 - 3252 \cdot \sin(120/2) / 1.05$$

$$= 910.948 \text{ mm.}$$

Bending Moment, M = Fh * d	=	19116.7734	Kg-m.
----------------------------	---	------------	-------

Bending Stress, Sb = (M * C1 / I)	=	16.1685	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	161.7334	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(103110 + 7113)904 / (2 \cdot 5700 \cdot 162))}$$

$$= 39.874 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (3250 + 22 + 22)$$



$$= 5705.375 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 5705 / (6 - 1)$$

$$= 1141.075 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1141 \cdot 904 / 2$$

$$= 5157.659 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 110224 / (5700 \cdot 904)$$

$$= 2.139 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 5158 \cdot 2.14$$

$$= 11032.755 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.2 + 160$$

$$= 249.911 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 11033 / 250$$

$$= 4.329 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.191E+05
Web	570.5	28.0	14.0	159.8	223650.7	0.647E+04
Totals	...	...	...	249.9	1927674.6	0.255E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 28$$

$$= 322.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 1927674 / 250$$

$$= 77.135 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(77.1, 350 - 77.1)$$

$$= 272.865 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(25545 / 250)}$$

$$= 101.101 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3600 - \cos(120/2) (3250 + 22 + 22) - 80$$

$$= 1873.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$



$$= 2 \cdot \sin(120/2) (3250 + 22 + 22)$$

$$= 5705.375 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 5700 - 50.8$$

$$= 5649.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (5705 - 5649) / 2$$

$$= 28.087 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(1873^2 + 28.1^2)}$$

$$= 1873.211 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 243)}$$

$$= 127.551$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1873 / 101$$

$$= 18.528$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 12124 / (2 \cdot 5700) \cdot 1141 \cdot 1873 / 2$$

$$= 1136.602 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18.5 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18.5)^2 / (2 \cdot 128^2)) 243 / (5/3 + 3 \cdot (18.5) / (8 \cdot 128) - (18.5)^3 / (8 \cdot 128^3))$$

$$= 139.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.33 / 139 + (1137 \cdot 273 / 255446192) / 162$$

$$= 0.105$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.246E+05
Web	1141.1	28.0	14.0	319.5	447301.4	0.495E+04
Totals	...	...	...	409.7	2151325.2	0.295E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2151325 / 410$$

$$= 52.515 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(52.5, \text{abs}(350 - 52.5))$$

$$= 297.485 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 3600 - \sqrt{(3294 + 22)^2 - (1141/2)^2} - 80$$

$$= 275.786 \text{ mm.}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{29533/410}$$

$$= 84.907 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 276/84.9$$

$$= 3.248$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 12124 / (2 \cdot 5700)$$

$$= 1.063 \text{ Kgf/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.06 \cdot 1141 \cdot 276$$

$$= 334.676 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 335/992772$$

$$= 3.306 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.25 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.25)^2 / (2 \cdot 128^2)) 243 /$$

$$(5/3 + 3 \cdot (3.25) / (8 \cdot 128) - (3.25^3) / (8 \cdot 128^3))$$

$$= 144.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 3.88/145 + (335 \cdot 297/29533)/162$$

$$= 0.047$$

*Review notes about nozzle loadings and supports in the Warnings report.***PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.**

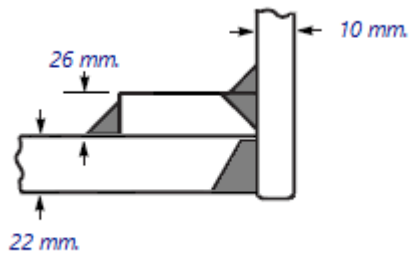
**Input, Nozzle Desc: M1-24"****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	90.70	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	26.0000	mm.
Weld leg size between Pad and Shell	Wp	14.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 305 / (90.7 \cdot 1 + 0.4 \cdot 3.5)$
 $= 1.1744 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4779 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1185.6000 mm.
Parallel to Vessel Wall, opening length	d	592.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.7/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.7/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 90.7/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

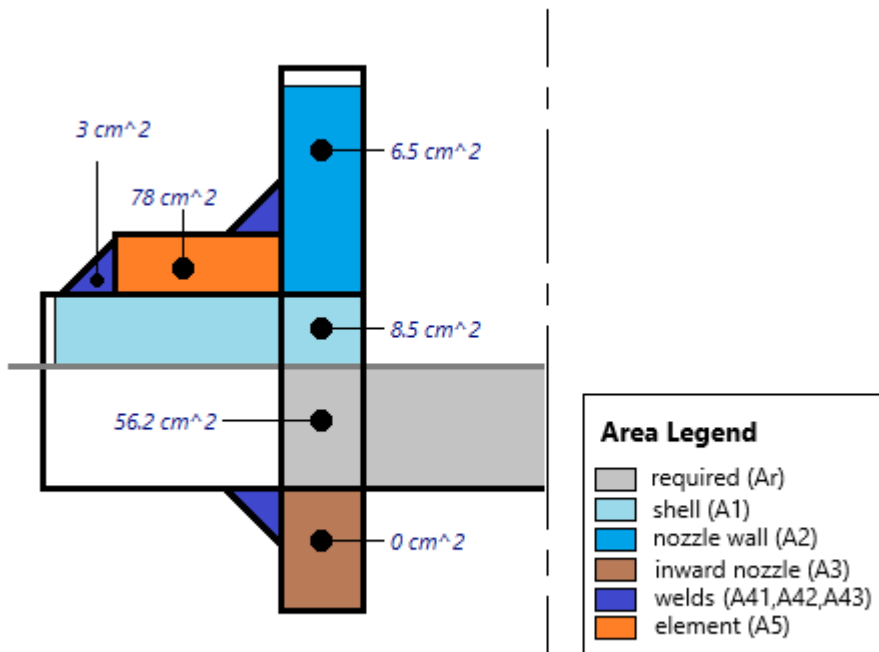


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	66.940	56.200	NA
Area in Shell	A1	53.991	8.531	NA
Area in Nozzle Wall	A2	6.792	6.507	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	77.998	77.998	NA
TOTAL AREA AVAILABLE	Atot	141.741	95.995	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	756.5377	26.0000 mm.
Based on given Pad Diameter:	909.6000	12.7346 mm.
Based on Shell or Nozzle Thickness:	991.6378	10.0000 mm.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(593 \cdot 19 \cdot 1 + 2 \cdot 8.4 \cdot 19 \cdot 1(1 - 1))$$

$$= 56.200 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 593(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 8.4$$

$$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$$

$$= 8.531 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 47) \cdot (8.4 - 1.48) \cdot 1$$

$$= 6.507 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 14^2 \cdot 1$$

$$= 2.960 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr4$$

$$= (910 - 610) \cdot 26 \cdot 1$$

$$= 77.998 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.0779 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.0779 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 22$, $t_r = 7.42$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 68 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/19.60 = 0.179

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 8.4000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.8800 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (56.2 - 8.53 + 2 \cdot 8.4 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 44312.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.51 + 78 + 2.96 - 0 \cdot 1) \cdot 90.7$
 $= 80894.98 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.51 + 0 + 1 + (3.43)) \cdot 90.7$
 $= 10112.43 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.51 + 0 + 2.96 + 78 + (3.43)) \cdot 90.7$
 $= 84064.68 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 43395. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 910 \cdot 14 \cdot 0.49 \cdot 90.7$
 $= 90651. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 301) \cdot (10 - 1.6) \cdot 0.7 \cdot 90.7$
 $= 51356. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 26 \cdot 0.74 \cdot 90.7$
 $= 170392. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (10 - 0) \cdot 0.74 \cdot 90.7$
 $= 65537. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (90651 + 51356) = 142008 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (43395 + 170392 + 65537 + 0) = 279324 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (90651 + 65537 + 0) = 156188 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 142007 Kgf, must exceed W = 44312 Kgf or W1 = 80894 Kgf

Path 2-2 = 279323 Kgf, must exceed W = 44312 Kgf or W2 = 10112 Kgf

Path 3-3 = 156187 Kgf, must exceed W = 44312 Kgf or W3 = 84064 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 7.9220 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 279.9218 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.668 %

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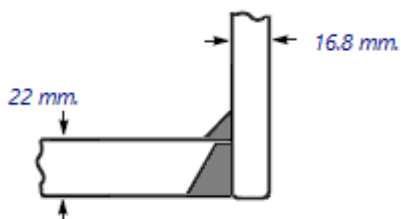
**Input, Nozzle Desc: K6-2"****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K6-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

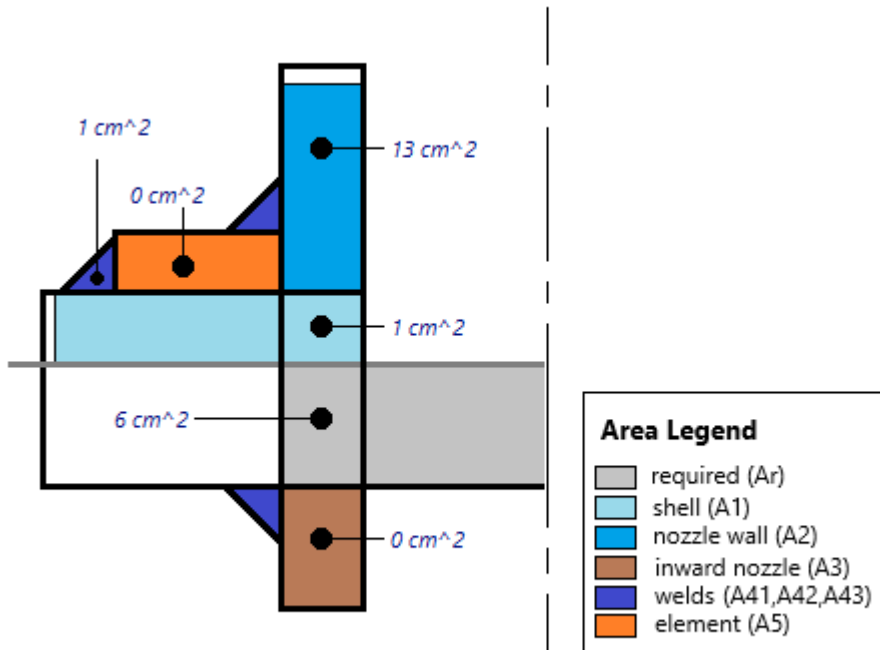


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1-1))$
 $= 5.977 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$
 $(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$
 $= 1.025 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2 / \sin(\alpha 3)$
 $= (2 \cdot 38)(15.2 - 0.43) / \sin(60)$
 $= 12.960 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 4984.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7$
 $= 12911.61 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (13 + 0 + 1 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 7015. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8$
 $= 12502. \text{ Kg}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W1 = 12911 \text{ Kg}$$

$$\text{Path 2-2} = 28648 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W2 = 18647 \text{ Kg}$$

$$\text{Path 3-3} = 28648 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W3 = 18647 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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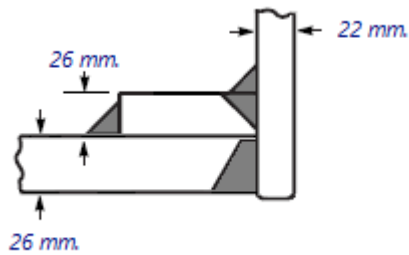
**Input, Nozzle Desc: N1A-40"****From: 20**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6500.00	mm.
Design Length of Section	L	3830.0938	mm.
Shell Finished (Minimum) Thickness	t	26.0000	mm.
Shell Internal Corrosion Allowance	cas	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	90.70	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		40.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1916.0001	mm.
Thickness of Pad	te	26.0000	mm.
Weld leg size between Pad and Shell	Wp	14.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm.
Reinforcing Pad Width		450.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1A-40"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	40.000 in.
Actual Thickness Used in Calculation	0.866 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3252 / (90.7 \cdot 1 - 0.6 \cdot 3.5)$
 $= 12.5771 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 508 / (90.7 \cdot 1 + 0.4 \cdot 3.5)$
 $= 1.9574 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 2.2800 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1950.3999 mm.
Parallel to Vessel Wall, opening length	d	975.2000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		61.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

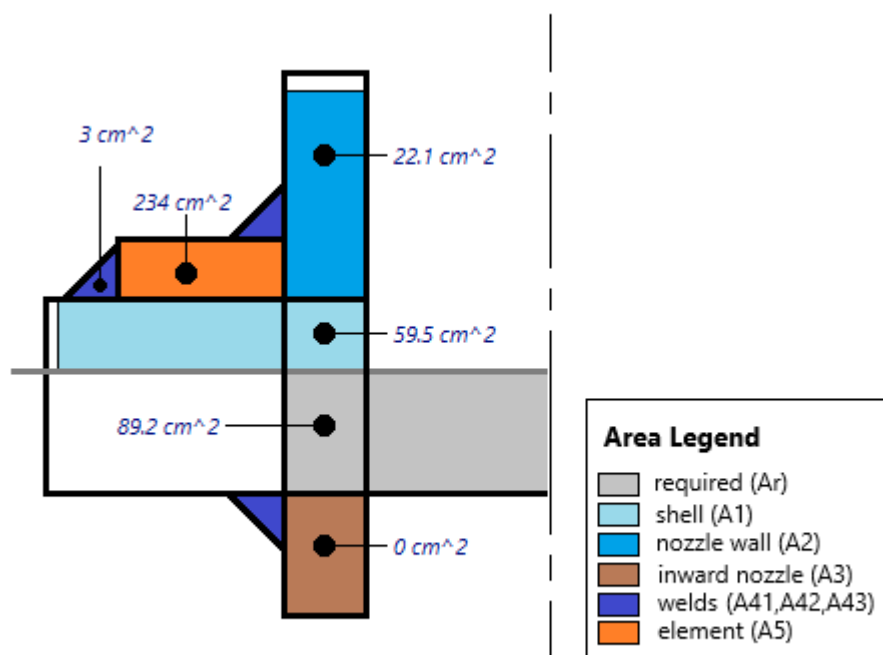


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	122.652	89.241	NA
Area in Shell	A1	115.297	59.466	NA
Area in Nozzle Wall	A2	22.499	22.106	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	234.000	234.000	NA
TOTAL AREA AVAILABLE	Atot	374.757	318.532	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1034.1118 26.0000 mm.
 Based on given Pad Diameter: 1916.0001 0.5232 mm.
 Based on Shell or Nozzle Thickness: 1037.4050 22.0000 mm.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (975 \cdot 18.3 \cdot 1 + 2 \cdot 20.4 \cdot 18.3 \cdot 1 (1 - 1))$
 $= 89.241 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 975 (1 \cdot 24.4 - 1 \cdot 18.3) - 2 \cdot 20.4$
 $(1 \cdot 24.4 - 1 \cdot 18.3) \cdot (1 - 1)$
 $= 59.466 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 61) \cdot (20.4 - 2.28) \cdot 1$$

$$= 22.106 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 14^2 \cdot 1$$

$$= 2.960 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr4$$

$$= (1916 - 1016) \cdot 26 \cdot 1$$

$$= 234.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.8800 \text{ mm.}$

Wall Thickness per UG16(b), $tr_{16b} = 4.1000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb1 = 14.1771 \text{ mm.}$

Wall Thickness $tb1 = \max(trb1, tr_{16b}) = 14.1771 \text{ mm.}$

Wall Thickness, shell/head, external pressure $trb2 = 5.3103 \text{ mm.}$

Wall Thickness $tb2 = \max(trb2, tr_{16b}) = 5.3103 \text{ mm.}$

Wall Thickness per table UG-45 $tb3 = 9.9312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.93, \max(14.2, 5.31)]$$

$$= 9.9312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.88, 9.93)$$

$$= 9.9312 \text{ mm.}$$

Available Nozzle Neck Thickness = 22.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.0,	Allowable	:	90.7 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.9 N./mm ²	Passed
Occasional	:	4.1,	Allowable	:	120.6 N./mm ²	Passed
Shear	:	8.8,	Allowable	:	63.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 22$, $t_r = 1.29$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.063$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A $15 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 22$, $t_r = 1.29$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.063$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B $-5 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 22$, $t_r = 1.29$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.063$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B $-5 \text{ }^\circ\text{C}$



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 26, tr = 8.26, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.34$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 22, tr = 1.29, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.063$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.50/51.00 = 0.069$

Weld Size Calculations, Description: N1A-40"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (89.2 - 59.5 + 2 \cdot 20.4 \cdot 1 \cdot (1 \cdot 24.4 - 18.3)) \cdot 90.7)$
 $= 29839.44 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (22.1 + 234 + 2.96 - 0 \cdot 1) \cdot 90.7$
 $= 239606.80 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (22.1 + 0 + 1 + (9.95)) \cdot 90.7$
 $= 30577.49 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (22.1 + 0 + 2.96 + 234 + (9.95)) \cdot 90.7$
 $= 248814.03 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$



$$= (3.14/2.0) \cdot 1016 \cdot 10 \cdot 0.49 \cdot 90.7$$

$$= 72325. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 1916 \cdot 14 \cdot 0.49 \cdot 90.7$$

$$= 190954. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 498) \cdot (22 - 1.6) \cdot 0.7 \cdot 90.7$$

$$= 206544. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 1016 \cdot 26 \cdot 0.74 \cdot 90.7$$

$$= 283993. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1016 \cdot (22 - 0) \cdot 0.74 \cdot 90.7$$

$$= 240302. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (190954 + 206544) = 397497 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (72325 + 283993 + 240302 + 0) = 596620 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (190954 + 240302 + 0) = 431256 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 397497 Kg, must exceed W = 29839 Kg	or W1 = 239606 Kg
Path 2-2 = 596619 Kg, must exceed W = 29839 Kg	or W2 = 30577 Kg
Path 3-3 = 431255 Kg, must exceed W = 29839 Kg	or W3 = 248814 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 39.9478 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 415.9477 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.213 \%$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 975.20 mm.

Input Echo, WRC297 Item 1, Description: N1A-40" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6500.000	mm.
Shell Thickness	26.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	90.70	N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	1.6000	mm.	
Nozzle Diameter	1016.000	mm.	
Nozzle Thickness	22.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	90.70	N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter of Reinforcing Pad	1916.000	mm.
Thickness of Reinforcing Pad	26.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	7648.38	Kgf
Circumferential Shear	Vc	5098.92	Kgf
Longitudinal Shear	Vl	5098.92	Kgf
Circumferential Moment	Mc	14271.47	Kg-m.
Longitudinal Moment	Ml	14271.47	Kg-m.
Torsional Moment	Mt	20387.82	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1016.0 + 2 \cdot 1.65 \cdot \sqrt{(3276.8(26.0 - 1.6))}$$

$$= 1949.112 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6553.600	mm.
Nozzle Outside Diameter	(d) =	1016.000	mm.
Vessel Thickness used	(T) =	50.400	mm.
Nozzle Thickness used	(t) =	20.400	mm.
T / t	=	2.471	
d / t	=	49.804	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.768$$

Nr/P	=	0.048
Mr/P	=	0.086
M0/P	=	0.056
N0/P	=	0.174
MrD/Mc	=	0.175
NrDL/Mc	=	0.090
M0d/Mc	=	0.137
N0DL/Mc	=	0.233
MrD/Ml	=	0.104
NrDL/Ml	=	0.059
M0D/Ml	=	0.070
N0DL/Ml	=	0.343

**Vessel Stresses (N./mm²):**

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	-34	31	48	23

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-15	15	-15	15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	-42	55	31	-5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	43	56	48	29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11
Outplane Stress Summary	-61	47	53	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-1	-1	-1	-1



Inplane Bending (P)	-15	15	-15	15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-56	56	56	-56
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	22	22	22	22

Inplane Stress Summary	-56	88	67	-15
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	61	88	68	18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	-13	-13	23	23
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-67	67	-67	67
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-148	148	148	-148
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-220	210	93	-68
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	7	7	4	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Outside	Inside	Outside	Inside
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Two * Max Shear Stress	221	224	93	92
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	-7	-7	17	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-67	67	-67	67
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-258	258	258	-258
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-331	320	204	-178
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	331	328	204	196
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1016.0 + 2 \cdot 1.65 \cdot \sqrt{(3263.8(26.0 - 1.6))}$$

$$= 1947.259 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6527.600 mm.
Nozzle Outside Diameter	(d) =	1916.000 mm.
Vessel Thickness used	(T) =	24.400 mm.
Nozzle Thickness used	(t) =	470.400 mm.
T / t	=	0.052
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.801$$



Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.023
MrD/Mc	=	0.241
NrDL/Mc	=	0.104
M0d/Mc	=	0.072
N0DL/Mc	=	0.046
MrD/M1	=	0.049
NrDL/M1	=	0.053
M0D/M1	=	0.015
N0DL/M1	=	0.024

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	46	46	46	46
Outplane Stress Summary	14	66	42	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-53	53	-53	53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-79	100	6	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	94	101	42	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-2	-2	-2	-2



Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-52	52	52	-52
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	-53	82	64	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-53	53	-53	53
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-177	177	177	-177
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	46	46	46

Inplane Stress Summary	-203	259	176	-70
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	203	259	177	70
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	22	22	22	22	22	22	22
Circ. Pl (SUS)	-23	-23	13	13	-5	-5	3	3
Circ. Q (SUS)	-31	31	13	-13	-71	71	41	-41
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-4	-4	2	2	-17	-17	7	7
Long. Q (SUS)	-48	48	18	-18	-53	53	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	22.0	22.0	22.0	22.0	22.0	22.0	22.0	22.0
Pm+Pl (SUS)	8.0	8.0	35.0	35.0	23.0	23.0	25.0	25.0
Pm+Pl+Q (Total)	41.4	55.2	48.2	27.3	59.7	88.1	66.3	18.6

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	90.70	Passed
Pm+Pl (SUS)	35.00	136.05	Passed
Pm+Pl+Q (TOTAL)	88.10	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	46	46	46	46	46	46	46
Circ. Pl (SUS)	-4	-4	0	0	-17	-17	7	7
Circ. Q (SUS)	-25	25	-3	3	-230	230	124	-124
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-11	-11	1	1	-7	-7	3	3
Long. Q (SUS)	-89	89	-17	17	-66	66	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	46.0	46.0	46.0	46.0	46.0	46.0	46.0	46.0
Pm+Pl (SUS)	42.0	42.0	46.0	46.0	29.0	29.0	53.0	53.0
Pm+Pl+Q (Total)	94.0	101.0	43.0	49.1	201.0	259.0	177.0	71.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	90.70	Passed
Pm+Pl (SUS)	53.00	136.05	Passed
Pm+Pl+Q (TOTAL)	259.01	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-23	-23	13	13	-17	-17	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-9	-9	7	7	-9	-9	7	7
Long. Q (SUS)	-215	215	81	-81	-325	325	191	-191
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0



Pm+Pl (SUS)	14.1	13.1	22.1	23.1	8.3	7.4	16.2	17.2
Pm+Pl+Q (Total)	220.2	223.4	92.4	93.5	330.1	327.2	202.3	197.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	90.70	Passed
Pm+Pl (SUS)	23.08	136.05	Passed
Pm+Pl+Q (TOTAL)	330.08	272.10	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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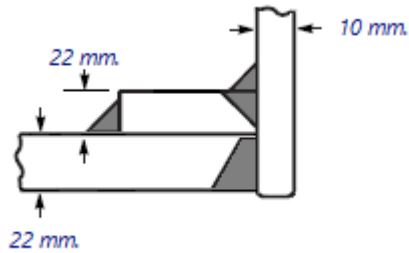
**Input, Nozzle Desc: M2-24"****From: 40**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6500.00	mm.
Design Length of Section	L	4481.4995	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	cas	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	90.70	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1009.6000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	14.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		200.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3252 / (90.7 \cdot 1 - 0.6 \cdot 3.5)$
 $= 12.5771 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 305 / (90.7 \cdot 1 + 0.4 \cdot 3.5)$
 $= 1.1744 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5906 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1185.6000 mm.
Parallel to Vessel Wall, opening length	d	592.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

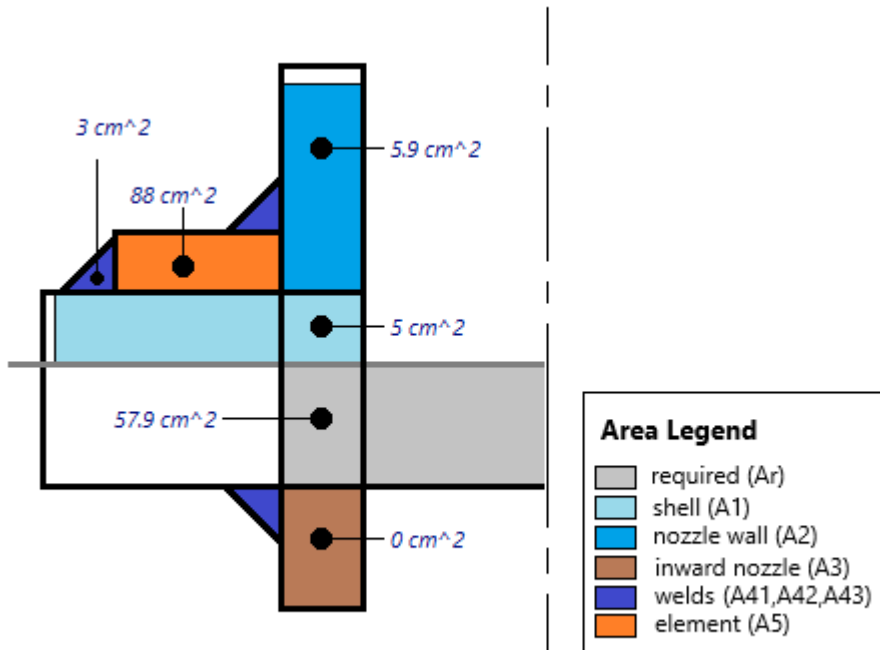


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	74.557	57.949	NA
Area in Shell	A1	46.374	5.032	NA
Area in Nozzle Wall	A2	6.214	5.856	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.960	2.960	NA
Area in Element	A5	87.998	87.998	NA
TOTAL AREA AVAILABLE	Atot	143.546	101.846	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 810.0640 22.0000 mm.
 Based on given Pad Diameter: 1009.6001 11.0255 mm.
 Based on Shell or Nozzle Thickness: 1050.6207 10.0000 mm.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (593 \cdot 19.6 \cdot 1 + 2 \cdot 8.4 \cdot 19.6 \cdot 1 (1 - 1))$
 $= 57.949 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 593 (1 \cdot 20.4 - 1 \cdot 19.6) - 2 \cdot 8.4$
 $(1 \cdot 20.4 - 1 \cdot 19.6) \cdot (1 - 1)$
 $= 5.032 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 43) \cdot (8.4 - 1.59) \cdot 1$$

$$= 5.856 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 14^2 \cdot 1$$

$$= 2.960 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr4$$

$$= (1010 - 610) \cdot 22 \cdot 1$$

$$= 87.998 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.1906 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.1906 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 22$, $t_r = 8.26$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.4$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 10$, $t_r = 0.77$, $c = 1.6 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/19.60 = 0.179

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 8.4000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.8800 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	9.8980 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (57.9 - 5.03 + 2 \cdot 8.4 \cdot 1 \cdot (1 \cdot 20.4 - 19.6)) \cdot 90.7)$
 $= 49074.10 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.86 + 88 + 2.96 - 0 \cdot 1) \cdot 90.7$
 $= 89541.90 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.86 + 0 + 1 + (3.43)) \cdot 90.7$
 $= 9510.67 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (5.86 + 0 + 2.96 + 88 + (3.43)) \cdot 90.7$
 $= 92711.60 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 43395. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1010 \cdot 14 \cdot 0.49 \cdot 90.7$
 $= 100617. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 301) \cdot (10 - 1.6) \cdot 0.7 \cdot 90.7$
 $= 51356. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 22 \cdot 0.74 \cdot 90.7$
 $= 144178. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (10 - 0) \cdot 0.74 \cdot 90.7$
 $= 65537. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (100617 + 51356) = 151974 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (43395 + 144178 + 65537 + 0) = 253109 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (100617 + 65537 + 0) = 166154 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 151973 Kgf, must exceed W = 49074 Kgf or W1 = 89541 Kgf

Path 2-2 = 253109 Kgf, must exceed W = 49074 Kgf or W2 = 9510 Kgf

Path 3-3 = 166153 Kgf, must exceed W = 49074 Kgf or W3 = 92711 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 14.3243 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 336.3243 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.668 %

This nozzle or its diameter limit intersects weld seam # [6] on this element. The distance between the nozzle and seam is, 315.40 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 592.80 mm.

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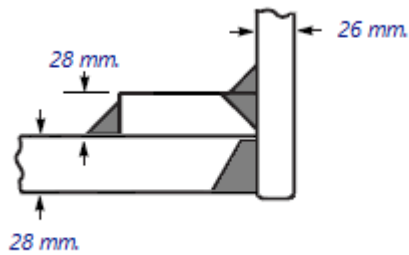
**Input, Nozzle Desc: N2-52"****From: 50**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6500.00	mm.
Design Length of Section	L	4481.4995	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10550.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	90.70	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		52.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	2320.8000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		500.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-52"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	52.000 in.
Actual Thickness Used in Calculation	1.024 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3252 / (90.7 \cdot 1 - 0.6 \cdot 3.5)$
 $= 12.5771 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 660 / (90.7 \cdot 1 + 0.4 \cdot 3.5)$
 $= 2.5446 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 2.8072 mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2544.0000 mm.
Parallel to Vessel Wall, opening length	d	1272.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		66.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

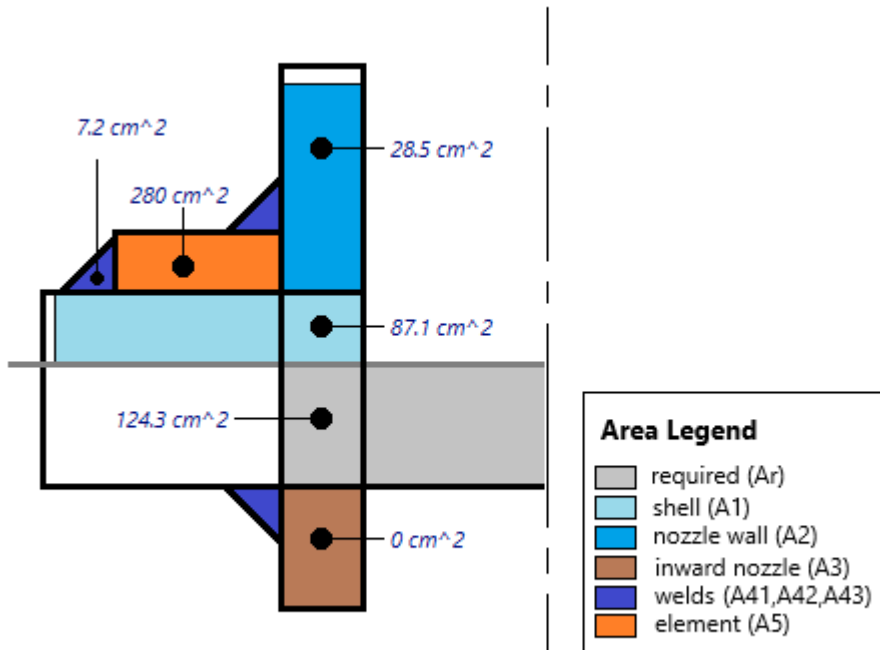


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	159.980	124.345	NA
Area in Shell	A1	175.828	87.118	NA
Area in Nozzle Wall	A2	28.848	28.502	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	7.240	7.240	NA
Area in Element	A5	279.993	279.993	NA
TOTAL AREA AVAILABLE	Atot	491.909	402.853	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 1326.1030 28.0000 mm.
Based on given Pad Diameter: 2320.8000 0.1485 mm.
Based on Shell or Nozzle Thickness: 1326.5110 26.0000 mm.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(1272 \cdot 19.6 \cdot 1 + 2 \cdot 24.4 \cdot 19.6 \cdot 1(1 - 1))$
 $= 124.345 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1272(1 \cdot 26.4 - 1 \cdot 19.6) - 2 \cdot 24.4$
 $(1 \cdot 26.4 - 1 \cdot 19.6) \cdot (1 - 1)$
 $= 87.118 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 66) \cdot (24.4 - 2.81) \cdot 1$$

$$= 28.502 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1 + 18^2 \cdot 1$$

$$= 7.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (2321 - 1321) \cdot 28 \cdot 1$$

$$= 279.993 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.4072 \text{ mm.}$

Wall Thickness per UG16(b), $tr_{16b} = 4.1000 \text{ mm.}$

Wall Thickness, shell/head, internal pressure $trb_1 = 14.1771 \text{ mm.}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1771 \text{ mm.}$

Wall Thickness, shell/head, external pressure $trb_2 = 5.3103 \text{ mm.}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3103 \text{ mm.}$

Wall Thickness per table UG-45 $tb_3 = 9.9312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.93, \max(14.2, 5.31)]$$

$$= 9.9312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.41, 9.93)$$

$$= 9.9312 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	16.0,	Allowable	:	90.7 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	269.8 N./mm ²	Passed
Occasional	:	4.5,	Allowable	:	120.6 N./mm ²	Passed
Shear	:	7.3,	Allowable	:	63.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 26$, $tr = 1.67$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.069$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 26$, $tr = 1.67$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.069$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 26$, $tr = 1.67$, $c = 1.6 \text{ mm.}$, $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.069$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
--	------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 28, tr = 8.26, c = 1.6 mm., E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.31$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 26, tr = 1.67, c = 1.6 mm., E* = 1
Thickness Ratio = $tr * E^* / (tg - c) = 0.069$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.50/51.00 = 0.069$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1908.0000 mm.
Parallel to Vessel Wall, opening length d 954.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 66.0000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure]	AR	82.901	82.901
Area Available in Shell	A1	43.559	43.559
Area Available in Nozzle Wall	A2	26.343	28.502
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	4.000	4.000
Area Available in Pad	A5	0.000	164.412
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	73.902	240.473

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

The flange is considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	104.0416	213.2000	15763.7549	98.161	25788.8262
Shell	111.5136	13.2000	64.7671	298.161	99200.4688
Pad	118.2720	40.4000	77.2710	270.961	86912.5391
Flange	277.6965	583.2734	7982.0742	271.912	213300.0938
Sum	611.5237				425201.91

Membrane stress per Appendix 1-7 (1 or 2) Sm 55.28 N./mm²



Allowable Membrane Stress	S	90.70	N./mm2
Bending stress at the inside of the Nozzle	Sb	16.90	N./mm2
Combined Membrane plus Bending Stress	Sm + Sb	72.18	N./mm2
Allowable Membrane + Bending Stress	1.5 * S	136.05	N./mm2
M.A.W.P for the Appendix 1-7 Stress calc		5.74	bars

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle =	43.960	cm2
Area from Shell =	77.506	cm2
Area from Pad =	82.203	cm2
Total Area, As =	203.668	cm2

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [R (Rn + tn + \sqrt{rm \cdot t}) + Rn (t + te + \sqrt{Rnm \cdot tn})] / As$$

$$= 3.5 [3252 (636 + 24.4 + 294) + 636 (180)] / 204$$

$$= 55.3 \text{ N./mm}^2$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= (Rn^3/6 + R \cdot Rn \cdot e) P$$

$$= (636^3/6 + 3252 \cdot 636 \cdot 298) 3.5$$

$$= 23538 \text{ Kg-m.}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t/2)) / I$$

$$= (23538 (298 + 26.4/2)) / 425202$$

$$= 16.903 \text{ N./mm}^2$$

Weld Size Calculations, Description: N2-52"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000	mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (124 - 87.1 + 2 \cdot 24.4 \cdot 1 \cdot (1 \cdot 26.4 - 19.6)) \cdot 90.7)$$

$$= 37521.45 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (28.5 + 280 + 7.24 - 0 \cdot 1) \cdot 90.7$$

$$= 292019.38 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (28.5 + 0 + 4 + (12.9)) \cdot 90.7$$

$$= 41975.68 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (28.5 + 0 + 7.24 + 280 + (12.9)) \cdot 90.7$$

$$= 303934.62 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$



$$= (3.14/2.0) \cdot 1321 \cdot 20 \cdot 0.49 \cdot 90.7$$

$$= 188045. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 2321 \cdot 18 \cdot 0.49 \cdot 90.7$$

$$= 297375. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 648) \cdot (26 - 1.6) \cdot 0.7 \cdot 90.7$$

$$= 321681. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 1321 \cdot 28 \cdot 0.74 \cdot 90.7$$

$$= 397581. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1321 \cdot (22 - 0) \cdot 0.74 \cdot 90.7$$

$$= 312392. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (297375 + 321681) = 619056 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (188045 + 397581 + 312392 + 0) = 898018 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (297375 + 312392 + 0) = 609768 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 619056 Kg, must exceed W = 37521 Kg	or W1 = 292019 Kg
Path 2-2 = 898018 Kg, must exceed W = 37521 Kg	or W2 = 41975 Kg
Path 3-3 = 609767 Kg, must exceed W = 37521 Kg	or W3 = 303934 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 67.8040 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 495.8040 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.008 \%$$

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 1210.74 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 1272.00 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2-52" :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.6000 mm.
Shell External Corrosion Allowance	0.0000 mm.



Shell Diameter	6500.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 90.70	N./mm2

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	1320.800	mm.
Nozzle Thickness	26.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 90.70	N./mm2

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter of Reinforcing Pad	2320.800	mm.
Thickness of Reinforcing Pad	28.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	9942.91	Kgf
Circumferential Shear	Vc	6628.59	Kgf
Longitudinal Shear	Vl	6628.59	Kgf
Circumferential Moment	Mc	24118.77	Kg-m.
Longitudinal Moment	Ml	24118.77	Kg-m.
Torsional Moment	Mt	34455.36	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1320.8 + 2 \cdot 1.65 \cdot \sqrt{(3278.8(28.0 - 1.6))}$$

$$= 2291.698 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6557.600	mm.
Nozzle Outside Diameter	(d) =	1320.800	mm.
Vessel Thickness used	(T) =	54.400	mm.
Nozzle Thickness used	(t) =	24.400	mm.
T / t	=	2.230	
d / t	=	54.131	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.211$$

Nr/P	=	0.041
Mr/P	=	0.077
M0/P	=	0.045
N0/P	=	0.156
MrD/Mc	=	0.174
NrDL/Mc	=	0.085
M0d/Mc	=	0.119
N0DL/Mc	=	0.229
MrD/Ml	=	0.088



NrDL/M1 = 0.048
MØD/M1 = 0.052
NØDL/M1 = 0.309

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-8	8	-8	8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	20	21	20	21
Outplane Stress Summary	-30	25	44	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-15	15	-15	15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-32	32	32	-32
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	-41	53	28	-4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	41	53	44	29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-8	8	-8	8
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	10	10	10	10
Outplane Stress Summary	-60	43	53	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-15	15	-15	15
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-63	63	63	-63
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	20	21	20	21
Inplane Stress Summary	-64	93	72	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	64	93	72	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11
Hoop Stress Summary	-13	-12	24	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-115	115	115	-115
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-174	168	72	-48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5



Shear Stress Summary	6	6	3	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	174	181	72	73

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11
Hoop Stress Summary	-8	-7	19	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-55	55	-55	55
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-238	238	238	-238
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-296	290	194	-170

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	5	5	5	5
Shear Stress Summary	3	3	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	296	298	194	190

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 1320.8 + 2 \cdot 1.65 \cdot \sqrt{(3264.8(28.0 - 1.6))}$$

$$= 2289.622 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6529.600 mm.

Nozzle Outside Diameter (d) = 2320.800 mm.

Vessel Thickness used (T) = 26.400 mm.

Nozzle Thickness used (t) = 524.400 mm.

T / t = 0.050



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.590$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.022
MrD/Mc	=	0.238
NrDL/Mc	=	0.099
M0d/Mc	=	0.069
N0DL/Mc	=	0.044
MrD/Ml	=	0.043
NrDL/Ml	=	0.044
M0D/Ml	=	0.013
N0DL/Ml	=	0.020

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-15	15	-15	15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	42	43	42	43
Outplane Stress Summary	9	64	38	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-59	59	-59	59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	-88	106	0	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	98	106	39	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-15	15	-15	15
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-64	88	69	-20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-59	59	-59	59
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-208	208	208	-208
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	42	43	42	43

Inplane Stress Summary	-246	290	199	-97
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	246	290	199	97
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	21	20	21	20	21	20	21
Circ. Pl (SUS)	-23	-23	13	13	-6	-6	4	4
Circ. Q (SUS)	-26	26	10	-10	-78	78	48	-48
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-3	-3	1	1	-18	-18	8	8
Long. Q (SUS)	-47	47	17	-17	-51	51	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	20.0	21.0	20.0	21.0	20.0	21.0	20.0	21.0
Pm+Pl (SUS)	10.0	9.0	33.0	34.0	22.0	23.0	24.0	25.0



Pm+Pl+Q (Total)	40.4	54.1	43.3	30.3	64.7	93.1	72.2	23.6
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	90.70	Passed
Pm+Pl (SUS)	34.00	136.05	Passed
Pm+Pl+Q (TOTAL)	93.08	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	42	43	42	43	42	43	42	43
Circ. Pl (SUS)	-5	-5	-1	-1	-20	-20	8	8
Circ. Q (SUS)	-26	26	-4	4	-267	267	149	-149
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-12	-12	0	0	-9	-9	3	3
Long. Q (SUS)	-96	96	-22	22	-75	75	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	42.0	43.0	42.0	43.0	42.0	43.0	42.0	43.0
Pm+Pl (SUS)	37.0	38.0	41.0	42.0	22.0	23.0	50.0	51.0
Pm+Pl+Q (Total)	98.0	105.0	38.1	46.3	245.0	290.0	199.0	98.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.00	90.70	Passed
Pm+Pl (SUS)	51.00	136.05	Passed
Pm+Pl+Q (TOTAL)	290.00	272.11	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-23	-23	13	13	-18	-18	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-170	170	60	-60	-293	293	183	-183

Front end KOD (V6502 ABC)

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: N2-52"

Nozl: 39

3:25pm

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HEXAGON

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	13.1	12.1	23.1	24.1	8.2	7.2	18.2	19.1
Pm+Pl+Q (Total)	172.2	180.4	72.3	72.4	295.1	298.1	195.2	190.4

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	90.70	Passed
Pm+Pl (SUS)	24.08	136.05	Passed
Pm+Pl+Q (TOTAL)	298.11	272.10	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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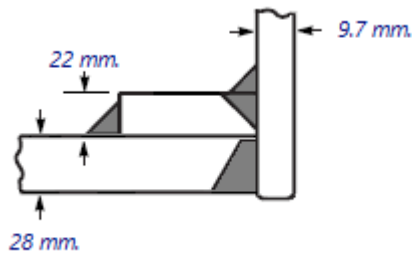
**Input, Nozzle Desc: N3-4"****From: 50**

Pressure for Reinforcement Calculations	P	3.544	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6500.00	mm.
Design Length of Section	L	4481.4995	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10550.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	80.76	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	314.3000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		100.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-4"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.54 \cdot 3252 / (90.7 \cdot 1 - 0.6 \cdot 3.54)$
 $= 12.7342 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.54 \cdot 57.1 / (80.8 \cdot 1 + 0.4 \cdot 3.54)$
 $= 0.2503 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6748 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0618 mm.
Parallel to Vessel Wall, opening length	d	98.0309 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.3364 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 80.8 / 90.7)$
 $= 0.890$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 80.8 / 90.7)$
 $= 0.890$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.89, 1)$
 $= 0.890$

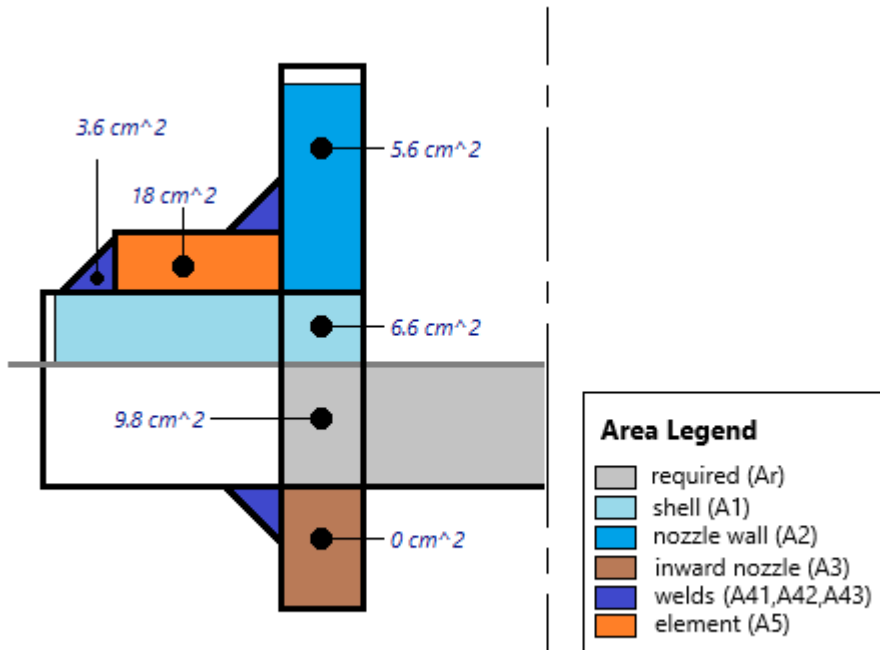


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12.711	9.757	NA
Area in Shell	A1	13.153	6.592	NA
Area in Nozzle Wall	A2	5.944	5.624	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.561	3.561	NA
Area in Element	A5	17.988	17.988	NA
TOTAL AREA AVAILABLE	Atot	40.646	33.765	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (98 \cdot 19.6 \cdot 1 + 2 \cdot 8.13 \cdot 19.6 \cdot 1 (1 - 0.89))$
 $= 9.757 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 98 (1 \cdot 26.4 - 1 \cdot 19.6) - 2 \cdot 8.13$
 $(1 \cdot 26.4 - 1 \cdot 19.6) \cdot (1 - 0.89)$
 $= 6.592 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.3) \cdot (8.13 - 0.67) \cdot 0.89$
 $= 5.624 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$



$$= 20^2 \cdot 0.89 + (0)^2 \cdot 0.89 + 0^2 \cdot 1$$

$$= 3.561 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (196 - 114) \cdot 22 \cdot 1$$

$$= 17.988 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2748 mm.
Wall Thickness per UG16(b),	tr16b = 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 14.3342 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 14.3342 mm.
Wall Thickness, shell/head, external pressure	trb2 = 5.3103 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.3103 mm.
Wall Thickness per table UG-45	tb3 = 6.8578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.86, \max(14.3, 5.31)]$$

$$= 6.8578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.27, 6.86)$$

$$= 6.8578 \text{ mm.}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.2,	Allowable	: 80.8 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 215.3 N./mm ²	Passed
Occasional	: 1.0,	Allowable	: 107.4 N./mm ²	Passed
Shear	: 21.9,	Allowable	: 56.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.17, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.021, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 0.17, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.021, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.73, tr = 0.17, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.021, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, c = 1.6 mm., E* = 1



Thickness Ratio = $tr * E^* / (tg - c) = 0.32$, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 9.73$, $tr = 0.17$, $c = 1.6$ mm., $E^* = 1$
Thickness Ratio = $tr * E^* / (tg - c) = 0.021$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.54/51.00 = 0.069$

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.1346 mm.
Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 * t_{min}$	$14.1400 = 0.7 * W_o$ mm.
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (9.76 - 6.59 + 2 \cdot 8.13 \cdot 0.89 \cdot (1 \cdot 26.4 - 19.6)) \cdot 90.7)$
 $= 3845.31$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.62 + 18 + 3.56 - 0 \cdot 0.89) \cdot 90.7$
 $= 25131.83$ Kg

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.62 + 0 + 3.56 + (3.82)) \cdot 90.7$
 $= 12032.21$ Kg

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (5.62 + 0 + 3.56 + 18 + (3.82)) \cdot 90.7$
 $= 28668.71$ Kg

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 114 \cdot 20 \cdot 0.49 \cdot 80.8$
 $= 14489$ Kg

Shear, Pad Element Weld [Spew]:



$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 314 \cdot 18 \cdot 0.49 \cdot 90.7 \\
 &= 40274. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.14 \cdot 53.1) \cdot (9.73 - 1.6) \cdot 0.7 \cdot 80.8 \\
 &= 7820. \text{ Kg}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 114 \cdot 22 \cdot 0.74 \cdot 90.7 \\
 &= 27034. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.14/2.0) \cdot 114 \cdot (22 - 0) \cdot 0.74 \cdot 90.7 \\
 &= 27034. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (40274 + 7820) = 48093 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (14489 + 27034 + 27034 + 0) = 68557 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (40274 + 27034 + 0) = 67308 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 48093	Kgf, must exceed W = 3845 Kg	or W1 = 25131 Kg
Path 2-2 = 68557	Kgf, must exceed W = 3845 Kg	or W2 = 12032 Kg
Path 3-3 = 67307	Kgf, must exceed W = 3845 Kg	or W3 = 28668 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 0.5025 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 428.5025 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 98.03 mm.

Input Echo, WRC297 Item 1, Description: N3-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6500.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	90.70 N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter Basis for Nozzle

OD



Nozzle Corrosion Allowance		1.6000	mm.
Nozzle Diameter		114.300	mm.
Nozzle Thickness		9.7345	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	80.76	N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter of Reinforcing Pad	314.300	mm.
Thickness of Reinforcing Pad	22.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	3.54	bars
Radial Load	P	764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (4.082) is < 5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(3275.8(28.0 - 1.6))}$$

$$= 1084.753 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6551.600	mm.
Nozzle Outside Diameter	(d) =	114.300	mm.
Vessel Thickness used	(T) =	48.400	mm.
Nozzle Thickness used	(t) =	8.135	mm.
T / t	=	5.950	
d / t	=	14.051	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.203$$

Nr/P	=	0.189
Mr/P	=	0.123
M0/P	=	0.333
N0/P	=	0.275
MrD/Mc	=	0.107
NrDL/Mc	=	0.140
M0d/Mc	=	0.309
N0DL/Mc	=	-0.142
MrD/Ml	=	0.108
NrDL/Ml	=	0.143
M0D/Ml	=	0.305
N0DL/Ml	=	-0.095

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	7	39	25	19
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	11	11	11	11

Inplane Stress Summary	4	16	12	10
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	8	39	25	19
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11

Outplane Stress Summary	-4	27	13	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	23	23	23	23
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Inplane Stress Summary	16	28	24	23
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
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Outplane Shear (Vl)	0	0	0	0
---------------------	---	---	---	---

Torsional Shear (Mt)	2	2	2	2
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Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	21	29	25	23
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (P)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (Mc)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (Mc)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (ML)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (ML)	0	0	0	0
-------------------	---	---	---	---

Hoop Pressure Stress	2	2	2	2
----------------------	---	---	---	---

Hoop Stress Summary	1	2	0	1
---------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-2	-2	-2	-2
---------------------	----	----	----	----

Axial Bending (P)	-19	19	-19	19
--------------------	-----	----	-----	----

Axial Membrane (Mc)	0	0	0	0
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Axial Bending (Mc)	0	0	0	0
--------------------	---	---	---	---

Axial Membrane (ML)	-20	-20	20	20
---------------------	-----	-----	----	----

Axial Bending (ML)	-40	40	40	-40
--------------------	-----	----	----	-----

Axial Pressure Stress	1	1	1	1
-----------------------	---	---	---	---

Axial Stress Summary	-81	36	40	-1
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
-------------------	---	---	----	----

Shear due to (Vl)	0	0	0	0
-------------------	---	---	---	---

Shear due to Torsion	11	11	11	11
----------------------	----	----	----	----

Shear Stress Summary	15	15	8	8
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	89	46	42	17
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	2	2	0	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-19	19	-19	19
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-41	41	41	-41
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-82	38	41	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	15	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	86	40	51	31

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{3264.8(28.0 - 1.6)}$$

$$= 1083.122 \text{ mm.}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 6529.600 \text{ mm.}$$

$$\text{Nozzle Outside Diameter (d)} = 314.300 \text{ mm.}$$

$$\text{Vessel Thickness used (T)} = 26.400 \text{ mm.}$$

$$\text{Nozzle Thickness used (t)} = 108.135 \text{ mm.}$$

$$T / t = 0.244$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.757$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$Nr/P = 0.122$$

$$Mr/P = 0.267$$

$$M0/P = 0.078$$



NØ/P = 0.135
 MrD/Mc = 0.314
 NrDL/Mc = 0.062
 MØd/Mc = 0.097
 NØDL/Mc = 0.031
 MrD/Ml = 0.231
 NrDL/Ml = 0.087
 MØD/Ml = 0.072
 NØDL/Ml = 0.093

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-5	5	-5	5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	43	43	43	43
Outplane Stress Summary	33	49	40	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-17	17	-17	17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-8	8	8	-8
Inplane Pressure Stress	21	21	21	21
Inplane Stress Summary	-6	45	12	29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	39	49	40	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-5	5	-5	5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	21	21	21	21
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Outplane Stress Summary	11	28	19	21
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-17	17	-17	17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-12	12	12	-12
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	43	43	43	43

Inplane Stress Summary	12	71	37	48
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	12	71	37	48
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	23	23	23	23	23	23	23
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-15	15	3	-3	-5	5	1	-1
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-5	5	1	-1	-15	15	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0
Pm+Pl (SUS)	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0
Pm+Pl+Q (Total)	9.2	38.2	26.3	20.4	22.4	29.2	24.4	22.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------



Pm (SUS)	23.00	90.70	Passed
Pm+Pl (SUS)	23.00	136.05	Passed
Pm+Pl+Q (TOTAL)	38.18	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	43	43	43	43	43	43	43
Circ. Pl (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Circ. Q (SUS)	-7	7	-3	3	-29	29	-5	5
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Long. Q (SUS)	-25	25	-9	9	-8	8	-2	2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	43.0	43.0	43.0	43.0	43.0	43.0	43.0	43.0
Pm+Pl (SUS)	42.0	42.0	42.0	42.0	42.0	42.0	42.0	42.0
Pm+Pl+Q (Total)	40.0	49.0	39.0	45.0	13.0	71.0	37.0	47.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.00	90.70	Passed
Pm+Pl (SUS)	42.00	136.05	Passed
Pm+Pl+Q (TOTAL)	71.00	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-22	-22	18	18	-22	-22	18	18
Long. Q (SUS)	-59	59	21	-21	-60	60	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	23.8	23.8	19.5	19.5	23.8	23.8	19.5	19.5
Pm+Pl+Q (Total)	86.6	45.6	41.6	16.5	84.5	40.7	48.0	28.4

**Nozzle Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	80.76	Passed
Pm+Pl (SUS)	23.77	121.14	Passed
Pm+Pl+Q (TOTAL)	86.65	242.27	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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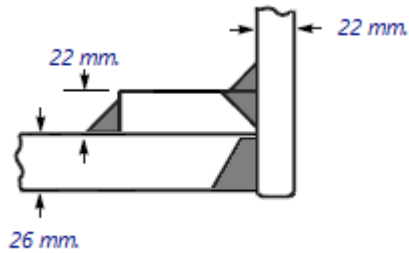
**Input, Nozzle Desc: N1B-40"****From: 70**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6500.00	mm.
Design Length of Section	L	3880.1240	mm.
Shell Finished (Minimum) Thickness	t	26.0000	mm.
Shell Internal Corrosion Allowance	cas	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		17000.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	90.70	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		40.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	90.70	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1916.0001	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		450.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1B-40"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	40.000 in.
Actual Thickness Used in Calculation	0.866 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3252 / (90.7 \cdot 1 - 0.6 \cdot 3.5)$
 $= 12.5771 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 508 / (90.7 \cdot 1 + 0.4 \cdot 3.5)$
 $= 1.9574 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 2.2800 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1950.3999 mm.
Parallel to Vessel Wall, opening length	d	975.2000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		61.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 90.7 / 90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

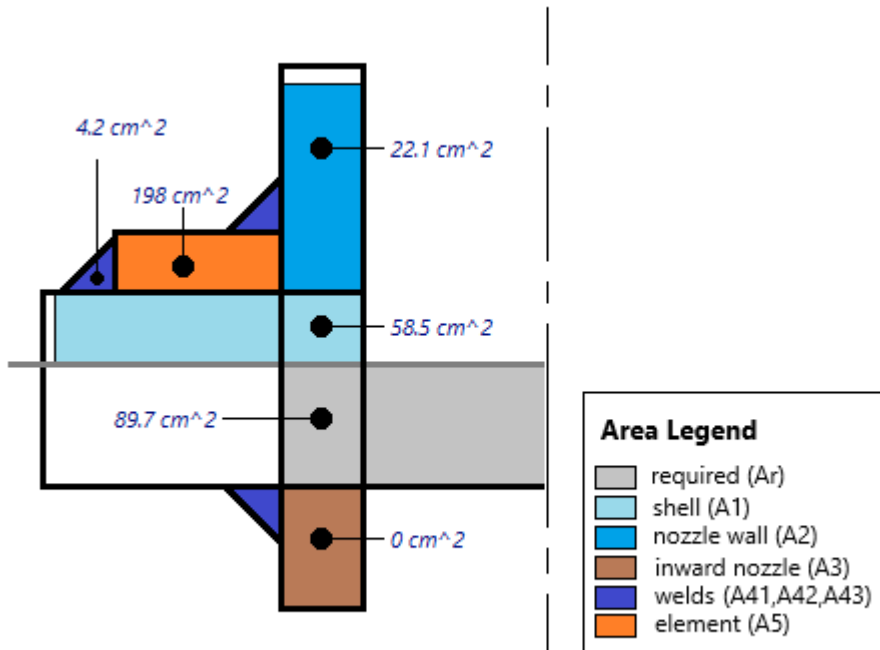


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	122.652	89.723	NA
Area in Shell	A1	115.297	58.502	NA
Area in Nozzle Wall	A2	22.499	22.106	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	4.233	4.233	NA
Area in Element	A5	197.995	197.995	NA
TOTAL AREA AVAILABLE	Atot	340.025	282.837	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1038.1895 22.0000 mm.
 Based on given Pad Diameter: 1916.0001 0.5424 mm.
 Based on Shell or Nozzle Thickness: 1038.1895 22.0000 mm.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (975 \cdot 18.4 \cdot 1 + 2 \cdot 20.4 \cdot 18.4 \cdot 1 (1 - 1))$
 $= 89.723 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 975 (1 \cdot 24.4 - 1 \cdot 18.4) - 2 \cdot 20.4$
 $(1 \cdot 24.4 - 1 \cdot 18.4) \cdot (1 - 1)$
 $= 58.502 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 61) \cdot (20.4 - 2.28) \cdot 1$$

$$= 22.106 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (1) \cdot 1 + (0) \cdot 1 + 323^2 \cdot 1$$

$$= 4.233 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot \text{fr4}$$

$$= (1916 - 1016) \cdot 22 \cdot 1$$

$$= 197.995 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8800 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 14.1771 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 14.1771 \text{ mm.}$
Wall Thickness, shell/head, external pressure	$trb2 = 5.3103 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 5.3103 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 9.9312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.93, \max(14.2, 5.31)]$$

$$= 9.9312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.88, 9.93)$$

$$= 9.9312 \text{ mm.}$$

Available Nozzle Neck Thickness = 22.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.0,	Allowable	:	90.7 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.9 N./mm ²	Passed
Occasional	:	4.1,	Allowable	:	120.6 N./mm ²	Passed
Shear	:	8.8,	Allowable	:	63.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

$$\text{Govrn. thk, } t_g = 22, t_r = 1.29, c = 1.6 \text{ mm.}, E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.063, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 22, t_r = 1.29, c = 1.6 \text{ mm.}, E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.063, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 22, t_r = 1.29, c = 1.6 \text{ mm.}, E^* = 1$$

$$\text{Thickness Ratio} = t_r \cdot E^* / (t_g - c) = 0.063, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.34$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 22, tr = 1.29, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.063$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50/51.00 = 0.069$

Weld Size Calculations, Description: N1B-40"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (89.7 - 58.5 + 2 \cdot 20.4 \cdot 1 \cdot (1 \cdot 24.4 - 18.4)) \cdot 90.7)$
 $= 31139.50 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (22.1 + 198 + 4.23 - 0 \cdot 1) \cdot 90.7$
 $= 207484.30 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (22.1 + 0 + 1 + (9.95)) \cdot 90.7$
 $= 30577.49 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (22.1 + 0 + 4.23 + 198 + (9.95)) \cdot 90.7$
 $= 216691.53 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\
 &= (3.14/2.0) \cdot 1016 \cdot 10 \cdot 0.49 \cdot 90.7 \\
 &= 72325. \text{ Kg}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 1916 \cdot 18 \cdot 0.49 \cdot 90.7 \\
 &= 245506. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.14 \cdot 498) \cdot (22 - 1.6) \cdot 0.7 \cdot 90.7 \\
 &= 206544. \text{ Kg}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 1016 \cdot 22 \cdot 0.74 \cdot 90.7 \\
 &= 240296. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.14/2.0) \cdot 1016 \cdot (22 - 0) \cdot 0.74 \cdot 90.7 \\
 &= 240302. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (245506 + 206544) = 452050 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (72325 + 240296 + 240302 + 0) = 552923 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (245506 + 240302 + 0) = 485808 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 452049 Kg, must exceed W = 31139 Kg	or W1 = 207484 Kg
Path 2-2 = 552922 Kg, must exceed W = 31139 Kg	or W2 = 30577 Kg
Path 3-3 = 485808 Kg, must exceed W = 31139 Kg	or W3 = 216691 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 39.9478 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 415.9477 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.213 \%$$

This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 922.67 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 975.20 mm.

Input Echo, WRC297 Item 1, Description: N1B-40" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6500.000	mm.
Shell Thickness	26.0000	mm.
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	90.70	N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		1.6000	mm.
Nozzle Diameter		1016.000	mm.
Nozzle Thickness		22.0000	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	90.70	N./mm ²

Using 3*S for Discontinuity Allowable (UG-23(e)) in Creep range.

Diameter of Reinforcing Pad	1916.000	mm.
Thickness of Reinforcing Pad	22.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	7648.38	Kgf
Circumferential Shear	Vc	5098.92	Kgf
Longitudinal Shear	Vl	5098.92	Kgf
Circumferential Moment	Mc	14271.47	Kg-m.
Longitudinal Moment	Ml	14271.47	Kg-m.
Torsional Moment	Mt	20387.82	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1016.0 + 2 \cdot 1.65 \cdot \sqrt{(3274.8(26.0 - 1.6))}$$

$$= 1948.827 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6549.600	mm.
Nozzle Outside Diameter	(d) =	1016.000	mm.
Vessel Thickness used	(T) =	46.400	mm.
Nozzle Thickness used	(t) =	20.400	mm.
T / t	=	2.275	
d / t	=	49.804	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.843$$

Nr/P	=	0.048
Mr/P	=	0.088
M0/P	=	0.053
N0/P	=	0.165
MrD/Mc	=	0.181
NrDL/Mc	=	0.090
M0d/Mc	=	0.129
N0DL/Mc	=	0.222
MrD/Ml	=	0.104
NrDL/Ml	=	0.058
M0D/Ml	=	0.064
N0DL/Ml	=	0.321

**Vessel Stresses (N./mm²):**

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-20	-20	20	20
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	24	24	24	24
Outplane Stress Summary	-37	34	52	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-18	18	-18	18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	12	12	12	12
Inplane Stress Summary	-51	65	35	-7

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	52	65	52	33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-11	11	-11	11
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-49	49	49	-49
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12
Outplane Stress Summary	-68	53	59	-17

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside



Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-18	18	-18	18
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-69	69	69	-69
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	24	24	24	24

Inplane Stress Summary	-70	105	79	-22
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	72	105	80	24
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10

Hoop Stress Summary	-16	-16	24	25
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-149	149	149	-149
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-224	213	93	-67
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	7	7	4	4
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Nozzle Stresses (N./mm²):



LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	224	230	93	93

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-5	-5	-5	-5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	-10	-9	18	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-69	69	-69	69
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-269	269	269	-269
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-343	333	213	-187

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	7	7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	344	343	213	207

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1016.0 + 2 \cdot 1.65 \cdot \sqrt{(3263.8(26.0 - 1.6))}$$

$$= 1947.259 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6527.600 mm.
Nozzle Outside Diameter	(d) =	1916.000 mm.
Vessel Thickness used	(T) =	24.400 mm.
Nozzle Thickness used	(t) =	470.400 mm.
T / t	=	0.052
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.801$$



Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.023
MrD/Mc	=	0.241
NrDL/Mc	=	0.104
M0d/Mc	=	0.072
N0DL/Mc	=	0.046
MrD/M1	=	0.049
NrDL/M1	=	0.053
M0D/M1	=	0.015
N0DL/M1	=	0.024

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	46	46	46	46
Outplane Stress Summary	14	66	42	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-53	53	-53	53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-79	100	6	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	94	101	42	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-52	52	52	-52
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	-53	82	64	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-53	53	-53	53
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-177	177	177	-177
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	46	46	46

Inplane Stress Summary	-203	259	176	-70
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	203	259	177	70
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	24	24	24	24	24	24	24
Circ. Pl (SUS)	-25	-25	15	15	-6	-6	4	4
Circ. Q (SUS)	-35	35	13	-13	-87	87	51	-51
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-4	-4	2	2	-19	-19	9	9
Long. Q (SUS)	-58	58	22	-22	-60	60	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	24.0	24.0	24.0	24.0	24.0	24.0	24.0	24.0
Pm+Pl (SUS)	9.0	9.0	39.0	39.0	25.0	25.0	28.0	28.0
Pm+Pl+Q (Total)	50.3	66.1	52.2	34.2	70.2	105.1	79.2	23.6

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	90.70	Passed
Pm+Pl (SUS)	39.00	136.05	Passed
Pm+Pl+Q (TOTAL)	105.08	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	46	46	46	46	46	46	46
Circ. Pl (SUS)	-4	-4	0	0	-17	-17	7	7
Circ. Q (SUS)	-25	25	-3	3	-230	230	124	-124
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-11	-11	1	1	-7	-7	3	3
Long. Q (SUS)	-89	89	-17	17	-66	66	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	46.0	46.0	46.0	46.0	46.0	46.0	46.0	46.0
Pm+Pl (SUS)	42.0	42.0	46.0	46.0	29.0	29.0	53.0	53.0
Pm+Pl+Q (Total)	94.0	101.0	43.0	49.1	201.0	259.0	177.0	71.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	90.70	Passed
Pm+Pl (SUS)	53.00	136.05	Passed
Pm+Pl+Q (TOTAL)	259.01	272.11	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-25	-25	15	15	-19	-19	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-9	-9	7	7	-9	-9	7	7
Long. Q (SUS)	-218	218	80	-80	-338	338	200	-200
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6

Front end KOD (V6502 ABC)

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: N1B-40"

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Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	16.1	15.1	24.1	25.1	10.2	9.2	18.1	19.1
Pm+Pl+Q (Total)	223.2	228.4	91.4	94.5	343.1	342.1	211.3	208.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	90.70	Passed
Pm+Pl (SUS)	25.07	136.05	Passed
Pm+Pl+Q (TOTAL)	343.08	272.10	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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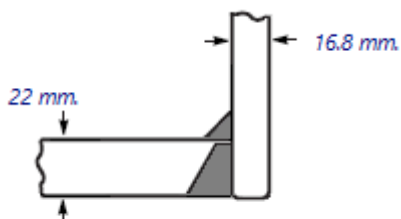
**Input, Nozzle Desc: K1A****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K1A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp([P / (S \cdot E)] - 1) \right)$ per Appendix 1-2 (a)(1)
 $= 27 \left(\exp([3.5 / (90.8 \cdot 1)] - 1) \right)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.9841 mm.
Parallel to Vessel Wall	Rn+tn+t	63.4920 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50 / 51.00 = 0.069$

Weld Size Calculations, Description: K1A

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0304 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.1446 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.1446 mm.
Wall Thickness, shell/head, external pressure	trb2	= 5.3058 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.3058 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.03, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K1A

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: K1A



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Noz1: 42 3:25pm Sep 16,2026



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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.3268 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.4547 mm.

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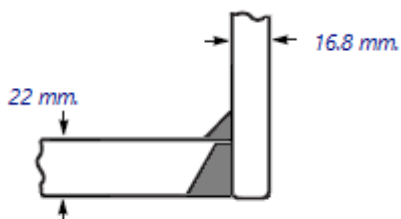
**Input, Nozzle Desc: K2A****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		350.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K2A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp([P / (S \cdot E)] - 1) \right)$ per Appendix 1-2 (a)(1)
 $= 27 \left(\exp([3.5 / (90.8 \cdot 1)] - 1) \right)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.9841 mm.
Parallel to Vessel Wall	Rn+tn+t	63.4920 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50 / 51.00 = 0.069$

Weld Size Calculations, Description: K2A

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0304 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.1446 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.1446 mm.
Wall Thickness, shell/head, external pressure	trb2	= 5.3058 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.3058 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.03, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K2A

Intermediate Calc. for nozzle/shell welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: K2A



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Nozl: 43 3:25pm Sep 16,2026



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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.3268 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.4547 mm.

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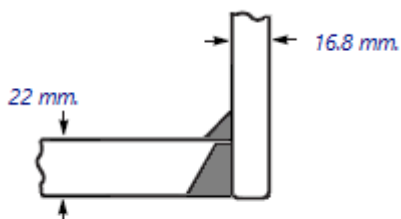
**Input, Nozzle Desc: K3A****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		10.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K3A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp([P / (S \cdot E)] - 1) \right)$ per Appendix 1-2 (a)(1)
 $= 27 \left(\exp([3.5 / (90.8 \cdot 1)] - 1) \right)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.9841 mm.
Parallel to Vessel Wall	Rn+tn+t	63.4920 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50 / 51.00 = 0.069$

Weld Size Calculations, Description: K3A

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0304 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.1446 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.1446 mm.
Wall Thickness, shell/head, external pressure	trb2	= 5.3058 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.3058 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.03, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K3A

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: K3A



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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.3268 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.4547 mm.

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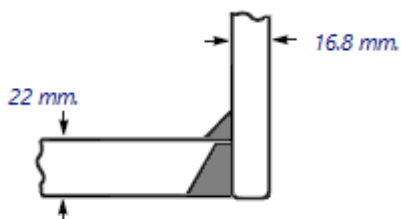
**Input, Nozzle Desc: K4A****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		20.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K4A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp\left(\frac{P}{S \cdot E}\right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 27 \left(\exp\left(\frac{3.5}{90.8 \cdot 1}\right) - 1 \right)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.9841 mm.
Parallel to Vessel Wall	Rn+tn+t	63.4920 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.50 / 51.00 = 0.069$

Weld Size Calculations, Description: K4A

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0304 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.1446 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.1446 mm.
Wall Thickness, shell/head, external pressure	trb2	= 5.3058 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.3058 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.03, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K4A

Intermediate Calc. for nozzle/shell welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: K4A



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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.3268 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.4547 mm.

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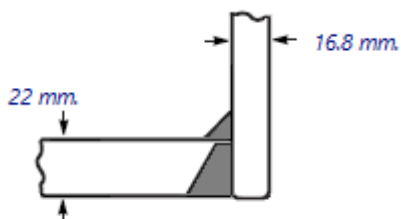
**Input, Nozzle Desc: K5A****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K5A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
= 11.2922 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp([P / (S_n \cdot E)] - 1) \right)$ per Appendix 1-2 (a)(1)
= $27 \left(\exp([3.5 / (90.8 \cdot 1)] - 1) \right)$
= 0.1043 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.9841	mm.
Parallel to Vessel Wall	Rn+tn+t	63.4920	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5A.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0045$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.50 / 51.00 = 0.069$

Weld Size Calculations, Description: K5A

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5A

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5A.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0304 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.1446 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.1446 mm.
Wall Thickness, shell/head, external pressure	trb2	= 5.3058 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.3058 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.03, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K5A

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Front end KOD (V6502 ABC).

PV Elite 28 U1 Licensee: SL User

FileName : Front end Flare KOD (V6502 ABC) 9%

Nozzle Calcs.: K5A



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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.3268 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.4547 mm.

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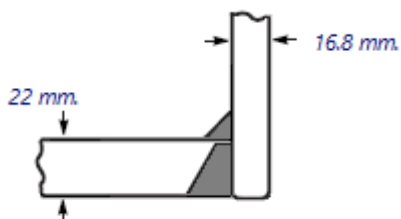
**Input, Nozzle Desc: K1B****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K1B**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

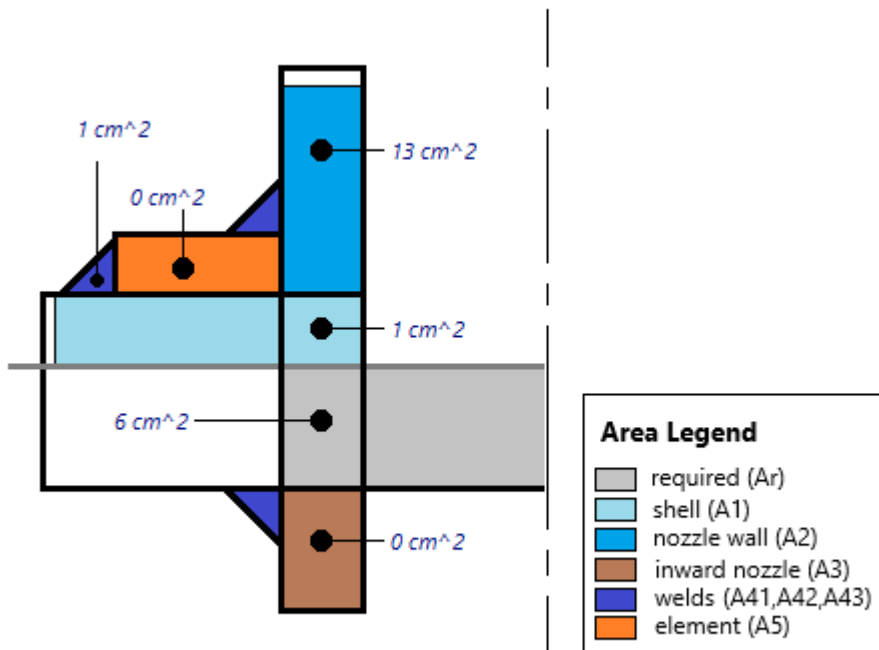


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1 - 1))$

= 5.977 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$

$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$

= 1.025 cm²

Area Available in Nozzle Projecting Outward [A2]:

= $(2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$

= $(2 \cdot 38)(15.2 - 0.43)/\sin(60)$

= 12.960 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K1B

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 4984.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7$
 $= 12911.61 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (13 + 0 + 1 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 7015. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8$
 $= 12502. \text{ Kg}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W1 = 12911 \text{ Kgf}$$

$$\text{Path 2-2} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W2 = 18647 \text{ Kgf}$$

$$\text{Path 3-3} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W3 = 18647 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K1B

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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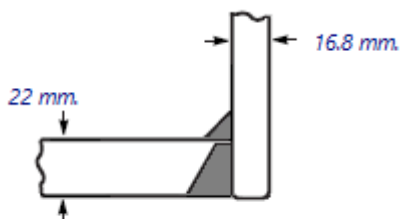
**Input, Nozzle Desc: K2B****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		190.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K2B**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

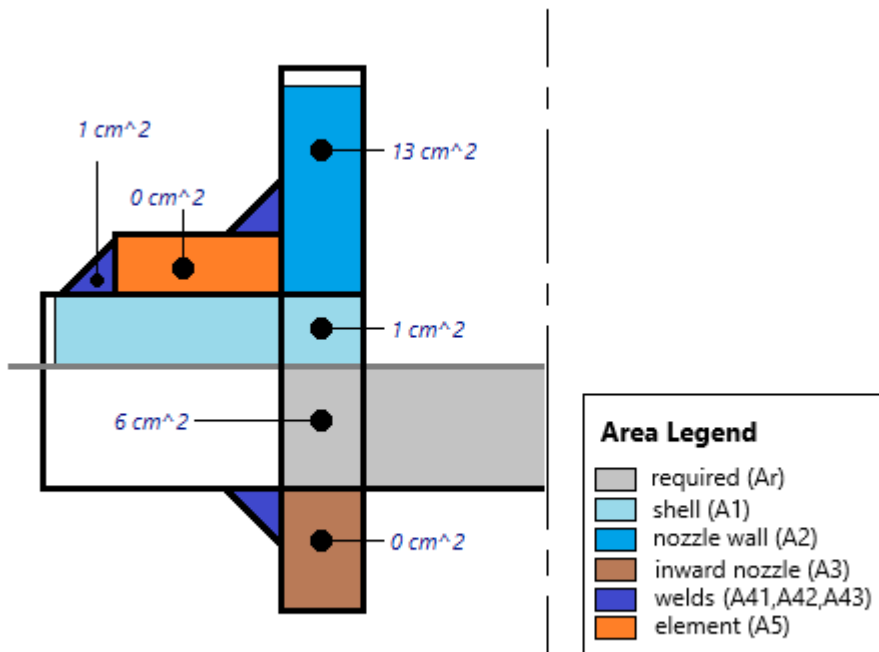


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1 - 1))$

= 5.977 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$

$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$

= 1.025 cm²

Area Available in Nozzle Projecting Outward [A2]:

= $(2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$

= $(2 \cdot 38)(15.2 - 0.43)/\sin(60)$

= 12.960 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K2B

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 4984.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7$
 $= 12911.61 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (13 + 0 + 1 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 7015. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8$
 $= 12502. \text{ Kg}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W1 = 12911 \text{ Kgf}$$

$$\text{Path 2-2} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W2 = 18647 \text{ Kgf}$$

$$\text{Path 3-3} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W3 = 18647 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K2B

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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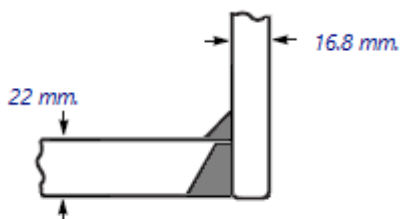
**Input, Nozzle Desc: K3B****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		170.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K3B**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

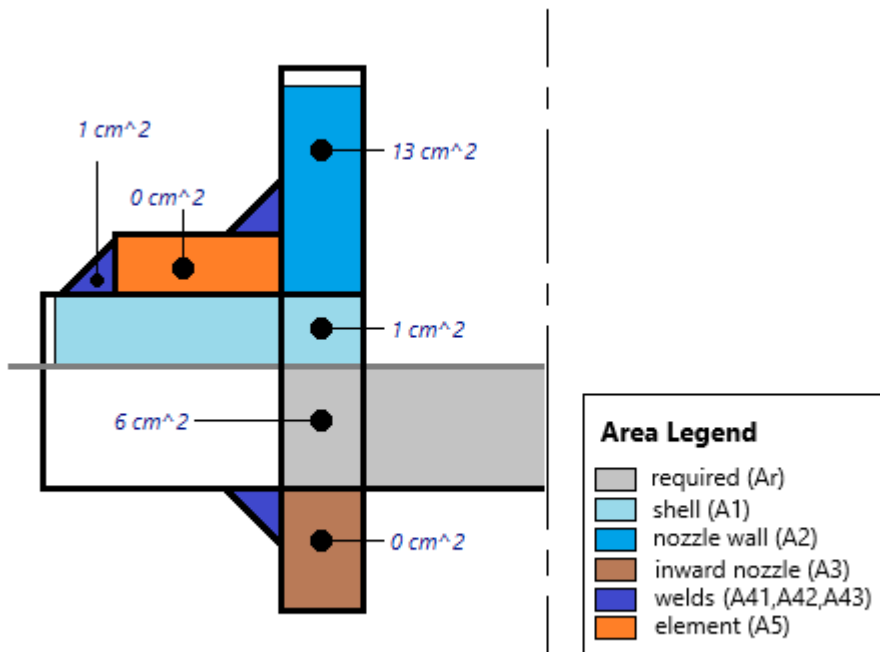


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1 - 1))$

= 5.977 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$

$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$

= 1.025 cm²

Area Available in Nozzle Projecting Outward [A2]:

= $(2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$

= $(2 \cdot 38)(15.2 - 0.43)/\sin(60)$

= 12.960 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K3B

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 4984.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7$
 $= 12911.61 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (13 + 0 + 1 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 7015. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8$
 $= 12502. \text{ Kg}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W1 = 12911 \text{ Kg}$$

$$\text{Path 2-2} = 28648 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W2 = 18647 \text{ Kg}$$

$$\text{Path 3-3} = 28648 \text{ Kg}, \text{ must exceed } W = 4984 \text{ Kg} \quad \text{or } W3 = 18647 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K3B.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K3B

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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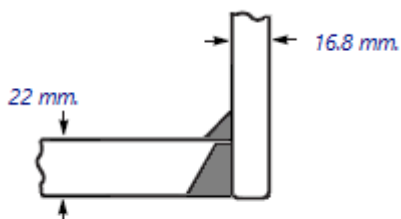
**Input, Nozzle Desc: K4B****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		160.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K4B**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

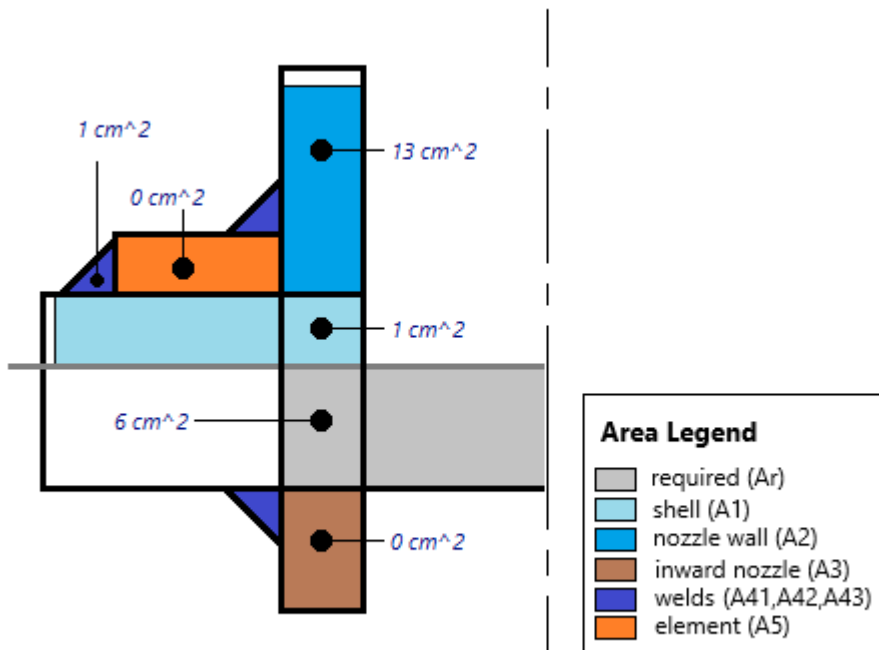


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1 - 1))$

= 5.977 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$

$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$

= 1.025 cm²

Area Available in Nozzle Projecting Outward [A2]:

= $(2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$

= $(2 \cdot 38)(15.2 - 0.43)/\sin(60)$

= 12.960 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K4B

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7)$
 $= 4984.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 $= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7$
 $= 12911.61 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (13 + 0 + 1 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 $= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7$
 $= 18647.39 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7$
 $= 7015. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8$
 $= 12502. \text{ Kg}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W1 = 12911 \text{ Kgf}$$

$$\text{Path 2-2} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W2 = 18647 \text{ Kgf}$$

$$\text{Path 3-3} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W3 = 18647 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K4B.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K4B

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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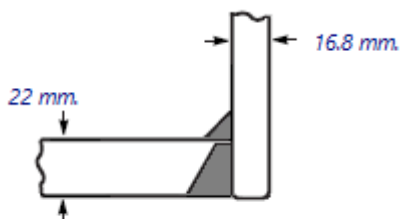
**Input, Nozzle Desc: K5B****From: 90**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	415	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	415	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	90.70	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	22.0000	mm.
Head Internal Corrosion Allowance	cas	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	2500.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	90.80	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		150.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K5B**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$
 $= 11.2922 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1) - 1)$
 $= 0.1043 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	134.2452 mm.
Parallel to Vessel Wall	Rn+tn+t	67.1226 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 90.8/90.7)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

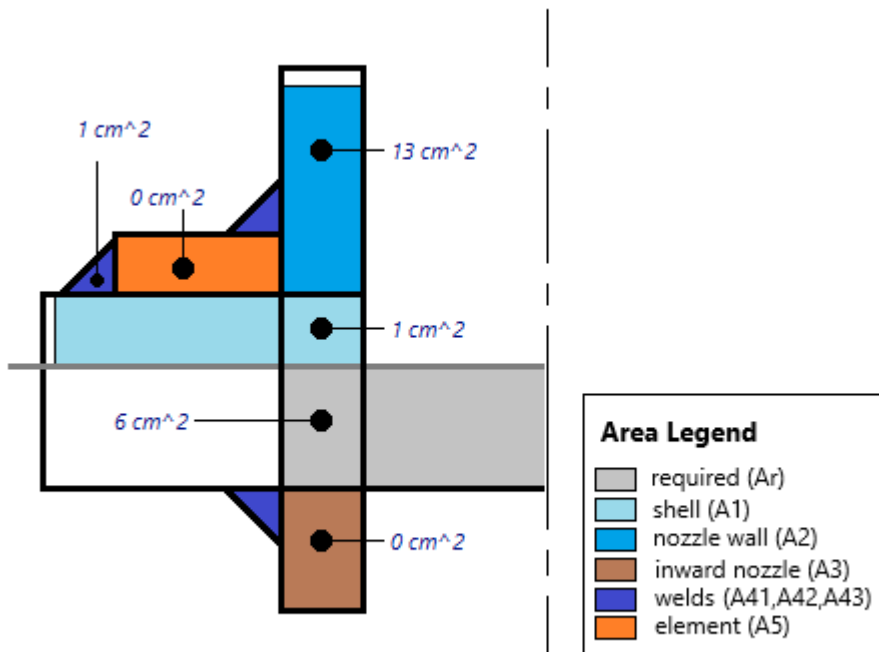


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.119	5.977	NA
Area in Shell	A1	6.485	1.025	NA
Area in Nozzle Wall	A2	13.246	12.960	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	20.731	14.985	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

58.93 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(63 \cdot 19 \cdot 1 + 2 \cdot 15.2 \cdot 19 \cdot 1(1 - 1))$

= 5.977 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $71.2(1 \cdot 20.4 - 1 \cdot 19) - 2 \cdot 15.2$

$(1 \cdot 20.4 - 1 \cdot 19) \cdot (1 - 1)$

= 1.025 cm²

Area Available in Nozzle Projecting Outward [A2]:

= $(2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$

= $(2 \cdot 38)(15.2 - 0.43)/\sin(60)$

= 12.960 cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)**

Govrn. thk, tg = 16.8, tr = 0.069, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.0045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 3.50/51.00 = 0.069

Weld Size Calculations, Description: K5B

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\
 &= \max(0, (5.98 - 1.02 + 2 \cdot 15.2 \cdot 1 \cdot (1 \cdot 20.4 - 19)) \cdot 90.7) \\
 &= 4984.99 \text{ Kg}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v \\
 &= (13 + 0 + 1 - 0 \cdot 1) \cdot 90.7 \\
 &= 12911.61 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\
 &= (13 + 0 + 1 + (6.2)) \cdot 90.7 \\
 &= 18647.39 \text{ Kg}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\
 &= (13 + 0 + 1 + 0 + (6.2)) \cdot 90.7 \\
 &= 18647.39 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 98.5 \cdot 10 \cdot 0.49 \cdot 90.7 \\
 &= 7015. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 40.4) \cdot (16.8 - 1.6) \cdot 0.7 \cdot 90.8 \\
 &= 12502. \text{ Kg}
 \end{aligned}$$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_o \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 98.5 \cdot (22 - 1.6) \cdot 0.74 \cdot 90.8$$

$$= 21634. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7015 + 12502) = 19517 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 0 + 21634 + 0) = 28649 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (7015 + 21634 + 0) = 28649 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 19516 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W1 = 12911 \text{ Kgf}$$

$$\text{Path 2-2} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W2 = 18647 \text{ Kgf}$$

$$\text{Path 3-3} = 28648 \text{ Kgf, must exceed } W = 4984 \text{ Kgf} \quad \text{or } W3 = 18647 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.

Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6503 \cdot 0.9 / (2 \cdot 90.7 \cdot 1 - 0.2 \cdot 3.5)$$

$$= 11.2922 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \cdot (\exp([3.5 / (90.8 \cdot 1)] - 1))$$

$$= 0.1043 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4304 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 125.2000 mm.

Parallel to Vessel Wall Rn+tn+t 62.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.0000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K5B.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles



that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.0304$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 4.1000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 14.1446$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 14.1446$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 5.3058$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.3058$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(14.1, 5.31)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(2.03, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K5B

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 20.0802 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 250.7626 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: